# Vorkurs Mathematik
-
# Introductory Course Mathematics
## Part 2 (continuous)

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1. Real numbers
2. Limit of a sequence, finite vs. infinite sums
3. Elementary functions
4. Continuity
5. Differentiation and Integration



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1 Real numbers
From natural to real
Properties of $\mathbb{R}$
Systems of equations and fields



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# New numbers for solving equations

- Many problems in Physics (but also in general) can be rephrased as (systems of) equations.

EG "Textbook free fall"

- Vertical velocity after $t$ seconds: $v(t) = v_0 - gt$ ($v_0$ upwards)
- Vertical position after $t$ seconds: $y(t) = y_0 + v_0 t - \frac{1}{2}gt^2$.

Q1: Determine $t$ so that $v(t) = 0$, i.e. solve $0 \stackrel{!}{=} v_0 - gt$.

Q2: Determine $t$ so that $y(t) = 0$, i.e. solve $0 \stackrel{!}{=} y_0 + v_0 t - \frac{1}{2}gt^2$.

- While the "laws" (postulates, hypotheses, assumptions) allow to easily compute $v(t)$ and $y(t)$ for a given $t$ ("forward problem"), both questions require to compute "backwards" ("inverse problem").
- Q1 leads to a linear equation $ax + b = 0$ which in general does not have a solution if we constrain $x$ to $\mathbb{N}_0$ e.g. consider $x + 1 = 0$ or $5x = 1$.

  One has to introduce both the concept of "take away/subtract/move left on the number line" and also "break the unit measure into $q$ equal parts" which leads to the extension of $\mathbb{N}$ to $\mathbb{Z}$ and subsequently $\mathbb{Q}$.

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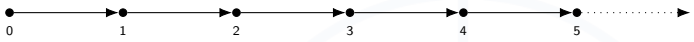
- While the "laws" (postulates, hypotheses, assumptions) allow to easily compute $v(t)$ and $y(t)$ for a given $t$ ("forward problem"), both questions require to compute "backwards" ("inverse problem").
- Q2 leads to a quadratic equation $ax^2 + bx + c = 0$ which in general does not have a solution even in $\mathbb{Q}$. E.g. consider $x^2 = 2$ (recall Euclid).

  But we can approximate $\sqrt{2}$ with arbitrary precision using only rational numbers. This leads to extension of $\mathbb{Q}$ to $\mathbb{R}$ as the numbers which can approximated with arbitrary precision using only rational numbers.

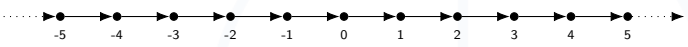
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# Integers and rational numbers

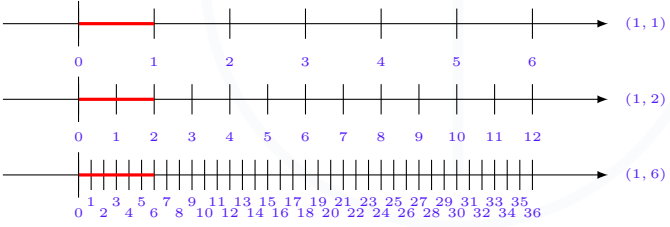
- $\mathbb{N}_0$ corresponds to the "number half-line" (see Peano axioms):




- The integers $\mathbb{Z} = \mathbb{N}_0 \cup \{-n \mid n \in \mathbb{N}\}$ extend $\mathbb{N}_0$ with the notion of "left of $0$":



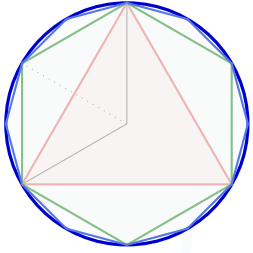

- The rational numbers $\mathbb{Q} = \{\frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{N}\}$ allow to break the unit $1$ of measure into $q$ equal parts ("fractions"):

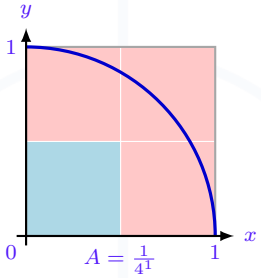



- Building up on $\mathbb{N}_0$ we can "construct (or implement)" both $\mathbb{Z}$ and $\mathbb{Q}$.

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# Approximating $\pi$ using straightedge and compass





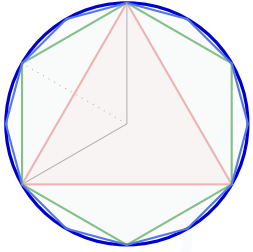

- Already in ancient times, irrational numbers like $\pi$ where approximated. E.g. purely constructive/geometric methods are:
  - Start with an equilateral triangle and then repeatedly bisect angles. But this uses (in the modern sense) irrational numbers.
  - Recursively subdivide the unit square (bisect the edges) and count. This yields a sequence of rational underapproximations $(a_i)_{i\in\mathbb{N}}$ with $a_i = 4 \cdot \frac{n_i}{4^i}$:

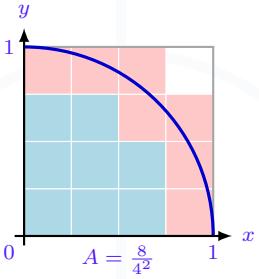
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    (The computer/machine does not mind doing this "mindless repetitive/mechanical" work. This is also one way how to "digitize" in general.)

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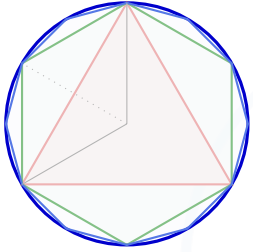

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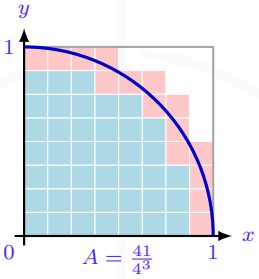
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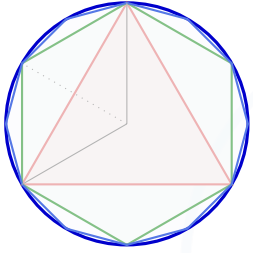

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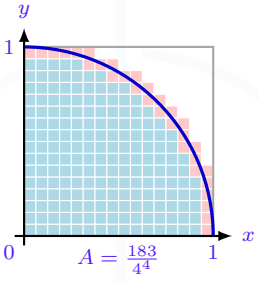
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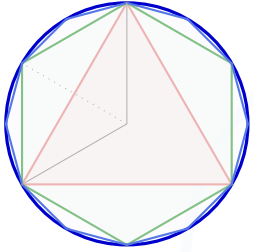

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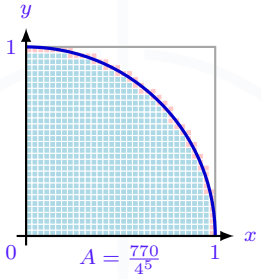
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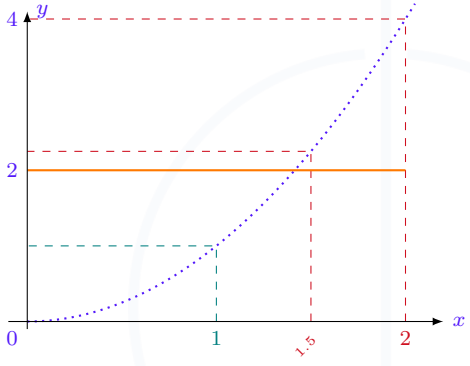

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# Approximating $\sqrt{2}$ and from $\mathbb{Q}$ to $\mathbb{R}$



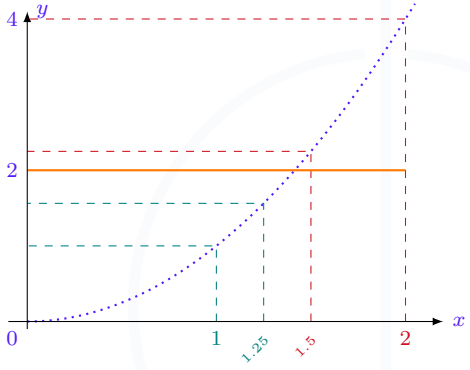

- Straightedge and compass also allow to square rational numbers (intercept theorem), so that bisection (binary search) can be implemented "geometrically" to obtain rational (under/over-)approximations of e.g. $\sqrt{2}$:

$$1 < \tfrac{5}{4} = 1.25 < \tfrac{11}{8} = 1.375 < \tfrac{45}{32} = 1.40625 < \tfrac{181}{128} = 1.4140625 < \ldots < \sqrt{2} < 2$$

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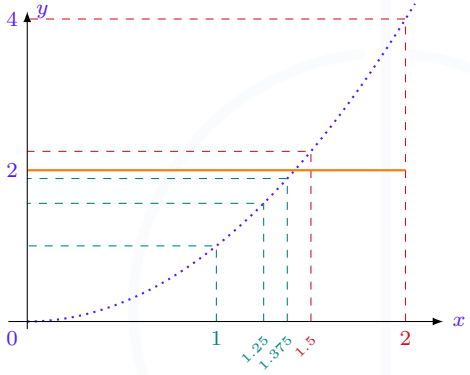

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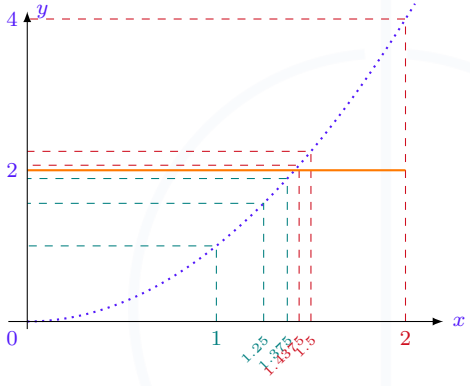

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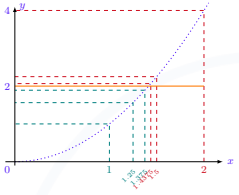

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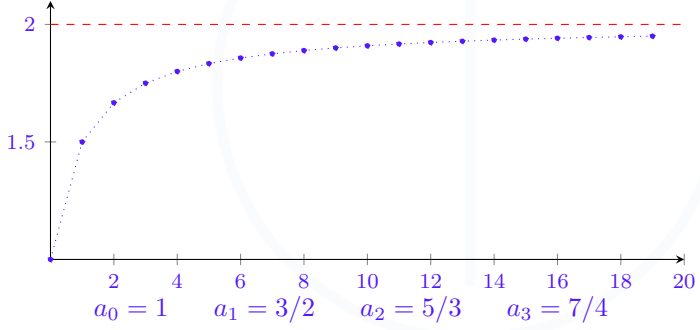

- The real numbers $\mathbb{R}$ are obtained from $\mathbb{Q}$ by accepting bounded monotonically increasing (or decreasing) sequences of rational numbers as definitions of potentially irrational numbers. (The main problem is to show that this is well-defined independent of the way of how we approximate; cf. analysis.)
- For many applications, we compute only with rational approximations anyways (see lim); the bound guarantees that we get increasingly precise approximations of a well-defined real number. (We can also compute "symbolically" using essentially logic to only relate the definitions.)
- Actually, we want to have a program that can calculate these approximations for us. But just as most real numbers are inconstructible using straightedge and compass (and finitely many steps), also most real numbers are uncomputable using a computer.

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# Construction of $\mathbb{R}$

- Idea: Every rational sequence $(a_i)_{i\in\mathbb{N}_0}$ ($a_i \in \mathbb{Q}$) that
  1. is monotonically increasing (i.e. always $a_i \leq a_{i+1}$) and
  2. bounded from above (i.e. there is some $b \in \mathbb{Q}$ s.t. always $a_i \leq b$)

  defines (determines) a number called its least upper bound (supremum):
- E.g. $a_i = 2 - \frac{1}{i+1}$ has $2$ as least upper bound.




$$|2 - a_i| = \left|2 - \left(2 - \tfrac{1}{i+1}\right)\right| = \tfrac{1}{i+1} \xrightarrow{i\to\infty} 0$$

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(xkcd, cf. Zeno)

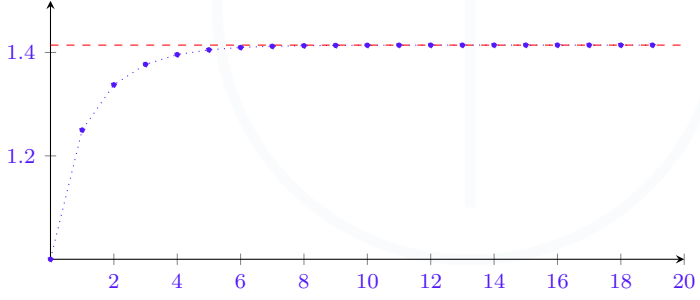
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- E.g. $a_{i+1} := a_i + \frac{2-a_i^2}{4a_i}$ with $a_0 := 1$

  (1) is strictly increasing, (2) is bounded by every $b \in \mathbb{Q}_{\geq 0}$ with $b^2 > 2$ from above, (3) exceeds every $c \in \mathbb{Q}_{\geq 0}$ with $c^2 < 2$ eventually.




$a_5 = 1.405085\ldots \quad a_{15} = 1.414204\ldots \quad a_{19} = 1.414213\ldots \quad \in \mathbb{Q}$

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defines (determines) a number called its least upper bound (supremum):

▷ $\mathbb{R}$ is thus constructed from $\mathbb{Q}$ by accepting also upper-bounded montonically increasing sequences as definitions (descriptions) of (irrational) numbers.

Multiple formalizations (constructions/definitions) exists.

But all of them are equivalent to the supremum property:

**Def** $\mathbb{R}$ is the (least) extension of $\mathbb{Q}$ so that every non-empty upper-bounded set $X \subseteq \mathbb{R}$ has a (unique) supremum $\sup X$ in $\mathbb{R}$.

**Def** $o \in \mathbb{R}$ is an upper bound of $X \subseteq \mathbb{R}$ if $x \leq o$ for all $x \in X$.

**Def** $s \in \mathbb{R}$ is the least upper bound/supremum of $X \subseteq \mathbb{R}$ if (i) $s$ is an upper bound of $X$ and (ii) $s \leq o$ for every upper bound $o$ of $X$.

($X = \{a_i \mid i \in \mathbb{N}_0\}$ is used instead of $(a_i)_{i\in\mathbb{N}_0}$.)

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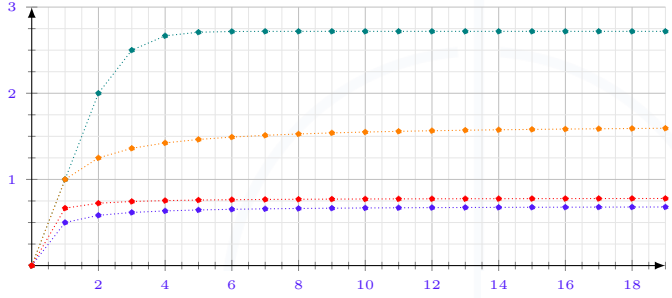
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- ▷ We only extend $\mathbb{Q}$ by as many new "numbers" as necessary so that every upper-bounded mon. increasing rational sequence has a least uppder bound.
- ⚠ I.e. if $x \in \mathbb{R} \setminus \mathbb{Q}$, then there is (mon. increasing upper-bounded) sequence of rational numbers that approximate $x$ (approximating $|x|$ suffices).
- ▷ Still there are more sequences of rational numbers than there are programs; i.e. not every $x \in \mathbb{R} \setminus \mathbb{Q}$ can be approximated by a program/algorithm.

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- Initial approximation $s_0 = 0$
  - Green: $s_{n+1} = s_n + \frac{1}{n!}$ with $\sup s_n = e \approx 2.718281828459045$.
  - Orange: $s_{n+1} = s_n + \frac{1}{(n+1)^2}$ with $\sup s_n = \frac{\pi^2}{6} \approx 1.6449340668482264$
  - Red: $s_{n+1} = s_n + \frac{1}{4n+1} - \frac{1}{4n+3}$ with $\sup s_n = \frac{\pi}{4} \approx 0.7853981633974483$.
  - Blue: $s_{n+1} = s_n + \frac{1}{2n+1} - \frac{1}{2n+2}$ with $\sup s_n = \ln(2) \approx 0.6931471805599453$
- All of the form: $s_n = \sum_{k=0}^{n} d_k$; leads to series (infinite sums; see later).

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# Plotting sequences in Python



```
import matplotlib.pyplot as plt
import math

n = 20  # change 'n': most of these approximations converge very slowly
approx_sqrt2 = [1]
approx_e = [0]
approx_pi2 = [0]
approx_pi = [0]
approx_log2 = [0]
for i in range(n):  # Colab tried to complete the code based on the variables names ... and failed.
  approx_sqrt2.append(approx_sqrt2[-1] + (2-pow(approx_sqrt2[-1],2))/(4*approx_sqrt2[-1]))
  approx_e.append(approx_e[-1] + 1/math.factorial(i))
  approx_pi2.append(approx_pi2[-1] + 6*pow(i+1,-2))
  approx_pi.append(approx_pi[-1] + 4 * (pow(4*i+1,-1) - pow(4*i+3,-1)))
  approx_log2.append(approx_log2[-1] + pow(2*i+1,-1)-pow(2*i+2,-1))

print(f"sqrt(2): {approx_sqrt2[-1]} vs {math.sqrt(2)}")
print(f"e: {approx_e[-1]} vs {math.exp(1)}")
print(f"pi^2: {approx_pi2[-1]} vs {math.pi ** 2}")
print(f"pi: {approx_pi[-1]} vs {math.pi}")
print(f"ln(2): {approx_log2[-1]} vs {math.log(2)}")

plt.plot(approx_sqrt2, label="sqrt(2)")
plt.plot(approx_e, label="e")
plt.plot(approx_pi2, label="pi^2")
plt.plot(approx_pi, label="pi")
plt.plot(approx_log2, label="ln(2)")
plt.legend()
plt.show()
```

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1 Real numbers
From natural to real
Properties of $\mathbb{R}$
Systems of equations and fields



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# Real numbers: field axioms

Thm Indepedent of the construction, the addition and multiplication of $\mathbb{Q}$ extend to $\mathbb{R}$ so that also for all $x, y, z \in \mathbb{R}$ (see the rules for computing with limits):

- Commutativity: $x + y = y + x$ and $x \cdot y = y \cdot x$
- Associativity: $x + (y + z) = (x + y) + z$ and $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- Neutal elements: There exist $0, 1 \in \mathbb{R}$ with $0 \neq 1$ s.t.

  $x + 0 = x = 0 + x$ and $x \cdot 1 = x = 1 \cdot x$.
- Inverse Elements:

  For every $x$ there exists $y$ with $x + y = 0$; this $y$ is denoted by $-x$.

  For every $x \neq 0$ there exists $y$ with $x \cdot y = 1$; this $y$ is denoted by $x^{-1}$.

  (Inverse elements are unique; see next slide.)
- Distributivity: $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$

These properties are also called field axioms ($\mathbb{Q}, \mathbb{R}$ are fields, but not $\mathbb{Z}, \mathbb{N}_0$).

- Standard convention: as long as no ambiguities arise

$$xy := x \cdot y,\ x - y := x + (-y),\ \tfrac{x}{y} := x/y := xy^{-1} := x \cdot y^{-1}$$

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# Real numbers: Properties of addition and multiplication

**Lem** (Reminder) For all $x, y, z, u, v \in \mathbb{R}$ we have (by virtue of the field axioms):

- Always $0 \cdot x = 0$.
- If $x + y = 0$ and $x + z = 0$, then $y = z$.
- Thus $-(-x) = x$.
- $(-x)y = -(xy) = x(-y)$.
- If $xy = 1$ and $xz = 1$, then $y = z$; always: $(x^{-1})^{-1} = x$
- $0 = -0$ und $1 = 1^{-1}$.
- $(xy)^{-1} = x^{-1}y^{-1}$
- $\frac{x}{y} + \frac{u}{v} = \frac{xv+uy}{vy}$ $(y \neq 0, v \neq 0)$
- If $xy = 0$, then $x = 0$ or $y = 0$.

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# Real numbers: Properties of addition and multiplication

**Lem** (Reminder) For all $x, y, z, u, v \in \mathbb{R}$ we have (by virtue of the field axioms):

- ▷ Always $0 \cdot x = 0$.

as $0 = 0 \cdot x + (-(0 \cdot x)) = (0 + 0) \cdot x + (-(0 \cdot x)) = 0 \cdot x + 0 \cdot x + (-(0 \cdot x)) = 0 \cdot x$

- ▷ If $x + y = 0$ and $x + z = 0$, then $y = z$.

as $y = 0 + y = (z + x) + y = z + (x + y) = z + 0 = z$

- ▷ Thus $-(-x) = x$.

as $(-x) + x = 0$ und $(-x) + (-(-x)) = 0$

- ▷ $(-x)y = -(xy) = x(-y)$.

as $xy + (-x)y = (x + (-x))y = 0y = 0$

- ▷ If $xy = 1$ and $xz = 1$, then $y = z$; always: $(x^{-1})^{-1} = x$ (cf. addition)
- ▷ $0 = -0$ und $1 = 1^{-1}$.

as $0 + 0 = 0$ and $1 \cdot 1 = 1$

- ▷ $(xy)^{-1} = x^{-1}y^{-1}$
- ▷ $\frac{x}{y} + \frac{u}{v} = \frac{xv+uy}{vy}$ $(y \neq 0, v \neq 0)$

as $xy^{-1} + uv^{-1} = xy^{-1}vv^{-1} + uv^{-1}yy^{-1} = (xv + uy)(vy)^{-1}$

- ▷ If $xy = 0$, then $x = 0$ or $y = 0$.

as Assume $x \neq 0 \neq y$. Then $1 = xy(x^{-1}y^{-1}) = 0(x^{-1}y^{-1}) = 0$; but $0 \neq 1$.

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# Real numbers: order axioms

**Def** $\leq, <$ are also extended from $\mathbb{N}_0, \mathbb{Z}, \mathbb{Q}$ to $\mathbb{R}$:

- Let $x = \sup X$ ($X \subseteq \mathbb{Q}$). Then $x \geq 0$ if there ex. $r \in X$ with $r \geq 0$, else $x \leq 0$.
- $x \leq y$ if $0 \leq y - x$
- $x < y$ if $x \leq y$ and $x \neq y$.

**Lem** For all $x, y, z, u, v \in \mathbb{R}$ (leads to order axioms):

- Either $x < y$ or $x = y$ or $y < x$ (trichotomy)
- For all $z$: $x < y$ **iff** $z + x < z + y$ (monotonicity of addition)
- For all $z > 0$: $x < y$ **iff** $z \cdot x < z \cdot y$ (monotonicity of multiplication)
- Supremum (least upper bound) property
  For every non-empty upper bounded set $X \subseteq \mathbb{R}$
  - i.e. there exists $b \in \mathbb{R}$ s.t. for all $x \in X$ always $x \leq b$ (short: $X \leq b$).

  its supremum (least upper bound) $\sup X \in \mathbb{R}$ exists.
  - i.e. $X \leq \sup X$ and for every $b \in \mathbb{R}$ with $M \leq b$ also $\sup X \leq b$.

  $\mathbb{R}, \mathbb{Q}$ are also called ordered fields (e.g. the complex numbers are a field that cannot be totally ordered).

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# Real numbers: properties of $\leq, <$



**Def** (Reminder) "$x > y$" stands for "$y < x$" (transposed relation).

**Lem** For all $x, y, z, u, v \in \mathbb{R}$:

- If $x < y$ and $z < 0$, then $z \cdot x > z \cdot y$.
- If $x < y$ and $u < v$, then $x + u < y + v$.
- $xy > 0$ **iff** ($x > 0$ and $y > 0$) or ($x < 0$ and $y < 0$)
- $x > 0$ **iff** $x^{-1} > 0$
- If $x \neq 0$, then $x^2 > 0$.
- If $0 < x < u$ and $0 < y < v$, then also $xy < uv$.
- If $y \neq 0$, then: $x/y > 0$ **iff** $xy > 0$
- If $x < y$ and $z > 0$, then $\frac{x}{z} < \frac{y}{z}$
- If $x > 0$ and $0 < y < z$, then $\frac{x}{z} < \frac{x}{y}$
- If $0 < x$ and $0 < y$, then: $x < y$ **iff** $x^2 < y^2$ **iff** $y^{-1} < x^{-1}$.

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# Example: Inequalities

▷ Example: Consider

$$\frac{x^2+y^2}{xy} > 2$$

If $xy < 0$, then

$$\frac{x^2+y^2}{xy} > 2 \text{ iff } x^2+y^2 < 2xy \text{ iff } x^2-2xy+y^2 < 0 \text{ iff } (x-y)^2 < 0$$

i.e. we obtain a contradiction.

So assume $xy > 0$. Then:

$$\frac{x^2+y^2}{xy} > 2 \text{ iff } x^2+y^2 > 2xy \text{ iff } x^2-2xy+y^2 > 0 \text{ iff } (x-y)^2 > 0 \text{ iff } x \neq y$$

If further $x > 0$, then also $y > 0$ and either $0 < x < y$ or $0 < y < x$.

Otherwise both $x < 0$ and $y < 0$ s.t. either $y < x < 0$ or $x < y < 0$.

Short: $x \neq y$ but same sign (next).

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**Def** Absolute value and sign (as on $\mathbb{Q}, \mathbb{Z}$):

$$|x| := \mathsf{sgn}(x) \cdot x \text{ with } \mathsf{sgn}(x) := \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

**Lem** For all $x, y \in \mathbb{R}$:

- $\min(x, y) = \frac{x+y}{2} - \frac{|x-y|}{2}$, $\max(x, y) = \frac{x+y}{2} + \frac{|x-y|}{2}$
- $|x - y| = \max(x, y) - \min(x, y)$
- $|xy| = |x|\,|y|$, specifically $\left|x^2\right| = |x|^2 = x^2$.
- $|xy| = 0$ **iff** $x = 0$ or $y = 0$
- $||x| - |y|| \leq |x \pm y| \leq |x| + |y|$ (same vs. opposite sign)




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**Def** Absolute value and sign (as on $\mathbb{Q}, \mathbb{Z}$):

$$|x| := \mathsf{sgn}(x) \cdot x \text{ with } \mathsf{sgn}(x) := \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

**Lem** For all $x, y \in \mathbb{R}$:

- $\min(x, y) = \frac{x+y}{2} - \frac{|x-y|}{2}$, $\max(x, y) = \frac{x+y}{2} + \frac{|x-y|}{2}$
- $|x - y| = \max(x, y) - \min(x, y)$
- $|xy| = |x|\,|y|$, specifically $|x^2| = |x|^2 = x^2$.
- $|xy| = 0$ **iff** $x = 0$ or $y = 0$
- $||x| - |y|| \leq |x \pm y| \leq |x| + |y|$ (same vs. opposite sign)

In particular, $|x - y|$ defines the standard metric on $\mathbb{R}$:

$$|x - y| \geq 0 \qquad |x - y| = |y - x| \qquad |x - y| = 0 \text{ iff } x = y$$

$$|x - y| = |x - z + z - y| \leq |x - z| + |z - y|$$

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# Real numbers: important subsets

**Def** Finite intervals for $a \leq b$ in $\mathbb{R}$:

- $(a, b) := ]a, b[ := \{x \in \mathbb{R} \mid a < x < b\}$ (open interval)
- $(a, b] := ]a, b] := \{x \in \mathbb{R} \mid a < x \leq b\}$ (left-open interval)
- $[a, b) := [a, b[ := \{x \in \mathbb{R} \mid a \leq x < b\}$ (right-open interval)
- $[a, b] := \{x \in \mathbb{R} \mid a \leq x \leq b\}$ (closed interval)

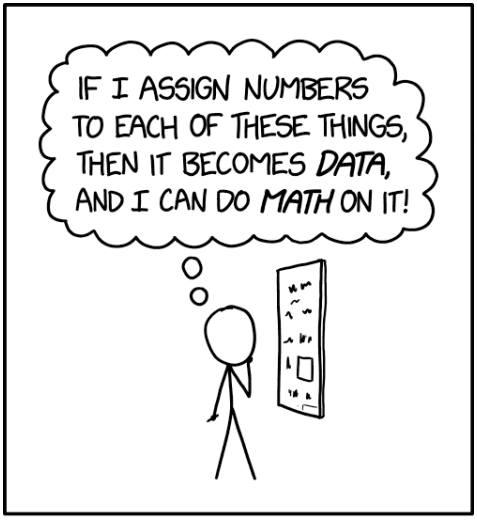
Infinite intervals:

- $\mathbb{R}^+ := \mathbb{R}_{>0} := (0, \infty) := ]0, \infty[ := \{x \in \mathbb{R} \mid 0 < x\}$ (positive)
- $\mathbb{R}_0^+ := \mathbb{R}_{\geq 0} := [0, \infty) := [0, \infty[ := \{x \in \mathbb{R} \mid 0 \leq x\}$ (nonnegative)
- $\mathbb{R}^- := \mathbb{R}_{<0} := (-\infty, 0) := ] - \infty, 0[ := \{x \in \mathbb{R} \mid x < 0\}$ (negative)
- $\mathbb{R}_0^- := \mathbb{R}_{\leq 0} := (-\infty, 0] := ] - \infty, 0] := \{x \in \mathbb{R} \mid x \leq 0\}$ (nonpositive)
- Analogously $[a, \infty)$, $(a, \infty)$, $(-\infty, b]$, $(-\infty, b)$.

⚠ (reminder) By construction/definition of $\mathbb{R}$: "$\mathbb{Q}$ is dense in $\mathbb{R}$."

For every $x \in \mathbb{R}$ and every $\varepsilon > 0$ there is some $r \in \mathbb{Q}$ mit $|x - r| < \varepsilon$.

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THE SAME BASIC IDEA UNDERLIES GÖDEL'S INCOMPLETENESS THEOREM AND ALL BAD DATA SCIENCE.

([xkcd])

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1 Real numbers
From natural to real
Properties of $\mathbb{R}$
Systems of equations and fields

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- Systems of equations are used as formal description of how quantities like position, speed, and acceleration depend on each other.
- Solving a system of equations can tell us e.g. how to control a physical system.

EG Assume we can choose the initial horizontal $v_x > 0$ and vertical speed $v_y > 0$ of an object starting at $x_0 = 0$ and $y_0 > 0$. (Further assumptions: earth is locally a plane with constant acceleration by gravity; no air drag.)

We want to reason how $v_x, v_y$ have to be chosen so that the object should travel $s_0 > 0$ units horizontally before hitting the ground. Formally:

| | | | | |
|---|---|---|---|---|
| (1) | $v_x \cdot t$ | $=$ | $s_0$ | |
| (2) | $y_0 + v_y t - \frac{1}{2} g t^2$ | $=$ | $0$ | |
| (3) | $t$ | $=$ | $\frac{s_0}{v_x}$ | solve (1) for $t$ assuming $v_x > 0$ |
| (4) | $v_y$ | $=$ | $\frac{1}{2} g t - \frac{y_0}{t}$ | solve (2) for $v_y$ assuming $t > 0$ |
| (5) | $v_y$ | $=$ | $\frac{1}{2} g \frac{s_0}{v_x} - \frac{y_0 v_x}{s_0}$ | replace $t$ in (4) by RHS of (3) |

As we assume that we can choose both $v_x > 0$ and $v_y > 0$, we "learn" that the smaller $v_x$, the longer the object must travel, and thus the higher $v_y$ has to be chosen – but we only need to solve linear equations to come to this conclusion.

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- ▷ Systems of equations are used as formal description of how quantities like position, speed, and acceleration depend on each other.

EG Assume now, $v_y > 0$ cannot be chosen by us. (Still: earth is locally a plane with constant acceleration by gravity; no air drag.)

The reasoning now becomes:

| | | | | |
|---|---|---|---|---|
| (1) | $v_x \cdot t$ | $=$ | $s_0$ | |
| (2) | $y_0 + v_y t - \frac{1}{2} g t^2$ | $=$ | $0$ | |
| (3) | $t$ | $=$ | $\frac{-v_y + \sqrt{v_y^2 - 2 g y_0}}{g}$ | \| solve (2) for $t$ assuming $t > 0$ |
| (4) | $v_x$ | $=$ | $\frac{s_0}{t}$ | \| solve (1) for $v_x$ assuming $t > 0$ |
| (5) | $v_x$ | $=$ | $\frac{s_0 g}{-v_y + \sqrt{v_y^2 - 2 g y_0}}$ | \| replace $v_x$ in (4) by RHS of (3) |

Now, we have to solve a quadratic equation to first determine the time $t$ the object will travel, and only then can we conclude how $v_x$ has to be chosen.

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# Systems of equations and fields (and algebra and algorithms)

- Systems of equations are used as formal description of how quantities like position, speed, and acceleration depend on each other.
- The general definition of a field (like $\mathbb{Q}, \mathbb{R}$) can be understood as simply requiring that we can solve linear equations; but a field does not guarantee that every quadratic equation has a solution.

EG There is no real number $x$ which satisfies $x^2 = -1$ as we have $x^2 > 0$ for every $x \in \mathbb{R}$ by the order axioms that we further impose on $\mathbb{R}$ besides the field axioms.

- We can "imagine" to extend $\mathbb{R}$ by a new number which we denote by the symbol $\mathrm{i}$ and by definition satisfies $\mathrm{i}^2 := -1$.

  Requiring that still the field axioms are satisfied leads to $\mathbb{C}$.

  E.g. we would still like to compute as follows if $a, b \in \mathbb{R}$:

$$(a + \mathrm{i}b)(a - \mathrm{i}b) = a^2 - \mathrm{i}^2 b^2 + \mathrm{i}(ab - ba) = a^2 + b^2$$

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# Systems of equations and fields (and algebra and algorithms)

- Systems of equations are used as formal description of how quantities like position, speed, and acceleration depend on each other.
- The Persian mathematician Al-Khwarizmi (780-850) published around 820 the Al-Jabr which discusses how to algorithmically solve quadratic equations by "rebalancing" both sides of an equation ("equal weight on both scales").

EG Recall how to solve a quadratic equation by "completing the square".

$$x^2 + bx + c = 0 \quad \textbf{iff} \quad (x + \tfrac{b}{2})^2 = \tfrac{b^2}{4} - c$$

(by adding $b^2/4 - c$ on both sides and then using distributivity.)

By the supremum property: $d = \frac{b^2}{4} - c$ has a square root in $\mathbb{R}$ **iff** $d \geq 0$.

- Our modern words "algebra" and "algorithm" originate from "Al-Jabr" and "Al-Khwarizmi", respectively. (Originally, "algorithms" where programs/recipes for humans how to compute with number including solving equations; "school math".)
- Till today, efficient algorithms for approximating solutions of systems of equations are an important research area.
- See also constraint solving/programming for the more general concept.

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**Thm** Over any field (like $\mathbb{Q}, \mathbb{R}$) a linear equation can be solved

$$a \cdot x + c = 0 \text{ iff } x = -c/a \text{ (for } a, c \in \mathbb{R} \text{ with } a \neq 0)$$

and thus also any system of linear equations using substitution:

▷ Example:

| | | | | |
|---|---|---|---|---|
| (1) | $2x + 3y$ | $=$ | $6$ | |
| (2) | $4x + 9y$ | $=$ | $15$ | |
| (3) | $x$ | $=$ | $3 - \frac{3}{2}y$ | \| solve (1) for $x$ |
| (4) | $4(3 - \frac{3}{2}y) + 9y$ | $=$ | $15$ | \| use (3) to substitute $x$ in (2) |
| (3) | $x$ | $=$ | $3 - \frac{3}{2}y$ | |
| (5) | $y$ | $=$ | $1$ | \| solve (4) for $y$ |
| (3) | $x$ | $=$ | $3 - \frac{3}{2}y$ | |
| (6) | $x$ | $=$ | $3 - \frac{3}{2}$ | \| use (5) to substitute $y$ in (3) |

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- When solving linear equation systems, each step is a necessary consequence from the assumption that the previous system has a solution .

EG If we obtain a contradictory system, e.g.

$$
\begin{array}{lllll}
(1) & 2x + 3y & = & 6 & \\
(2) & 4x + 6y & = & 15 & \\
\hline
(3) & x & = & 3 - \frac{3}{2}y & \mid \text{solve (1) for } x \\
(4) & 4(3 - \frac{3}{2}y) + 6y & = & 15 & \mid \text{use (3) to substitute } x \text{ in (2)} \\
\hline
(3) & x & = & 3 - \frac{3}{2}y & \mid \text{solve (1) for } x \\
(5) & 12 & = & 15 & \mid \text{simplify (4)}
\end{array}
$$

then the original assumption that there is a solution must have been wrong.

EG If an equation (constraint) becomes trivially true, we can drop it:

$$
\begin{array}{lllll}
(1) & 2x + 3y & = & 6 & \\
(2) & 4x + 6y & = & 12 & \\
\hline
(3) & x & = & 3 - \frac{3}{2}y & \mid \text{solve (1) for } x \\
(4) & 4(3 - \frac{3}{2}y) + 6y & = & 12 & \mid \text{use (3) to substitute } x \text{ in (2)} \\
\hline
(3) & x & = & 3 - \frac{3}{2}y & \mid \text{solve (1) for } x \\
(5) & 12 & = & 12 & \mid \text{simplify (4)} \\
\hline
(3) & x & = & 3 - \frac{3}{2}y & \mid \text{solve (1) for } x
\end{array}
$$

Here, we only need to ensure that $(3)$ is true by e.g. first choosing $y$.

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1. Real numbers
2. Limit of a sequence, finite vs. infinite sums
3. Elementary functions
4. Continuity
5. Differentiation and Integration



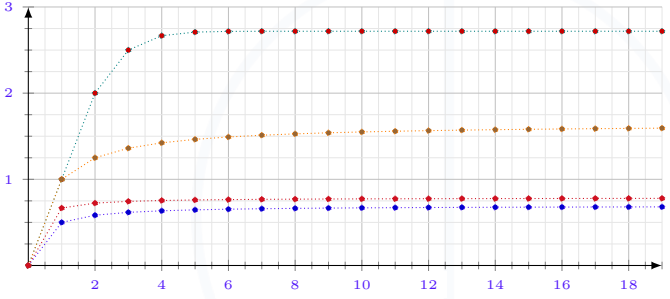
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2 Limit of a sequence, finite vs. infinite sums
Convergence and Divergence
Properties, Calculating with Limits
Finite vs. infinite sums (series)



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- By construction of $\mathbb{R}$, every monotonically increasing upper-bounded sequence $(a_n)_{n\in\mathbb{N}_0}$ (with $a_n \in \mathbb{R}$) has a supremum $a := \sup_{n\in\mathbb{N}_0} a_n \in \mathbb{R}$.




- This $a := \sup a_n$ is also called the limit of $(a_n)_{n\in\mathbb{N}_0}$;
  $(a_n)_{n\in\mathbb{N}_0}$ is said to converge to $a$.

⚠ Considering only monotone increasing sequences can be restrictive or complicate things.

"limit" and "convergence" are thus defined for general sequences.

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# Monotonicity and Boundedness

**Def** The sequence $(a_n)_{n\in\mathbb{N}_0}$ (with $a_n \in \mathbb{R}$) ist

- ▷ montonically increasing **:iff** $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}_0$.

  montonically decreasing **:iff** $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}_0$.
- ▷ strictly montonically increasing **:iff** $a_n < a_{n+1}$ for all $n \in \mathbb{N}_0$.

  strictly montonically decreasing **:iff** $a_n > a_{n+1}$ for all $n \in \mathbb{N}_0$.
- ▷ upper bounded **:iff** there ex. a $c \in \mathbb{R}$ s.t. $a_n \leq c$ for all $n \in \mathbb{N}_0$.

  lower bounded **:iff** there ex. a $c \in \mathbb{R}$ s.t. $a_n \geq c$ for all $n \in \mathbb{N}_0$.
- ▷ bounded **:iff** there ex. a $c \in \mathbb{R}$ s.t. $|a_n| \leq c$ for all $n \in \mathbb{N}_0$.

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**Def** The sequence $(a_n)_{n\in\mathbb{N}_0}$ (with $a_n \in \mathbb{R}$)

- converges to a given $a \in \mathbb{R}$ (in $\mathbb{R}$) **:iff** for every $\varepsilon > 0$ there is an $n_0 \in \mathbb{N}_0$ s.t. for all $n \geq n_0$ always $|a - a_n| < \varepsilon$ holds.

  (I.e. for every given error bound $\varepsilon > 0$ there existes a time $n_0$ from which on all approximations deviate from $a$ by less than $\varepsilon$; this allows to compute with convergent sequences.)

  $a$ is the limit of $(a_n)_{n\in\mathbb{N}_0}$ (short: $a = \lim_{n\to\infty} a_n$).

  (One can show that there is at most one limit; later)

- is convergent (in $\mathbb{R}$) **:iff** it converges to some $a \in \mathbb{R}$.
- is divergent (in $\mathbb{R}$) **:iff** it is not convergent in $\mathbb{R}$. Special cases
  - tends to infinity **:iff** for every $C > 0$ there ex. an $n_0 \in \mathbb{N}_0$ s.t. $a_n > C$ for all $n \geq n_0$ (short: $\lim_{n\to\infty} a_n = \infty$)
  - tends to minus infinity **:iff** for every $C < 0$ there ex. an $n_0 \in \mathbb{N}_0$ s.t. $a_n < C$ for all $n \geq n_0$ (short: $\lim_{n\to\infty} a_n = -\infty$)

⚠ $(a_n)_{n\in\mathbb{N}_0}$ is convergent **iff** $(a_n)_{n\geq n_0}$ is convergent for some $n_0 \in \mathbb{N}$.

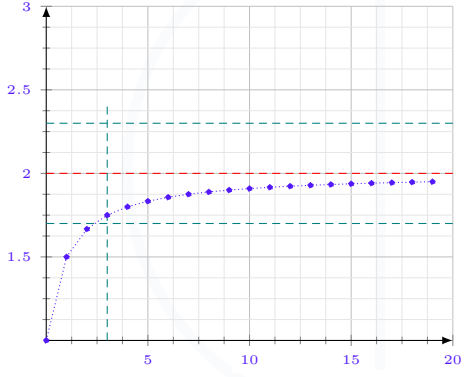
(I.e. any finite prefix can be discarded.)

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# Limit: example

EG $a_i = 2 - \frac{1}{1+i}$ with $\lim_{i\to\infty} a_i = 2$

For every given (error bound) $\varepsilon > 0$ almost all (i.e. except for finitely many) $a_i$ have to be located within the "band" $(2-\varepsilon, 2+\varepsilon)$.



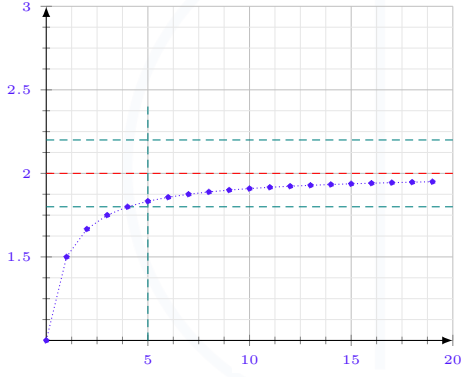

$$|2 - a_i| < \varepsilon \text{ iff } \frac{1}{i+1} < \varepsilon \text{ iff } i + 1 > \frac{1}{\varepsilon} \text{ iff } i > \frac{1}{\varepsilon} - 1$$

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# Limit: example

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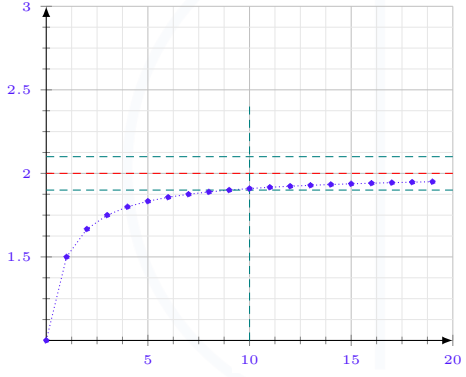

$|2 - a_i| < \varepsilon$ **iff** $\frac{1}{i+1} < \varepsilon$ **iff** $i + 1 > \frac{1}{\varepsilon}$ **iff** $i > \frac{1}{\varepsilon} - 1$

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# Limit: example

EG $a_i = 2 - \frac{1}{1+i}$ with $\lim_{i\to\infty} a_i = 2$

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# Limit: example – proof (*)

EG $a_i = 2 - \frac{1}{1+i}$ with $\lim_{i\to\infty} a_i = 2$

For every given (error bound) $\varepsilon > 0$ almost all (i.e. except for finitely many) $a_i$ have to be located within the "band" $(2-\varepsilon, 2+\varepsilon)$.

- Structure:

  For all $\varepsilon \in \mathbb{R}_{>0}$ (there exists $i_0 \in \mathbb{N}$ s.t. (for all $i \geq i_0$: $|a_i - 2| < \varepsilon$))

- Proof:
  - "Examiner" fixes $\varepsilon \in \mathbb{R}_{>0}$.
  - "Examinee" needs to determine $i_0 \in \mathbb{N}_0$ s.t. all $|a_i - 2| < \varepsilon$ holds for all $i \geq i_0$.

    Rewriting the requirement yields.

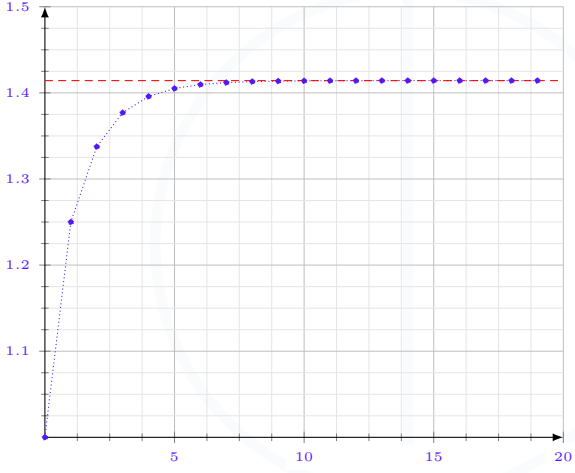
    $$|2 - a_i| < \varepsilon \text{ iff } \frac{1}{i+1} < \varepsilon \text{ iff } i + 1 > \frac{1}{\varepsilon} \text{ iff } i > \frac{1}{\varepsilon} - 1$$

    E.g. for $\varepsilon = \frac{1}{k}$ with $k \in \mathbb{N}$ choose $i_0 = k$.

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# Convergence, Divergence: Examples

EG $a_0 = 1$ and $a_{i+1} = a_i + \frac{2-a_i^2}{4a_i}$ with $\lim_{i\to\infty} a_i = \sqrt{2}$

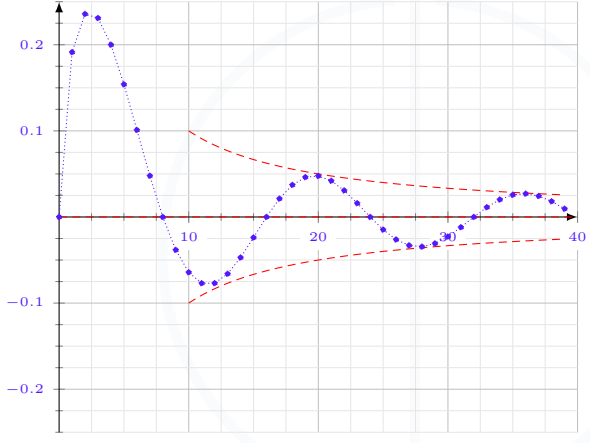



strictly monotonically increasing, bounded, convergent

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# Convergence, Divergence: Examples

EG $a_i = \frac{1}{1+i}\sin(\frac{\pi}{8}i)$ with $\lim_{i\to\infty} a_i = 0$

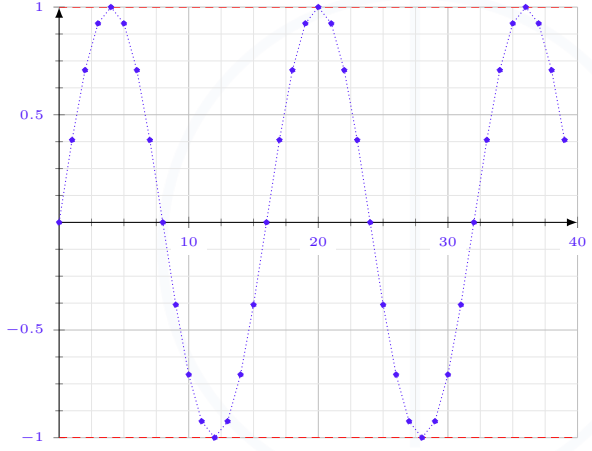



bounded, convergent, not monotone

(sandwiched between $1/n$ and $-1/n$.)

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EG $a_i = \sin(\frac{\pi}{8} i)$



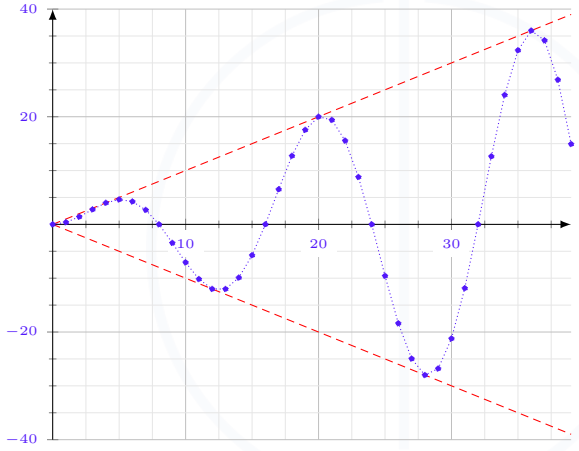

bounded, divergent, not monotone

(Does not define a unique number.)

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# Convergence, Divergence: Examples

EG $a_i = i \cdot \sin(\frac{\pi}{8} i)$

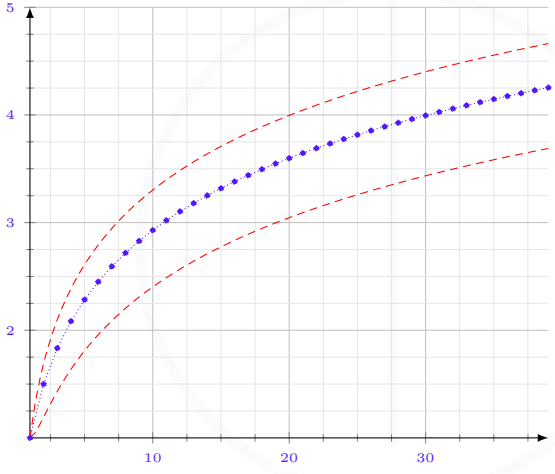



unbounded, divergent, not monotone

(Does not define a unique number.)

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EG $a_0 = 0$ and $a_{i+1} = a_i + \frac{1}{i+1}$



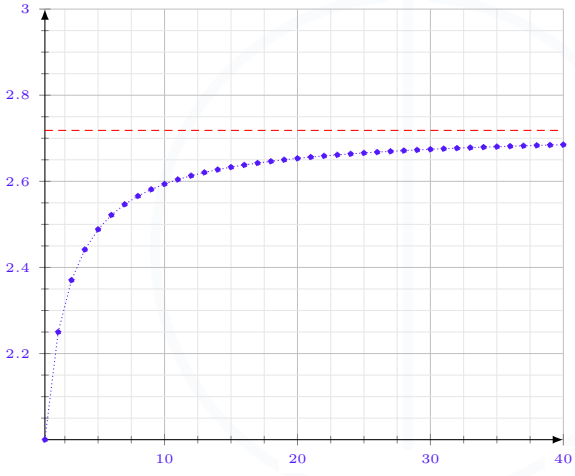

unbounded, tends to (positive) infinity, monotone

$(\ln(n) + 1/n \leq \sum_{k=0}^{n} \frac{1}{k} \leq \ln(n) + 1)$

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# Convergence, Divergence: Examples

EG $a_i = (1 + \frac{1}{i})^i$ with $e := \lim_{i \to \infty} a_i = 2.718281828459045...$

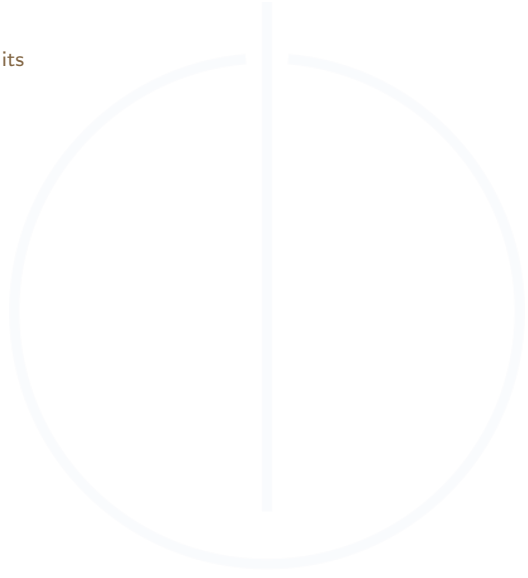



strictly monotonically increasing, bounded, convergent

(cf. eg. l'Hospital)

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2. Limit of a sequence, finite vs. infinite sums
   - Convergence and Divergence
   - Properties, Calculating with Limits
   - Finite vs. infinite sums (series)



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**Lem** A sequence $(a_n)_{n\in\mathbb{N}_0}$ has at most one limit.

- ▷ Assume there are two limits $a, b \in \mathbb{R}$. Then by definition:
  For every $\varepsilon > 0$ there is some $n_0$ (*) s.t. for all $n \geq n_0$ both $|a - a_n| < \varepsilon$ and $|b - a_n| < \varepsilon$, and thus also $|a - b| = |a - a_n + a_n - b| \leq |a_n - a| + |a_n - b| < 2\varepsilon$.
  As we can choose $\varepsilon > 0$ arbitrarily small, we have to have $a = b$.
  (*: if $|a - a_n| < \varepsilon$ for all $n \geq n_0'$ and $|b - b_n| < \varepsilon$ for all $n \geq n_0''$, then we can choose $n_0 = \max(n_0', n_0'')$; here the "for all $n \geq n_0$" is used.)

**Lem** If $(a_n)_{n\to\infty}$ is monotonically increasing and upper-bounded by $C$,
then $\lim_{n\in\mathbb{N}_0} a_n = \sup_{n\in\mathbb{N}_0} a_n \leq C$.
If $(a_n)_{n\to\infty}$ is monotonically decreasing and lower-bounded by $C$,
then $\lim_{n\in\mathbb{N}_0} a_n = \inf_{n\in\mathbb{N}_0} a_n = -\sup_{n\in\mathbb{N}_0}(-a_n) \geq C$.

- ▷ By definition/construction of $\mathbb{R}$.
- ▷ Suffices if monotone starting from some index $n_0$ (discard prefix).
- ▷ If $(a_n)_{n\in\mathbb{N}_0}$ converges to $a$, then it contains a monotone (either increasing or decreasing) subsequence $(a_{i_n})_{n\in\mathbb{N}_0}$ which (trivially) converges also $a$.
- ▷ Further criteria: Cauchy, Sandwich theorem

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# Rules for calculating with limits

**Lem** Let $\lim_{i\to\infty} a_i = a$ and $\lim_{i\to\infty} b_i = b$ be convergent. Then:

- $\lim_{i\to\infty}(a_i \pm b_i) = a \pm b$
- $\lim_{i\to\infty}(a_i b_i) = ab$
- If $b \neq 0$, then $\lim_{i\to\infty} \frac{a_i}{b_i} = \frac{a}{b}$ (as there ex. some $n_0$ s.t. $b_i \neq 0$ for all $i \geq n_0$).
- For every $b \in \mathbb{R}$: $\lim_{i\to\infty}(a_i + b) = a + b$ and $\lim_{i\to\infty}(a_i b) = ab$.
  (Special case for $b_i := b$ a constant sequence.)

(Again makes use of the "for all $n \geq n_0$" in the definition.)

- These rules allow to compute with limits without knowing the proof/reason why.

**EG** Assume $b > 0$ and $b_i > 0$ for $i$ sufficiently large.

$$\left|\frac{a}{b} - \frac{a_i}{b_i}\right| = \frac{|ab_i - ba_i|}{bb_i} = \frac{|ab_i - ab + ab - ba_i|}{bb_i + bb - bb} \leq \underbrace{\frac{|a(b_i - b)| + |b(a_i - a)|}{b^2 + b(b_i - b)}}_{\xrightarrow{i\to\infty} 0}$$

⚠ As every $x \in \mathbb{R}$ is the limit of a rational sequence $(x_n)_{n\in\mathbb{N}_0}$ this extends addition and multiplication from $\mathbb{Q}$ to $\mathbb{R}$ (and shows that these are continuous).
In particular, if we have programs which compute the approximations $a_n, b_n \in \mathbb{Q}$, we can compute/calculate with the programs.

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2. Limit of a sequence, finite vs. infinite sums
   - Convergence and Divergence
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- A "natural number" in a given base $b \in \mathbb{N}$ is actually a sequence of digits

$$d_l d_{l-1} \dots d_1 d_0 \qquad \text{with} \quad d_i \in \mathbb{Z}_b = \{0, \dots, b-1\}$$

which represents the natural number:

$$\sum_{i=0}^{l} d_i \cdot b_i^i = d_l b^l + d_{l-1} b^{l-1} + \dots + d_1 b^1 + d_0 b^0$$

(Nowadays, we are used to write numbers starting with the "most-significant digit" on the left, but that is a purely "cosmetic" choice.)

EG Consider the "word" $123456$. In base $10$ it has the value $123456$ but in base $7$ it has the value $22875$ (in base $10$).

To better distinguish between the word "$123456$" and its value in a given base often notation like

$$(123456)_7 = (22875)_{10} \quad (123456)_8 = (42798)_{10} \quad (123456)_9 = (74733)_{10}$$

is used (as long as we only use base $10$, this is of course not necessary).

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- A "natural number" in a given base $b \in \mathbb{N}$ is actually a sequence of digits

$$d_l d_{l-1} \dots d_1 d_0 \qquad \text{with} \quad d_i \in \mathbb{Z}_b = \{0, \dots, b-1\}$$

which represents the natural number:

$$\sum_{i=0}^{l} d_i \cdot b_i^i = d_l b^l + d_{l-1} b^{l-1} + \dots + d_1 b^1 + d_0 b^0$$

(Nowadays, we are used to write numbers starting with the "most-significant digit" on the left, but that is a purely "cosmetic" choice.)

- To convert a natural number $n \in \mathbb{N}_0$ into some given base $b$, we can inductively strip away the least-significant digit.
  - (IB) If $n = 0$: return $0$ (or the empty word "").
  - (IS) If $n > 0$:
    compute $d := (n \bmod b) := n - \lfloor \frac{n}{b} \rfloor$ (floor/round down);
    convert $n' := \frac{n-d_0}{b}$ to obtain $d_l d_{l-1} \dots d_0$ with $(d_l d_{l-1} \dots d_0)_b = n'$;
    return $d_l d_{l-1} \dots d_1 d_0 d$;

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# Finite and infinite sums (series)

- We can also represent the fractional part of a real number a given base $b \in \mathbb{N}$.
- A word "$.d_1 d_2 d_3 \dots d_l$" ($d_i \in \mathbb{Z}_b$) is now representing the number

$$(.d_1 d_2 \dots d_l)_b := \sum_{i=1}^{l} d_i \cdot b^{-i} = d_1 \cdot b^{-1} + d_2 \cdot b^{-2} + \dots + b_l \cdot b^{-l}$$

By the geometric sum $\sum_{i=0}^{l} x^i = \frac{1-x^{l+1}}{1-x}$ ($|x| \neq 1$) we have

$$0 \leq (.d_1 d_2 \dots d_l)_b \leq 1 - b^{-(l+1)}$$

EG $(.10)_2 = (.1)_2 = (.5)_{10} = (0.50)_{10}$ (trailing zeros do not matter).

- Some rational $x \in (0, 1)$ do not have a finite representation w.r.t. some base $b$.

EG $(.3)_{10} < (.33)_{10} < (0.3...3)_{10} < (.1)_3 < (.3 \dots 34)_{10} < (.334)_{10} < (.34)_{10}$.

- An irrational $x \in (0, 1)$ cannot have a finite representation w.r.t. any base $b$ as $(.d_1 d_2 \dots d_l)_b$ is always rational.

We thus need to allow infinite sequences of digits.

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# Finite and infinite sums (series)

- We can also represent the fractional part of a real number a given base $b \in \mathbb{N}$.
- Also an infinite sequence of digits has a well-defined meaning

$$0 \leq (.d_1 d_2 d_3 \ldots)_b := \sum_{i=0}^{\infty} d_i \cdot b^{-i} := \lim_{l \to \infty} (.d_1 d_2 \ldots d_l)_b \leq 1$$

  As this is a bounded monotonically increasing sequence s.t. the limit always exists. (This was our original motivation for $\mathbb{R}$.)

- Given some $0 < x < 1$ we can also compute inductively a representation w.r.t. a given base $b$; this time, we inductively determine the most-significant digit:

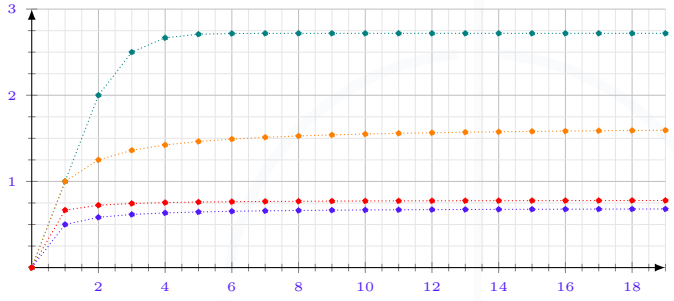
  While $x > 0$: compute $d := \lfloor b \cdot x \rfloor$; output $d$; continue with $x := b \cdot x - d$;

EG $(.1)_3 = (0.\overline{3})_{10} = (.\overline{01})_2$.

- Let $x_l, d_1, \ldots, d_l$ be the value of $x$ and the digits output after $l$ iterations:
  - $0 \leq x_0 - (.d_1 \ldots d_l)_b = x_l \cdot b^{-(l+1)} < b^{-(l+1)}$.
  - If $x$ is rational, then either eventually $x_l = 0$ or $x_i = x_l$ for some $i < l$ (leading to an "ultimately periodic" sequence).

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# Approximation of numbers by means of infinite sums (reminder)




EG Infinite sums as limits of finite sums arise e.g. when approximating irrational numbers by adding up "vanishing" corrections.

- Green: $s_{n+1} = s_n + \frac{1}{n!}$ with $\sup s_n = e \approx 2.718281828459045$.
- Orange: $s_{n+1} = s_n + \frac{1}{(n+1)^2}$ with $\sup s_n = \frac{\pi^2}{6} \approx 1.6449340668482264$
- Red: $s_{n+1} = s_n + \frac{1}{4n+1} - \frac{1}{4n+3}$ with $\sup s_n = \frac{\pi}{4} \approx 0.7853981633974483$.
- Blue: $s_{n+1} = s_n + \frac{1}{2n+1} - \frac{1}{2n+2}$ with $\sup s_n = \ln(2) \approx 0.6931471805599453$

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**Def** The sum of finitely many numbers $a_1, \ldots, a_n \in \mathbb{R}$ ($n \in \mathbb{N}_0$) is denoted by

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + \ldots + a_n$$

Formally, this is a recursive/inductive definition:

- If $n = 0$, then $\sum_{i=1}^{n} a_i := 0$, else $\sum_{i=1}^{n} a_i := a_n + \sum_{i=1}^{n-1} a_i$.
- The summation index/variable $i$ can be chosen arbitrarily (if no ambiguities arise), and the index can be shifted $\sum_{i=0}^{n} a_i = \sum_{k=k_0}^{n+k_0} a_{k-k_0}$.

**Def** The sum of an infinite sequence $(a_i)_{i\in\mathbb{N}}$ ($a_i \in \mathbb{R}$)

$$\sum_{i=1}^{\infty} a_i := \lim_{n\to\infty} \sum_{i=1}^{n} a_i$$

is called a series and its value is defined as the limit of the sequence $(s_n)_{n\in\mathbb{N}_0}$

$$s_n = a_1 + \ldots + a_n = s_{n-1} + a_n$$

of its partial sums in the given order (if it exists).

⚠ Computation with finite sums is unproblematic, but infinite sums (series) can be problematic.

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# Examples: finite sums

$$\sum_{k=0}^{n} k = \sum_{k=1}^{n} k = 1 + 2 + \ldots + n = \tfrac{n(n+1)}{2}$$

$$\sum_{k=0}^{n} (2k+1) = \sum_{k=1}^{n+1} (2k-1) = 1 + 3 + 5 + \ldots + (2n-1) = 2\sum_{k=1}^{n} k - n = n^2$$

$$\sum_{k=0}^{n} k^2 = \sum_{k=1}^{n} k^2 = 1 + 4 + 9 + \ldots + n^2 = \tfrac{n(n+1)(2n+1)}{6}$$

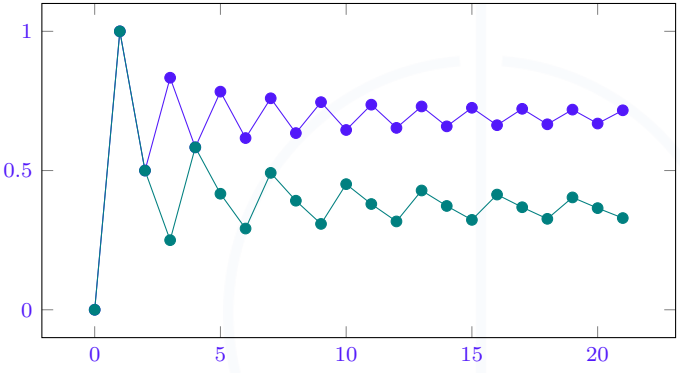
$$\sum_{k=0}^{n} x^k = 1 + x + x^2 + \ldots + x^n = \begin{cases} \frac{1-x^{n+1}}{1-x} & \text{if } x \neq 1 \\ n+1 & \text{else} \end{cases}$$

$$H_n := \sum_{k=1}^{n} \tfrac{1}{k} = 1 + \tfrac{1}{2} + \tfrac{1}{3} + \ldots + \tfrac{1}{n} \in (\tfrac{1}{n} + \log(n), 1 + \log(n))$$

(Harmonic number: see integral later and Block stacking problem.)

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# Example: Rearrangement of the alternating harmonic series




$$\sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i} = \ln(2) \quad \text{vs} \quad \sum_{i=1}^{\infty} \left( \frac{1}{2i-1} - \frac{1}{4i-2} - \frac{1}{4i} \right) = \frac{\ln(2)}{2}$$

- blue: $\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \ldots$
- green: $\frac{1}{1} - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \ldots$

(Many other rearrangements: $1 + \frac{1}{3} + \frac{1}{5} + \ldots + \frac{1}{2n+1} - \frac{1}{2} + \frac{1}{2n+3} + \ldots.$)

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# Example: Rearranging the alternating harmonic series



```
import matplotlib.pyplot as plt
import math
from fractions import Fraction
def a(n: int): # 1/1, -1/2, 1/3, -1/4, 1/5, -1/6, 1/7, -1/8, 1/9
  return Fraction((-1) ** n, (n+1))
# the more odd (even) indices we move to the front, the larger (smaller) the limit
def b(n: int): # 1/1, -1/2, -1/4, 1/3, -1/6, -1/8, 1/5, ...
  k, r = divmod(n,3)
  if r == 0: return Fraction(1,2*k+1)
  if r == 1: return Fraction(-1,4*k+2)
  return Fraction(-1,4*k+4)
def c(n: int): # 1/1, 1/3, -1/2, 1/5, 1/7, -1/4, 1/9, 1/11, -1/8
  k, r = divmod(n,3)
  if r == 0: return Fraction(1,4*k+1)
  if r == 1: return Fraction(1,4*k+3)
  return Fraction(-1,2*k+2)

n_max = 20
for expr in [a,b,c]:
  partial_sums = [0]
  for n in range(n_max+1):
    partial_sums.append(partial_sums[-1]+float(expr(n)))
  plt.plot(partial_sums)
for limit in [math.log(2), math.log(2)/2, 3*math.log(2)/2]:
  plt.plot([0,n_max],[limit,limit], ':')
plt.show()
```

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**Thm** Riemann rearrangement theorem:

If a series $\sum_{i=0}^{\infty} a_i$ converges for the given order, but rearranging the series changes the limit (e.g. the alternating harmonic series), then for any given $x \in \mathbb{R} \cup \{\pm\infty\}$ there is an rearrangement $\sigma\colon \mathbb{N}_0 \xrightarrow{\text{bij}} \mathbb{N}_0$ s.t. $\sum_{i=0}^{\infty} a_{\sigma(i)} = x$.

(I.e. the meaning/value of a series in general depends on the specific sequence.)

▷ Sill, some series always converge to the same value even if we rearrange them:

For the real numbers, these series are characterized by their absolute convergence (cf. Levy-Steinitz theorem):

**Def** The series $\sum_{i=0}^{\infty} a_i$ converges absolutely if $\sum_{i=0}^{\infty} |a_i|$ converges.

(This requires that the "corrections" $a_i + a_{i+1} + \dots$ must vanish.)

⚠ For now remember: adding infinitely many numbers can be problematic.

(Rules for computing with finitely many numbers do not apply anymore in general.)

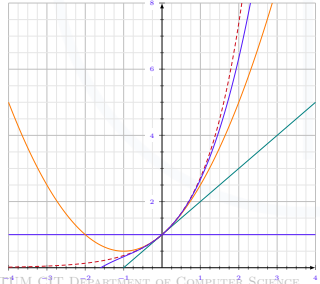
But with absolutely convergent series we can compute almost as with finite sums as they always define the same number.

In particular, absolutely convergent series can be used to define and approximate important elementary functions like $\exp, \sin, \cos$.

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# Examples of important absolutely convergent (power) series

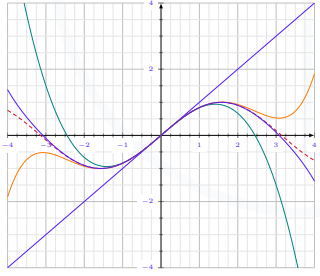
- $\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \ldots = (1-x)^{-1}$ for all $|x| < 1$.
- $\sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{6} + \ldots = \exp(x) = e^x$ for all $x \in \mathbb{R}$.
- $\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = \frac{x}{1} - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} \pm \ldots = \sin(x)$ for all $x \in \mathbb{R}$.
- $\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} \pm \ldots = \cos(x)$ for all $x \in \mathbb{R}$.
- $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \pm \ldots = \ln(1+x)$ for all $|x| < 1$.
- $\sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2} \frac{x^{2k+1}}{2^{2k}(2k+1)} = x + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{x^5}{5} + \ldots = \arcsin(x)$ for all $|x| \leq 1$.



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# Examples of important absolutely convergent (power) series

- $\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \ldots = (1-x)^{-1}$ for all $|x| < 1$.
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# Absolute convergence of $\exp(x)$ (and how we can compute with $e, \sin, \cos$ on the computer) (*)

**Lem** $\exp(x) = \sum_{i=0}^{\infty} \frac{x^k}{k!}$ converges absolutely for every $x \in \mathbb{R}$.

- ▷ Fix any $x \in \mathbb{R} \setminus \{0\}$; then choose e.g. $k_0 \geq 2\,|x|$ s.t.

$$\frac{|x|^{k_0+i}}{(k_0+i)!} = \frac{|x|^{k_0}}{k_0!} \cdot \frac{|x|}{k_0+1} \cdot \ldots \cdot \frac{|x|}{k_0+i} \leq \frac{|x|^{k_0}}{k_0!} \cdot 2^{-i}$$

Thus the partial sums are bounded ($n \geq k_0$):

$$\sum_{k<k_0} \frac{|x|^k}{k!} + \sum_{k=k_0}^{n} \frac{|x|^k}{k!} \leq \sum_{k<k_0} \frac{|x|^k}{k!} + \frac{|x|^{k_0}}{k_0!} \sum_{i=0}^{n-k_0} 2^{-i} \leq \sum_{k<k_0} \frac{|x|^k}{k!} + 2\,\frac{|x|^{k_0}}{k_0!}$$

(the geometric sum/series is everywhere)

and thus $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ is absolutely convergent for every given $x \in \mathbb{R}$.

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**Lem** If $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} b_k x^k$ are absolutely, then for all $x \in \mathbb{R}$

$$\left(\sum_{k=0}^{\infty} a_k x^k\right) \cdot \left(\sum_{k=0}^{\infty} b_k x^k\right) = \sum_{k=0}^{\infty} \left(\sum_{i=0}^{k} a_i b_{k-i}\right) x^k$$

(Called Cauchy product/convolution; extended from polynomials; see analysis later.)

**Cor** Without proof:

1. $\exp(x+y) = \exp(x) \cdot \exp(y)$ (binomial theorem).
2. $\exp(x)^{-1} = \exp(-x)$ (as $1 = \exp(0) = \exp(x + (-x)) = \exp(x) \cdot \exp(-x)$).
3. $\exp(1)^{1/n} = \exp(1/n)$ for $n \in \mathbb{N}$ (as $\exp(1) = \exp(n \cdot 1/n) = \exp(1/n)^n$)
4. $\exp(p/q) = \exp(1)^{p/q}$ for all $p/q \in \mathbb{Q}$ ($q > 0$)
5. $\exp(1)^x = \exp(x)$ for all $x \in \mathbb{R}$ (by continutiy).

**Def** $e := \exp(1)$ is called Euler's number ($e = 2.7182818284\ldots$).

**Cor** $e^x = \exp(x)$ for all $x \in \mathbb{R}$.

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(Called Cauchy product/convolution; extended from polynomials; see analysis later.)

⚠ Similarly, one can show that

$$\sin(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \qquad \cos(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

are absolutely convergent on $\mathbb{R}$ and that they satisfy the addition theorems

$$\sin(x+y) = \sin(x)\cdot\cos(y)+\cos(x)\cdot\sin(y) \qquad \cos(x+y) = \cos(x)\cos(y)-\sin(x)\sin(y)$$

which consequently allows to show that the series indeed coincide with the geometric definitions of $\sin$ and $\cos$.

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1. Real numbers
2. Limit of a sequence, finite vs. infinite sums
3. Elementary functions
4. Continuity
5. Differentiation and Integration



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# Exponentiation: integer exponents/powers

**Def** For $n \in \mathbb{N}_0$ and $x \in \mathbb{R}$ the $n$-th power of $x$ is defined as

$$x^0 := 1 \qquad x^{n+1} = x^n \cdot x \qquad \text{(sometimes also } 0^0 := 0 \text{ or undef.)}$$

(i.e. $x^n$ is the product of $n$ copies of $x$.)

Negative powers (exponents) are defined for $x \neq 0$ as powers of the inverse:

$$x^{-n} := (x^{-1})^n = \left(\tfrac{1}{x}\right)^n$$

(i.e. $x^{-n}$ is the product of $n$ copies of the mult. inverse of $x$.)

- Examples:
  - $0.5^2 = 1 \cdot 0.5 \cdot 0.5 = 0.25$
  - $0.5^{-3} = 1 \cdot 0.5^{-1} \cdot 0.5^{-1} \cdot 0.5^{-1} = 1 \cdot 2 \cdot 2 \cdot 2 = 8 = 2^3$

⚠ Integer powers/exponents simply count the number of copies of $x$ resp. $x^{-1}$.

As the inverse is unique, we have for $x \neq 0$:

$$(x^{-1})^{-1} = x \qquad (x^{-1})^n = (x^n)^{-1} = \tfrac{1}{x^n}$$

and thus for all $m, n \in \mathbb{Z}$:

$$x^m \cdot x^n = x^{m+n} \qquad (x^m)^n = x^{mn}$$

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# Principal $n$-th root, rational and real powers

**Lem** For every $y \in \mathbb{R}_{\geq 0}$ and every $n \in \mathbb{N}$ the equation $x^n = y$ has a unique solution in $\mathbb{R}_{\geq 0}$ called the principal $n$-th root of $y$ denoted by $\sqrt[n]{y}$ or $y^{1/n}$.

(In general, any solution of $x^n = y$ is called an $n$-th root of $y$.)

- Existence: $y^{1/n} := \sqrt[n]{y} := \sup\{z \in \mathbb{R}_{\geq 0} \mid z^n \leq y\}$.

  Uniqueness: If $0 \leq z_1 < z_2$, then $0 \leq z_1^n < z_2^n$ (monotonicity).
- $\mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, x \mapsto \sqrt[n]{x}$ as inverse of $\mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, x \mapsto x^n$.

**Def** A rational power $y^{m/n}$ ($m \in \mathbb{Z}, n \in \mathbb{N}$) is defined for $y \in \mathbb{R}_{>0}$ by

$$y^{m/n} := (y^{1/n})^m = (y^m)^{1/n} \quad (n \in \mathbb{N})$$

($\sqrt[n]{y^m} = (y^m)^{1/n} = (((y^{1/n})^n)^m)^{1/n} = (((y^{1/n})^m)^n)^{1/n} = (y^{1/n})^m = (\sqrt[n]{y})^m$)

- Examples:
  - $4^{1/2} = (2^2)^{1/2} = 2^{2 \cdot 1/2} = 2^1 = 2$
  - $(4^{-1})^{1/2} = (2^{-2})^{1/2} = 2^{-1} = (4^{1/2})^{-1} = 4^{-1/2}$
  - $0.25^{-3/2} = (1 \cdot 0.25^{1/2} \cdot 0.25^{1/2} \cdot 0.25^{1/2})^{-1} = (2^{-3})^{-1} = 2^3 = 8$

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**Lem** If $r \in \mathbb{R}$, choose any sequence $(r_n)_{n\in\mathbb{N}}$ with $\lim_{n\to\infty} r_n = r$ and $r_n \in \mathbb{Q}$, then

$$x^r := \lim_{n\to\infty} x^{r_n}$$

is well-defined for all $x \in \mathbb{R}_{>0}$ (i.e. the value $x^r$ is independent of the concrete choice of $(r_n)_{n\mathbb{N}_0}$, see also "continuity"; alternatively, $x^r := \exp(r \cdot \ln(x))$ for $x \in \mathbb{R}_{>0}$).

- ▷ Example: $2^\pi = 8.824977\ldots$

  $2^3 = 8, \quad 2^{3.1} = 8.57419..., \quad 2^{3.14} = 8.81524..., \quad 2^{3.141} = 8.82135...$

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# Rational and real powers: properties

**Lem** For all $a, b \in \mathbb{R}_{>0}$ and $r, s \in \mathbb{R}$ we have (extends from $\mathbb{Q}$, cf. Analysis)

$$a^r a^s = a^{r+s} \quad \frac{a^r}{a^s} = a^{r-s} \quad (a^r)^s = a^{rs} = (a^s)^r \quad a^r b^r = (ab)^r \quad \frac{a^r}{b^r} = \left(\frac{a}{b}\right)^r$$

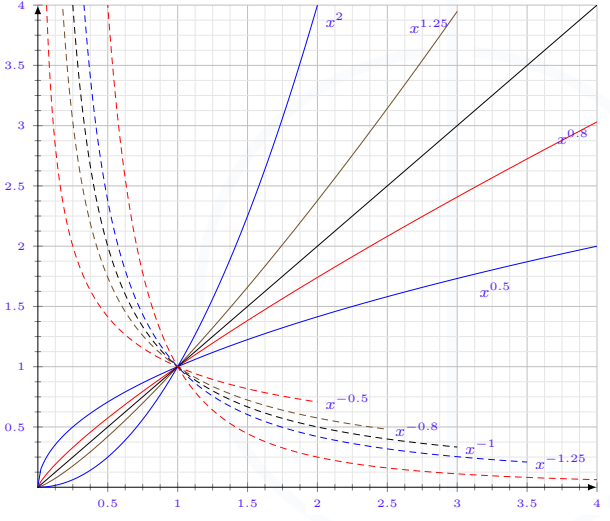
In particular $a^{p/q} = a^{p \cdot 1/q} = (a^p)^{1/q} = (a^{1/q})^p$.

- Examples:
  - $2^{0.3} 2^{2/3} = 2^{9/30} 2^{20/30} = (2^{1/30})^9 (2^{1/30})^{20} = (2^{1/30})^{29} = 2^{29/30}$
  - $2^{0.3} \cdot 3^{2/3} = 2^{9/30} 3^{20/30} = (2^9 3^{20})^{1/30}$
  - $(\sqrt[5]{3})^2 = (3^{1/5})^2 = 3^{2/5} = \sqrt[5]{3^2}$

**Lem** As multiplication is monotone (if $0 < x < y$ and $0 < u < v$, then $0 < ux < vy$):

- $p_r \colon \mathbb{R}_{>0} \to \mathbb{R}, x \mapsto x^r$ is strictly mon. increasing if $r > 0$.
  $p_r \colon \mathbb{R}_{>0} \to \mathbb{R}, x \mapsto x^r$ is strictly mon. decreasing if $r < 0$.
  (By definition $p_{1/r}(x) = p_r^{-1}(x)$ (swap $(x, x^r)$, same monotonicity), while $p_{-r}(x) = p_r(1/x)$ (reverses monotonicity).)
- $\mathsf{pow}_b \colon \mathbb{R} \to \mathbb{R}_{\geq 0}, x \mapsto b^x$ is strictly mon. increasing if $b > 1$.
  $\mathsf{pow}_b \colon \mathbb{R} \to \mathbb{R}_{\geq 0}, x \mapsto b^x$ is strictly mon. decreasing if $0 < b < 1$.
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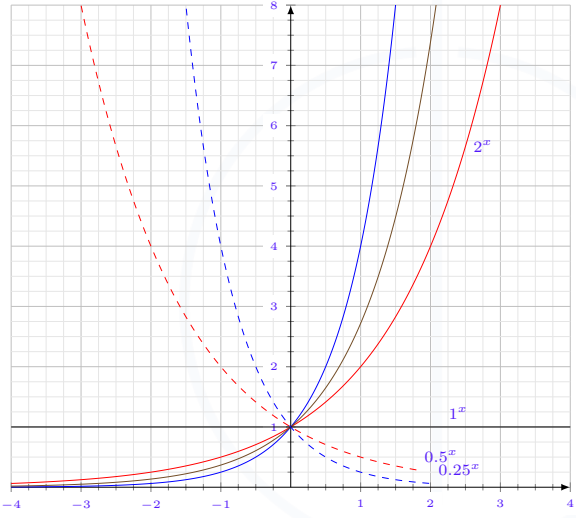
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($p_r(x) = x^r$ for fixed exponent $r$ with $p_r(1/x) = p_{-r}(x)$; if $r > 0$ ($r < 0$), strictly mon. increasing (decreasing); inverse $p_r^{-1}(x) = p_{1/r}(x)$ for $r \neq 0$.)

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# Exponentiation: monotonicity wrt. exponent




($\mathrm{pow}_b(x) = b^x$ for fixed base $b > 0$ with $\mathrm{pow}_b(x) = \mathrm{pow}_{1/b}(-x)$; if $b > 1$ ($0 < b < 1$), strictly mon. increasing (decreasing); inverse: logarithm.)

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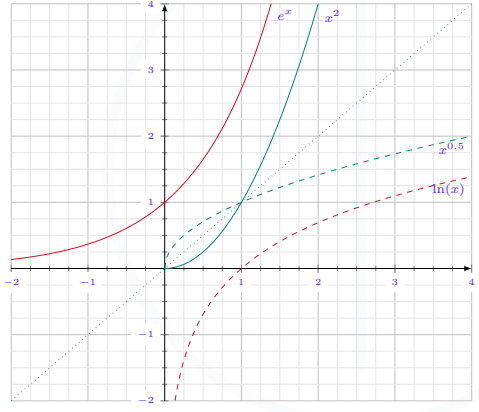


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# Exponentiation: inverse functions

**Lem** As $p_r(x) = x^r$ (for $r \in \mathbb{R}_{>0}$) and $\mathrm{pow}_b(x) = b^x$ (for $b \in \mathbb{R}_{>0} \setminus \{1\}$) are strictly mon. increasing on $\mathbb{R}_{\geq 0}$ resp. $\mathbb{R}$, their inverse functions exist.

(They are further continuous, so their image is $\mathbb{R}_{\geq 0}$ resp. $\mathbb{R}_{>0}$.)




(Reminder: $f^{-1}$ by means of reflection at $y = x$ if $f$ is bijective.)

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# Exponentiation: inverse functions

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(They are further continuous, so their image is $\mathbb{R}_{\geq 0}$ resp. $\mathbb{R}_{>0}$.)

- For fixed exponent $r > 0$ the inverse function of

$$p_r \colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, x \mapsto x^r$$

is just $p_{1/r}(y) = y^{1/r}$.

This is also true for $r < 0$ except that we need to rule out $0$.

**Def** For fixed base $b \in \mathbb{R}_{>0} \setminus \{1\}$ the inverse function of

$$\mathsf{pow}_b \colon \mathbb{R} \to \mathbb{R}_{>0}, x \mapsto b^x$$

is called logarithm to the base $b$ $\log_b \colon \mathbb{R}_{>0} \to \mathbb{R}$.

Usually $\log := \log_e$.

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**Lem** From the rules for computing in the exponent ($x, y > 0$) it follows:

$$\log_b(xy) = \log_b(x) + \log_b(y) \quad \log_b \frac{x}{y} = \log_b x - \log_b y \quad \log_b x^y = y \log_b x$$

Further $\log_b \colon \mathbb{R}_{>0} \to \mathbb{R}$ is strictly monotonically increasing for $b > 1$ with

$$\log_b x < 0 \text{ for } x < 1 \text{ and } \log_b x > 0 \text{ for } x > 1 \text{ and } \log_{1/b} x = -\log_b x$$

**EG** $\frac{x}{y} = x \cdot y^{-1} = b^{\log_b(x)} (b^{\log_b(y)})^{-1} = b^{\log_b(x) - \log_b(y)}$.

I.e. $\log_b(x/y) = \log_b(x) - \log_b(y)$ (if $x, y > 0$).

**EG** $\log_e(z) = \log_e(b^{\log_b(z)}) = \log_e(b) \cdot \log_b(z)$ i.e. $\log_b(z) = \frac{\log_e(z)}{\log_e b}$.

Thus $\log_b(z)$ is mon. increasing if $b > 1$ and mon. decreasing if $0 < b < 1$.

In particular $\log_{1/b}(z) = \frac{\log_e(z)}{\log_e(1/b)} = -\log_b(z)$.

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**Def** The logarithm to Euler's number $e$

$$e := \sum_{k=0}^{\infty} \frac{1}{k!} = \lim_{n\to\infty} (1 + \tfrac{1}{n})^n = 2.71828182845904523536028747135 2\ldots$$

is called natural logarithm and denoted by

$$\ln(x) := \log(x) := \log_e(x)$$

I.e. $x = \ln(e^x)$ for all $x \in \mathbb{R}$ and $y = e^{\ln(y)}$ for all $y \in \mathbb{R}_{>0}$.

Specifically we have $\ln(z) = \ln(b^{\log_b(z)}) = \ln(b) \cdot \log_b(z)$ s.t.

$$\log_b(z) = \frac{\ln(z)}{\ln(b)}$$

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# Polynomial and rational functions

**Def** A polynomial is an arithmetic expression built only from variables, constants, addition, subtraction, and multiplication ("programs" defined on every field or ring).

**Fct** Every polynomial can be expanded into a finite sum of finite products of constants and variables; e.g., in the case of a single variable $X$ as $\sum_{i=0}^{n} a_i X^i$.

**EG** $(1+X)(1+X) = 1+2X+X^2$.

**Lem** A nonzero polynomial $\sum_{i=0}^{n} a_i X^i$ with $a_i \in \mathbb{R}$ can have at most $n$ roots in $\mathbb{R}$.

**Def** Rational expressions extend polynomials by further allowing division.

**Fct** A rational expression can always be written as a fraction of two polynomials.

**EG** $(1+1/X)/(1+X) = \frac{1+X}{X+X^2}$

⚠ Polynomials and rational expressions are not functions: a function always requires a domain on which the expression can be evaluated.

**EG** $\frac{1+X}{X+X^2}$ defines a function $f\colon \mathbb{R} \setminus \{0,-1\} \to \mathbb{R}, x \mapsto \frac{1+x}{x+x^2}$. By canceling the common factor $1+X$, we obtain $\frac{1}{X}$, which defines the function $g\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}, x \mapsto \frac{1}{x}$ with $g(x) = f(x)$ for all $x \in \mathbb{R} \setminus \{0,-1\}$ ($g$ extends $f$).

▷ We discuss rational functions in more detail in the Python part.

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3 Elementary functions

Powers and Exponentation

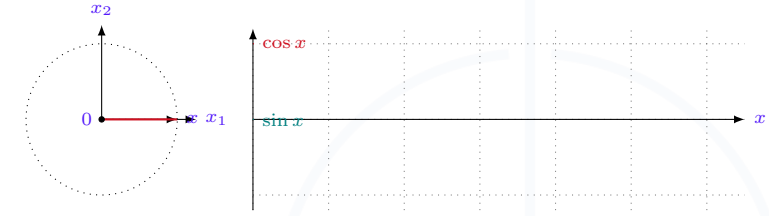
Logarithm

Polynomial and rational functions

Trigonometric Functions

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# Sine and cosine




**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

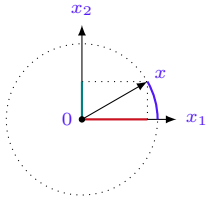
  From rad(ians) to deg(ree): $x \mapsto \frac{360°}{2\pi} x = \frac{180°}{\pi} x$

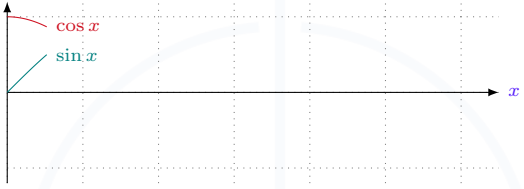
- $\sin(x)$ and $\cos(x)$ are the orthogonal projections of the point at $x$ rad.

  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 0.52359$ rad $(30.0°)$
$\sin(x) \approx 0.49997$
$\cos(x) \approx 0.86601$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

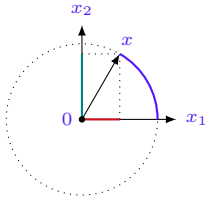
  From rad(ians) to deg(ree): $x \mapsto \frac{360°}{2\pi} x = \frac{180°}{\pi} x$

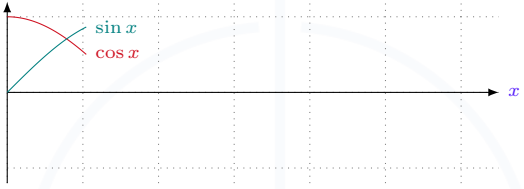
- $\sin(x)$ and $\cos(x)$ are the orthogonal projections of the point at $x$ rad.

  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 1.04718$ rad $(60.0°)$
$\sin(x) \approx 0.866$
$\cos(x) \approx 0.5$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

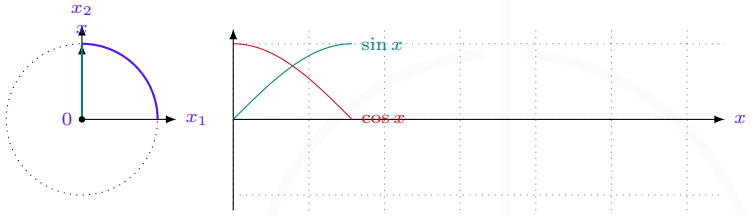
  From rad(ians) to deg(ree): $x \mapsto \frac{360°}{2\pi} x = \frac{180°}{\pi} x$

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  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 1.57077$ rad $(90.0°)$
$\sin(x) \approx 0.99998$
$\cos(x) \approx 0.00002$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

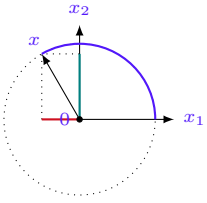
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# Sine and cosine







$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 2.09436$ rad $(120.0°)$
$\sin(x) \approx 0.86603$
$\cos(x) \approx -0.49995$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

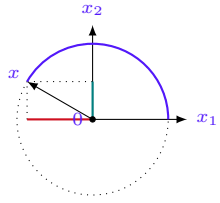
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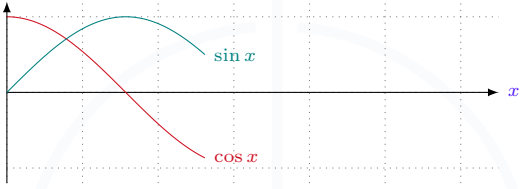
- $\sin(x)$ and $\cos(x)$ are the orthogonal projections of the point at $x$ rad.

  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 2.61795$ rad $(150.0°)$
$\sin(x) \approx 0.50003$
$\cos(x) \approx -0.86598$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

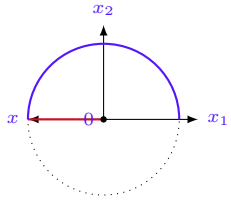
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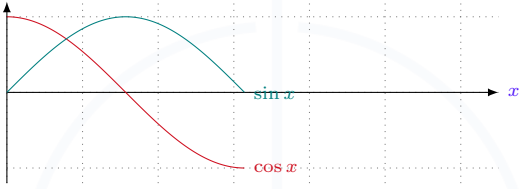
- $\sin(x)$ and $\cos(x)$ are the orthogonal projections of the point at $x$ rad.

  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 3.14154$ rad $(180.0°)$
$\sin(x) \approx 0.00005$
$\cos(x) \approx -0.99998$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

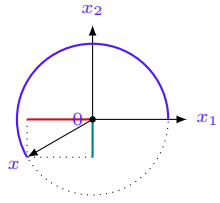
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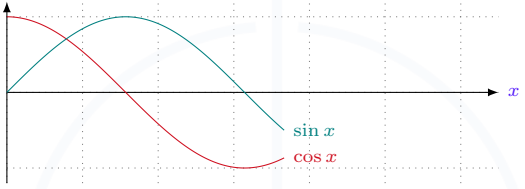
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  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 3.66513$ rad $(210.0°)$
$\sin(x) \approx -0.49994$
$\cos(x) \approx -0.86604$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

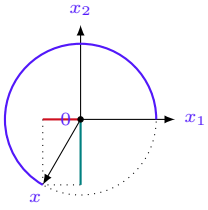
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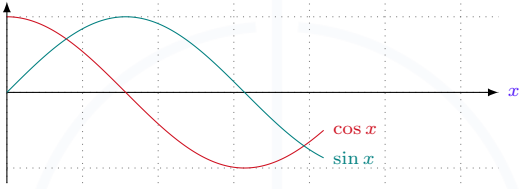
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  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 4.18872$ rad $(240.0°)$
$\sin(x) \approx -0.86598$
$\cos(x) \approx -0.50005$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

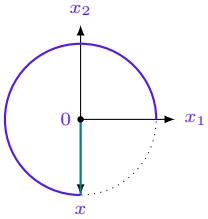
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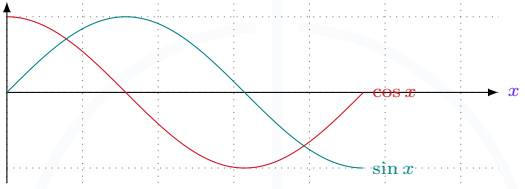
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  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 4.71231$ rad $(270.0°)$
$\sin(x) \approx -0.99998$
$\cos(x) \approx -0.00008$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

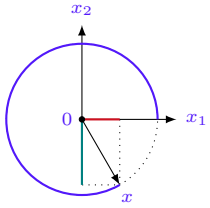
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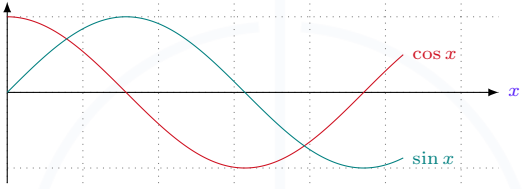
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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 5.2359$ rad $(300.0°)$
$\sin(x) \approx -0.86604$
$\cos(x) \approx 0.49992$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

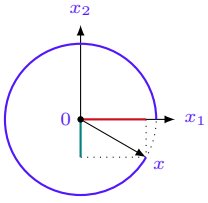
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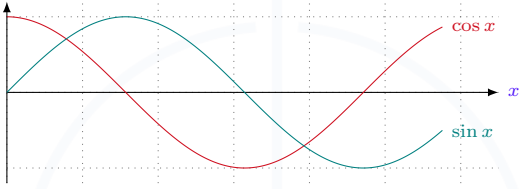
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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 5.75949$ rad $(330.0°)$
$\sin(x) \approx -0.50006$
$\cos(x) \approx 0.86597$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

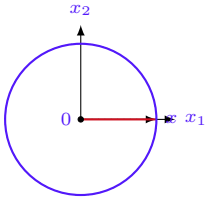
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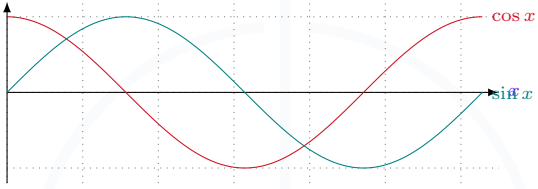
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  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 6.28308$ rad $(360.0°)$
$\sin(x) \approx -0.00009$
$\cos(x) \approx 0.99998$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

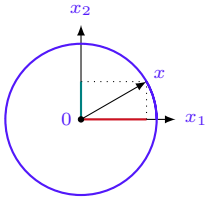
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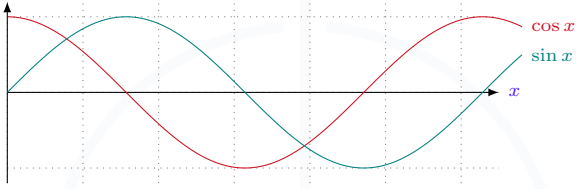
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  (lengths of the opposite/adjacent legs of the corresponding right triangle.)

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# Sine and cosine




$\sin(x)^2 + \cos(x)^2 = 1$
$x \approx 6.80667$ rad $(390.0°)$
$\sin(x) \approx 0.4999$
$\cos(x) \approx 0.86607$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

($\pi$: ratio circumference to diameter.)

- In most programming languages $\sin(x), \cos(x)$ assume that $x$ is measured in radians (arc length):

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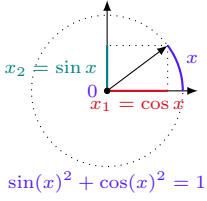
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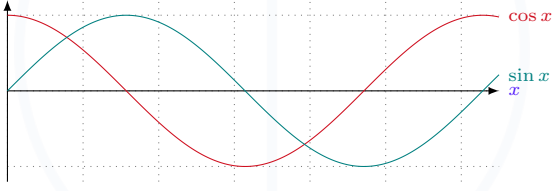
**Lem** If $x$ is given in rad, we have:

- $\sin(x) = \sum_{k=0}^{\infty}(-1)^k \frac{x^{2k+1}}{(2k+1)!} = \frac{x}{1} - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} \pm \dots$
- $\cos(x) = \sum_{k=0}^{\infty}(-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} \pm \dots$

**Def** $2\pi$ is the circumference of the unit circle $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$.

With $x$ as arc length (radians) on the unit length ($\frac{360}{2\pi} \cdot x$ in degree):




with $\sin(-x) = -\sin(x)$ (odd $x^{2k+1}$) and $\cos(-x) = \cos(x)$ (even $x^{2k}$).

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# Sine and cosine

**Lem** If $x$ is given in rad, we have:

- $\sin(x) = \sum_{k=0}^{\infty}(-1)^k \frac{x^{2k+1}}{(2k+1)!} = \frac{x}{1} - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} \pm \dots$
- $\cos(x) = \sum_{k=0}^{\infty}(-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} \pm \dots$

**Lem** Addition theorems: For all $x, y \in \mathbb{R}$

- $\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y)$
- $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$

- Definition of $\sin$ and $\cos$:

  The addition theorems can be proved using e.g., the power series and their absolute convergence (see "Cauchy product"), but also using only their geometric definition (see e.g., here or linearity of rotations).

  The addition theorems allow to show the continuity and differentiability of $\sin$ and $\cos$ and subsequently derive the power series from the geometric definition.

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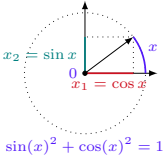
# Sine and cosine

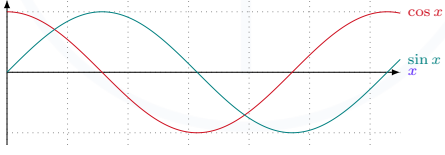
**Lem** If $x$ is given in rad, we have:

- $\sin(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = \frac{x}{1} - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} \pm \dots$
- $\cos(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} \pm \dots$

**Lem** For all $x \in \mathbb{R}$:

- $\sin(x)^2 + \cos(x)^2 = 1$ (unit circle, Pythagoras)
- $\sin(x + 2\pi) = \sin(x)$ and $\cos(x + 2\pi) = \cos(x)$ (period $2\pi$)
- $\sin(-x) = -\sin(x)$ and $\cos(-x) = \cos(x)$ (odd/even; see power series)
- $\sin(x) = \cos(\frac{\pi}{2} - x)$ and $\cos(x) = \sin(\frac{\pi}{2} - x)$ (swap axis)
- $\sin(x) = \cos(x - \frac{\pi}{2})$ and $\cos(x) = \sin(x + \frac{\pi}{2})$ (phase)




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# Sine and cosine

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**Lem** $\sin(x)$ is strictly mon. increasing on $[-\pi/2, \pi/2]$.
$\cos(x)$ is strictly mon. decreasing on $[0, \pi]$.

Hence, the inverse functions exists:

$\arcsin(x) := \sin^{-1}(x)$ (from $[-1,1]$ to $[-\pi/2, \pi/2]$)
$\arccos(x) := \cos^{-1}(x)$ (from $[-1,1]$ to $[0,\pi]$)

($\arcsin(x) + \arccos(x) = \pi/2$ (sum of angles in right triangle))

**Def** $\tan(x) := \frac{\sin(x)}{\cos(x)}$ for $x \in \mathbb{R} \setminus \{\frac{\pi}{2} + \pi k \mid k \in \mathbb{Z}\}$.

$\cot(x) := \frac{\cos(x)}{\sin(x)} = \tan(\frac{\pi}{2} - x)$ for $x \in \mathbb{R} \setminus \{\pi k \mid k \in \mathbb{Z}\}$.

$\tan(x)$ is strictly mon. increasing on $(-\pi/2, \pi/2)$,
its inverse $\tan^{-1}(x)$ on $(-\pi/2, \pi/2)$ is denoted by $\arctan(x)$.

(See also the animation on wikipedia.)

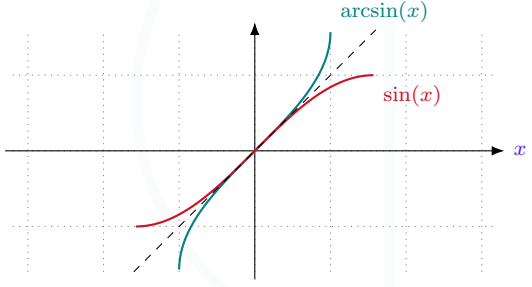
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- Example: $\sin$ and $\arcsin$



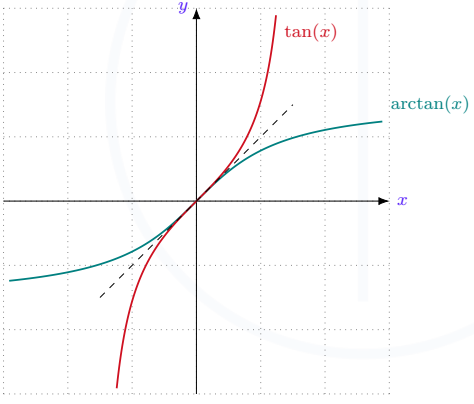

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- Example: $\tan$ and $\arctan$




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# Elementary functions on Python

```
import matplotlib.pyplot as plt
import math

elementary_functions = (lambda x: pow(2,x), lambda x: pow(3,x), math.sqrt, math.exp, math.log, math.log2, math.si

N = 100
x_min = 0.1 # Python reports 'math domain error' if 'x_min=0' as 'log' is undefined at 0
x_max = 2
xs = [x_min + k*(x_max-x_min)/N for k in range(N)]
for f in elementary_functions:
  ys = [f(x) for x in xs]
  plt.plot(xs, ys)
plt.show()
```

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1. Real numbers
2. Limit of a sequence, finite vs. infinite sums
3. Elementary functions
4. Continuity
5. Differentiation and Integration



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4 Continuity
Definition
Properties



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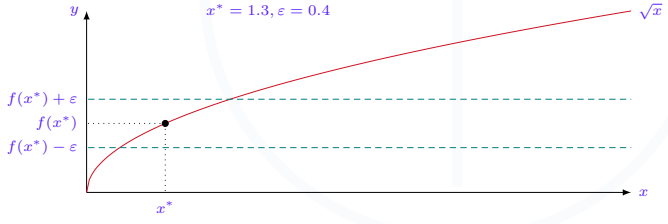
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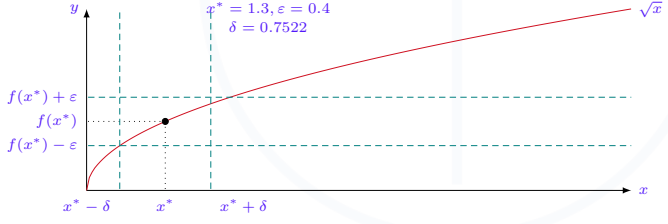
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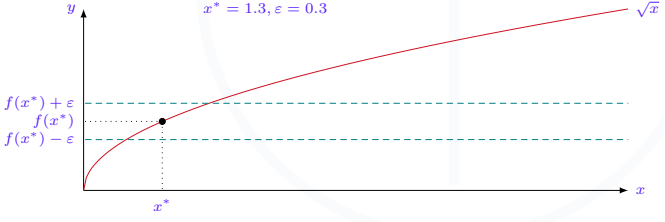
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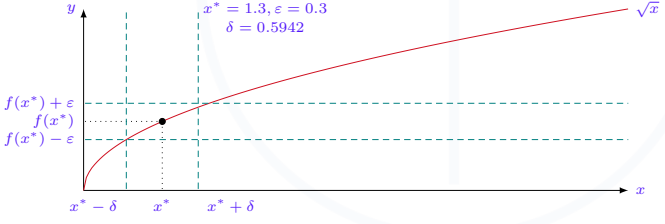
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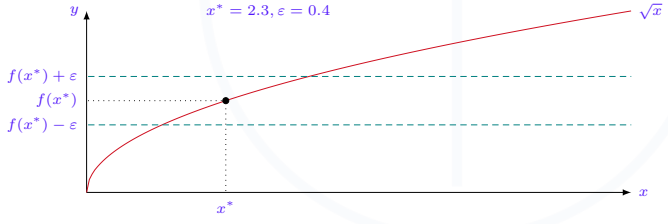
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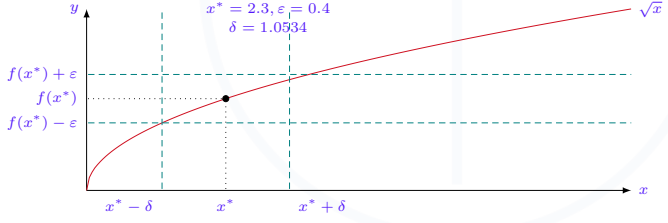
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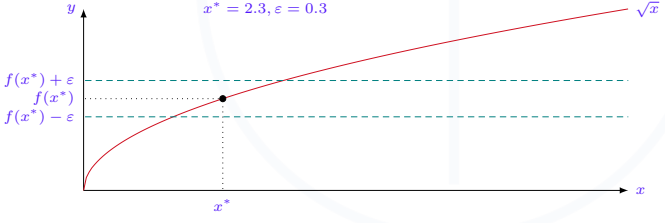
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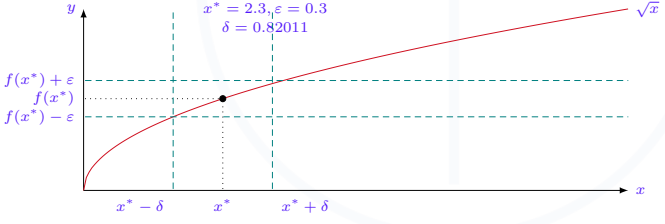
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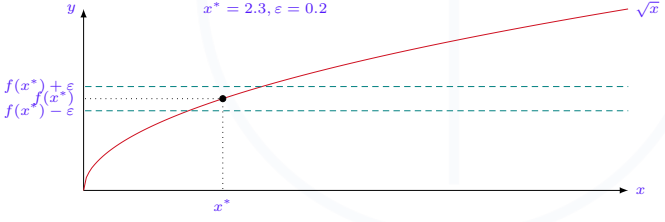
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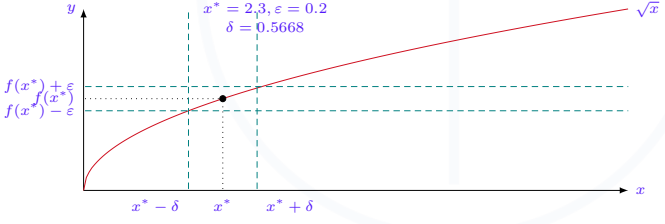
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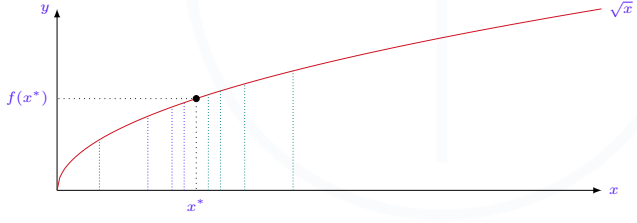

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- ▷ Negation: There is some $\varepsilon > 0$ s.t. for all $\delta > 0$ there is some $x \in I$ with $|x - x^*| < \delta$ and $|f(x) - f(x^*)| \geq \varepsilon$ **iff** there exists a sequence $(x_i)_{i\in\mathbb{N}_0}$ in $I$ with $\lim_{i\to\infty} x_i = x^*$ and $\lim_{i\to\infty} f(x_i) \neq f(x^*)$.

⚠ I.e. the limit can be taken before or after applying $f$: we can also evaluate $f$ on approximations for $x^*$ to obtain approximations of $f(x^*)$.

(Very useful if we can only approximate $x^* \in \mathbb{R}$.)



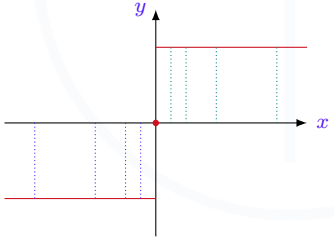

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(Very useful if we can only approximate $x^* \in \mathbb{R}$.)

**EG** $f(x) = c$ constant ($I = (-\infty, \infty) = \mathbb{R}$)

- ▷ For every given $x^* \in \mathbb{R}$ and $\varepsilon > 0$ any $\delta > 0$ can be chosen as

$$|f(x) - f(x^*)| = |c - c| = 0 < \varepsilon$$

Alternatively: Let $\lim_{i\to\infty} x_i = x^*$. Then (trivially)

$$\lim_{i\to\infty} f(x_i) = \lim_{i\to\infty} c = c = f(x^*)$$

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# Continuity: Definition using Sequences

**Lem** A function $f\colon I \to \mathbb{R}$ is continuous at $x^* \in \mathbb{R}$ ($I \subseteq \mathbb{R}$ interval) **iff** for every sequence $(x_i)_{i\in\mathbb{N}_0}$ in $I$ with $\lim_{i\to\infty} x_i = x^*$ also $\lim_{i\to\infty} f(x_i) = f(x^*)$; short: $\lim_{i\to\infty} f(x_i) = f(\lim_{i\to\infty} x_i)$.

- Negation: There is some $\varepsilon > 0$ s.t. for all $\delta > 0$ there is some $x \in I$ with $|x - x^*| < \delta$ and $|f(x) - f(x^*)| \geq \varepsilon$ **iff** there exists a sequence $(x_i)_{i\in\mathbb{N}_0}$ in $I$ with $\lim_{i\to\infty} x_i = x^*$ and $\lim_{i\to\infty} f(x_i) \neq f(x^*)$.

⚠ I.e. the limit can be taken before or after applying $f$: we can also evaluate $f$ on approximations for $x^*$ to obtain approximations of $f(x^*)$.

(Very useful if we can only approximate $x^* \in \mathbb{R}$.)

**EG** $f(x) = x$ ($I = (-\infty, \infty) = \mathbb{R}$)

- For every given $x^* \in \mathbb{R}$ and $\varepsilon > 0$ choose $\delta = \varepsilon$ as for $|x - x^*| < \delta$

$$|f(x) - f(x^*)| = |x - x^*| < \delta = \varepsilon$$

Alternatively: Let $\lim_{i\to\infty} x_i = x^*$. Then

$$\lim_{i\to\infty} f(x_i) = \lim_{i\to\infty} x_i = x^* = f(x^*)$$

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(Very useful if we can only approximate $x^* \in \mathbb{R}$.)

**EG** $f(x) = |x|$ ($I = (-\infty, \infty) = \mathbb{R}$)

- For every given $x^* \in \mathbb{R}$ and $\varepsilon > 0$ choose $\delta = \varepsilon$ as for $|x - x^*| < \delta$

$$|f(x) - f(x^*)| = ||x| - |x^*|| \leq |x - x^*| < \delta = \varepsilon$$

Alternatively: Let $\lim_{i\to\infty} x_i = x^*$. Then

$$|f(x_i) - f(x^*)| = ||x_i| - |x^*|| \leq |x_i - x^*| \xrightarrow{i\to\infty} 0$$

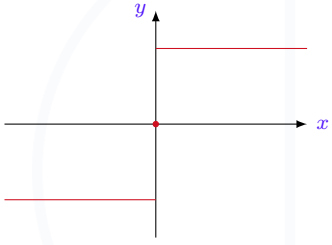
(Reminder: $||x| - |y|| \leq |x - y| \leq |x| + |y|$)

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# Example: Discontinuous functions (*)

- $f$ is not continuous at $x^* \in I$ **iff** there exists a sequence $(x_i)_{i \in \mathbb{N}_0}$ in $I$ with $\lim_{i \to \infty} x_i = x^*$ and $\lim_{i \to \infty} f(x_i) \neq f(x^*)$.

EG $f(x) = \text{sgn}(x)$ is continuous in every $x^* \in \mathbb{R} \setminus \{0\}$, but not at $x^* = 0$.




- Given $x^* \neq 0$, we can choose $\delta = |x^*|$ for every $\varepsilon > 0$.
- Given $x^* = 0$, for every $0 < \varepsilon < 1$ there is no suitable $\delta > 0$.

  E.g. for $x_i := \frac{1}{i+1}$ we have $f(x_i) = 1$ s.t.

  $\lim_{i \to \infty} x_i = 0$ but $\lim_{i \to \infty} f(x_i) = 1 \neq 0 = f(0)$.

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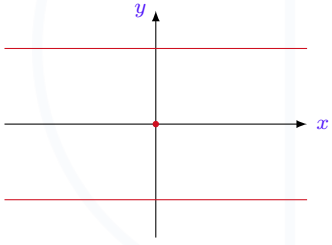
# Example: Discontinuous functions (*)

- $f$ is not continuous at $x^* \in I$ **iff** there exists a sequence $(x_i)_{i \in \mathbb{N}_0}$ in $I$ with $\lim_{i \to \infty} x_i = x^*$ and $\lim_{i \to \infty} f(x_i) \neq f(x^*)$.

EG As $\mathbb{Q}$ is dense in $\mathbb{R}$,

$$f \colon \mathbb{R} \to \mathbb{R}, x \mapsto \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

is not continuous in any $x^* \in \mathbb{R}$.




- E.g. fix any irrational $x^* \in \mathbb{R} \setminus \mathbb{Q}$. Then there exists a rational sequence $(x_i)_{i \in \mathbb{N}_0}$ with $\lim_{i \to \infty} x_i = x^*$ s.t. $-1 = f(x^*) = f(\lim_{i \to \infty} x_i) \neq \lim_{i \to \infty} f(x_i) = 1$
- If $x^* \in \mathbb{Q}$, then choose e.g. $x_i = x^* + \frac{\sqrt{2}}{i+1} \in \mathbb{R} \setminus \mathbb{Q}$.

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**Def** $f: I \to \mathbb{R}$ is continuous on the interval $I \subseteq \mathbb{R}$ **:iff** $f$ is continuous in every $x^* \in I$.

⚠ Continuous functions can be approximated by simple functions as they do not "jump": e.g. for any given error bound $\varepsilon > 0$ and point $x^*$ we can approximate $f$ on $x \in (x^* - \delta, x^* + \delta) \cap I$ by $f(x^*) \pm \varepsilon$.

▷ If $f$ is continuous on a closed (compact) interval $I = [a, b]$, then for any given $\varepsilon > 0$ we can determine an $\delta_\varepsilon > 0$, s.t. $|f(x^*) - f(x)| < \varepsilon$ for all $|x - x^*| < \varepsilon$ independently of $x^*$.

(See e.g. uniform continuity (animation).)

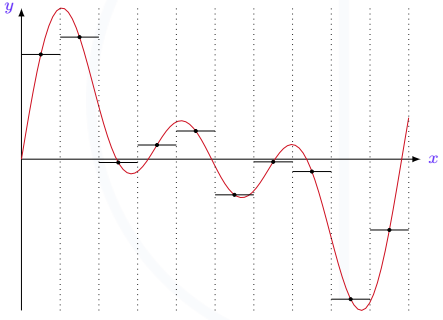
I.e. we may "sample" $f$ at finitely many points in $[a, b]$ at a distance of less than $\delta_\varepsilon$ and represent $f$ by these finitely many samples with an error less than $\varepsilon$ – which allows to approximately represent $f$ on the computer (e.g. images/video, audio signals, imaging procedures for medical applications, ...)

See also integral.

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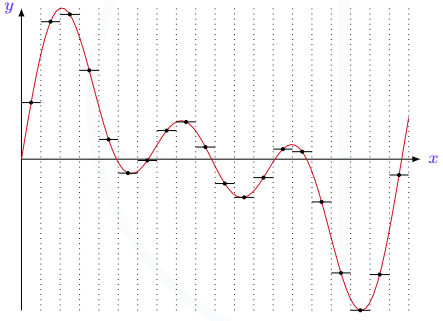

("digization of images, video, audio, continuous signals, ...")

⚠ Another way to view this: we can "learn" a continuous function $f\colon [a, b] \to \mathbb{R}$.

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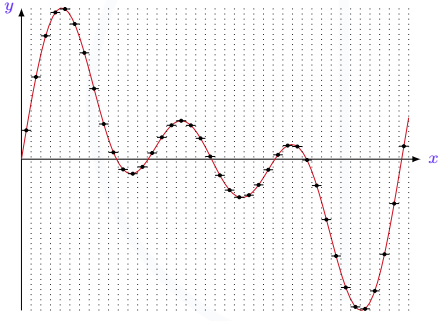

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4 Continuity

- Definition
- Properties

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# Rules for computing with continuous functions

**Lem** Using the rules for computing with limits, one can show:
If $f$ and $g$ are both continuous in $x^* \in I$ and $\lim_{i\to\infty} x_i = x^*$, then

- $\lim_{i\to\infty}(f \pm g)(x_i) = f(x^*) \pm g(x^*) = (f \pm g)(x^*)$
- $\lim_{i\to\infty}(f \cdot g)(x_i) = f(x^*) \cdot g(x^*) = (f \cdot g)(x^*)$
- $\lim_{i\to\infty}(f/g)(x_i) = \frac{f(x^*)}{g(x^*)} = (f/g)(x^*)$ if $g(x^*) \neq 0$.
- $\lim_{i\to\infty} |f(x_i)| = |f(x^*)|$.

(i.e. their sum, difference, product, quotient, absolute value are all continuous.)

If $g$ is continuous at $f(x^*)$, then

$$\lim_{i\to\infty} g(f(x_i)) = g(f(x^*))$$

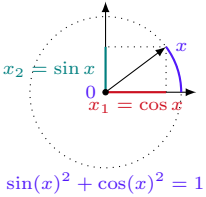
i.e. their composition is also continuous.

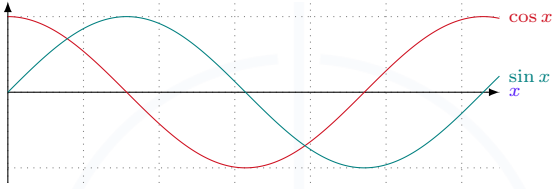
(Again, very useful for computing with continuous functions, they stay "well-behaved".)

**Cor** Thus, every polynomial represents a continuous function on $\mathbb{R}$ as $f(x) = \text{const}$ and $f(x) = x$ are continuous functions on $\mathbb{R}$.

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# Continuity of sin and cos




- Using either the representation as series or elementary geometry:
  - $\sin(x) \leq x$ on $[0, \infty]$.
  - $\cos(x) \geq 1 - x$ resp. $x \geq 1 - \cos(x)$ on $[0, \infty]$.

By virtue of the addition theorems:

$$\begin{aligned} |\cos(x+h) - \cos(x)| &= |\cos(x)(\cos(h) - 1) - \sin(x)\sin(h)| \\ &\leq |\cos(x)| \cdot |h| + |\sin(x)| \cdot |h| \leq 2|h| \xrightarrow{h \to 0} 0 \\ |\sin(x+h) - \sin(x)| &= |\sin(x)(\cos(h) - 1) + \sin(h)\cos(x)| \\ &\leq |\sin(x)| \cdot |h| + |\cos(x)| \cdot |h| \leq 2|h| \xrightarrow{h \to 0} 0 \end{aligned}$$

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# Important continuous functions

**Lem** Each of the following function is continuous on its domain:

- Every polynomial function $f\colon \mathbb{R} \to \mathbb{R}, x \mapsto a_0 + a_1 x + \ldots + a_n x^n$.
- The absolute value $|x|$ on $\mathbb{R}$.
- $e^x$, $\sin(x)$, $\cos(x)$ on $\mathbb{R}$
- $\ln(x)$ on $\mathbb{R}_{>0}$, $\sqrt[k]{x} = x^{1/k}$ on $\mathbb{R}_{\geq 0}$ (see inverse of strictly monotone functions)
- For every fixed $b > 0$ in general $b^x$ on $\mathbb{R}$ and $\log_b(x)$ on $\mathbb{R}_{>0}$.

(See the lecture on analysis for proofs.)

- (*) As $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ converges absolutely on $\mathbb{R}$, one can show that $\exp(x)$ is continuous in $x^* = 0$.

  This already means that $\exp(x)$ is continuous on $\mathbb{R}$, as with $\lim_{i\to\infty} x_i = x^*$ also

  $\lim_{i\to\infty} e^{x_i} = \lim_{i\to\infty} e^{x^*} e^{x_i - x^*} = e^x \lim_{i\to\infty} e^{x_i - x^*} = e^{x^*} \cdot 1$ ($x^*$ fixed)

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# Intermediate value theorem (w/o proof)

**Thm** For every continuous funktion $f\colon [a,b] \to \mathbb{R}$ we have ($[a,b]$ important):

Intermediate value theorem ("continuous functions do not jump")

For every $y$ between $f(a)$ and $f(b)$, there is some $x \in [a,b]$ with $f(x) = y$.

Short: $[\min(f(a), f(b)), \max(f(a), f(b))] \subseteq f([a,b])$

(If $f$ is monotone, then $[\min(f(a), f(b)), \max(f(a), f(b))] = f([a,b])$.)

Bolzano's theorem (consequence)

If $f(a) < 0 < f(b)$, then $f$ has a zero/root in $(a,b)$.

- ▷ Allows to approximate zeros using bisection (binary search).

Minimum and Maximum

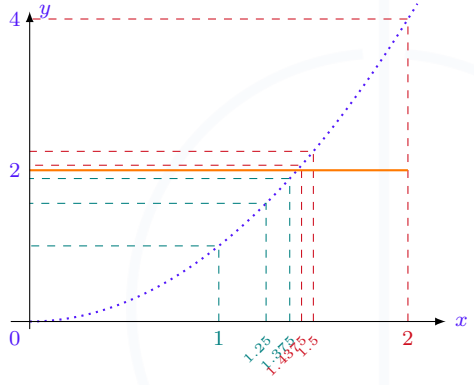
$f$ is bounded on $[a,b]$ and $\min f([a,b]), \max f([a,b])$ exist.

Fixed-point theorem for continuous self-maps

If $f([a,b]) \subseteq [a,b]$, then there exists an $x \in [a,b]$ with $f(x) = x$.

Such an $x$ is called a fixed-point of $f$.

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- As $x^n$ is a polynomial, the function $f_c\colon \mathbb{R}_{\geq 0} \to \mathbb{R}, x \mapsto x^n - c$ is continuous and strictly monotonically increasing for every constant $c \in \mathbb{R}$.
- If $c \geq 0$, then $f_c(0) = -c \leq 0$ and $f_c(b) > 0$ for some sufficiently large $b$.
  (If $c > 1$, we can choose $b = c$, else $b = 1$ suffices.)
- Thus, we can approximate the only zero $\sqrt[n]{c} = f_0^{-1}(c) = f_c^{-1}(0)$ of $f_c(x)$ in $(0, b)$ by using "$f_c(x_i) > 0$?" to guide bisection (binary search) as $f_c(x) = x^n$ "cannot" jump over $c$. (This computes in fact the binary representation.)

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# Inverse of strictly monotone functions (w/o proof)

**Thm** If $f\colon I \to \mathbb{R}$ is strictly mon. increasing (decreasing) on the interval $I \subseteq \mathbb{R}$, then

$f^{-1}\colon f(I) \to I$ exists and is also stricly mon. increasing (decreasing) and continuous.

If $f$ is also continuous on $I$, then $f(I)$ is also an interval.

- ▷ Strict monotonicity guarantees already that $f\colon I \xrightarrow{\text{bij}} f(I)$.
- ▷ If $f$ is not continuous on $I$, then $f(I)$ is not an interval, but $f^{-1}$ is still continuous in every $y^* \in f(I)$.

**Cor** As $f(x) = \exp(x)$ is continuous and strictly mon. increasing on $\mathbb{R} \to \mathbb{R}_{>0}$, $f(\mathbb{R}) = \mathbb{R}_{>0}$ and $f^{-1}(x) = \ln(x)$ is continuous on $\mathbb{R}_{>0} \to \mathbb{R}$.

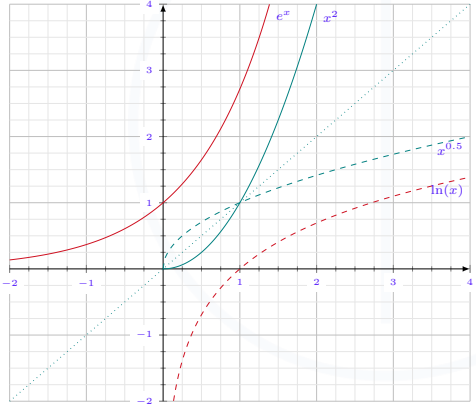
As $f(x) = x^n$ is continuous and strictly mon. increasing on $\mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, $f(\mathbb{R}_{\geq 0}) = \mathbb{R}_{\geq 0}$ and $f^{-1}(x) = \sqrt[n]{x}$ is continuous on $\mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$.

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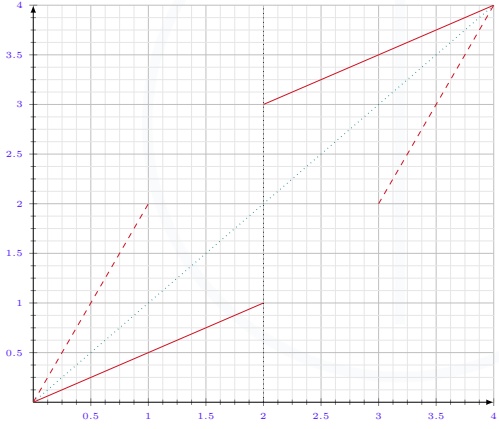



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1. Real numbers
2. Limit of a sequence, finite vs. infinite sums
3. Elementary functions
4. Continuity
5. Differentiation and Integration



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5 Differentiation and Integration
Derivative
Differentiation Rules
Antiderivative
Integration and Fundamental Theorem of Calculus
L'Hospital (*)
Taylor series, extrema (*)
Newton's Method (*)

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# Derivative

- A continuous function can be approximated by a constant $f(x^*)$ on a sufficiently small interval $(x^* - \delta, x^* + \delta)$ for a given error bound $\varepsilon$.
- We can also try to approximate $f$ at $x^*$ by means of a line $c \cdot (x - x^*) + f(x^*)$.
  With $h := x - x^*$ the error is then:

$$e(h) := f(x^* + h) - (c \cdot h + f(x^*)) = \left( \frac{f(x^* + h) - f(x^*)}{h} - c \right) h$$

  If we require, that $e(h)$ vanishes faster than $h$, i.e.

$$0 = \lim_{h \to 0} \frac{e(h)}{h} = \lim_{h \to 0} \left( \frac{f(x^*+h) - f(x^*)}{h} - c \right)$$

  then we obtain the derivative of $f$ at $x^*$ if the limit exists:

$$c = \lim_{h \to 0} \frac{f(x^*+h) - f(x^*)}{h} =: \frac{d}{dx} f(x^*) =: f'(x^*)$$

  As the limit $c$ is uniquely determined

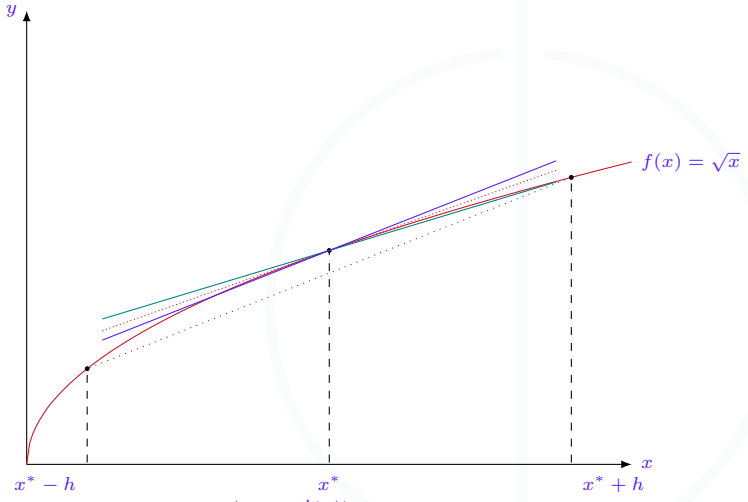
$$f(x^* + h) \approx f'(x^*) \cdot h + f(x^*)$$

  is the best approximation of all possible lines through $(x^*, f(x^*))$.

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# Example: Derivative of $\sqrt{x}$





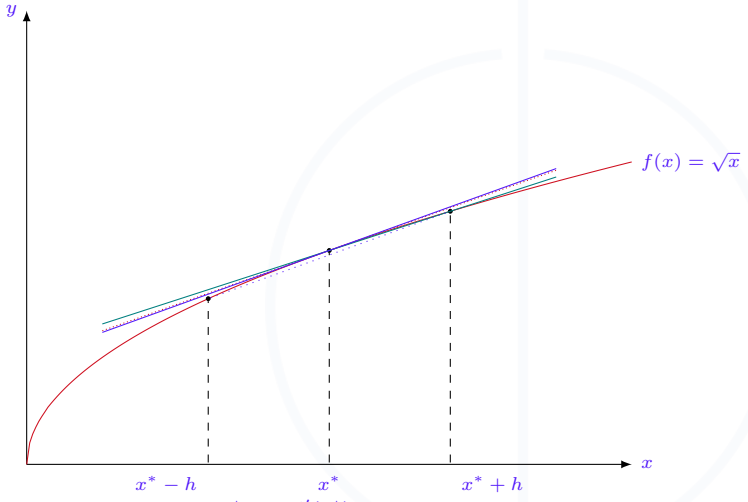

$x^* = 2, f'(x^*) = 0.3536$

$h = 1.6, \frac{f(x^*+h)-f(x^*)}{h} = 0.302, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.3952$

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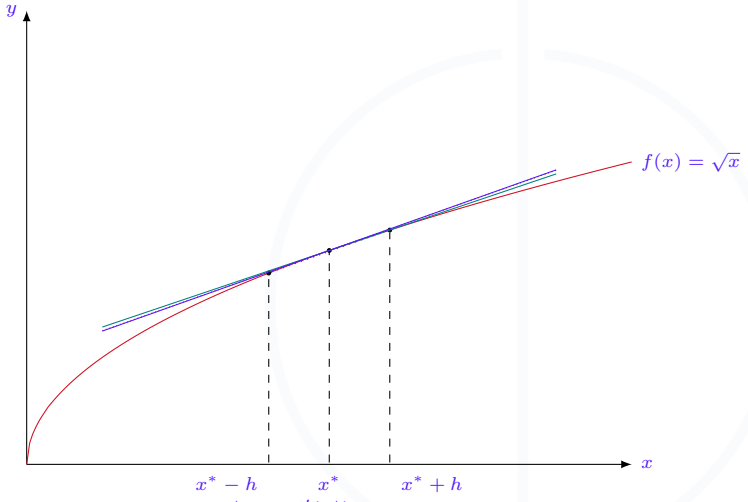

$x^* = 2, f'(x^*) = 0.3536$

$h = 0.8, \frac{f(x^*+h)-f(x^*)}{h} = 0.3239, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.3611$

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# Example: Derivative of $\sqrt{x}$





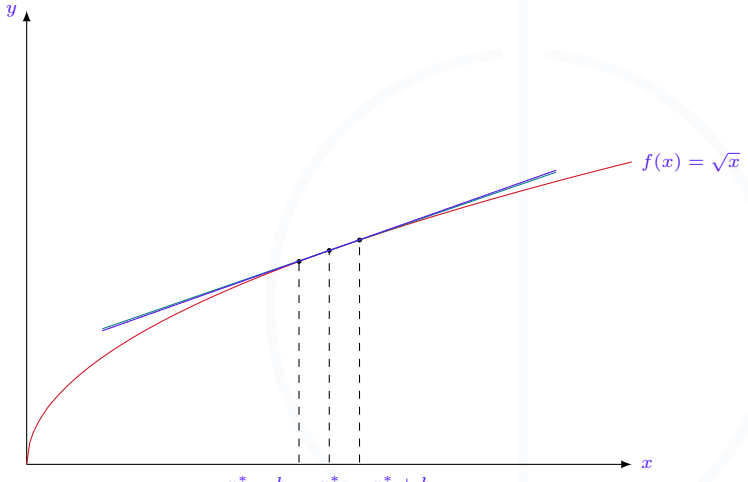

$x^* = 2, f'(x^*) = 0.3536$

$h = 0.4, \frac{f(x^*+h)-f(x^*)}{h} = 0.3375, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.35533$

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# Example: Derivative of $\sqrt{x}$





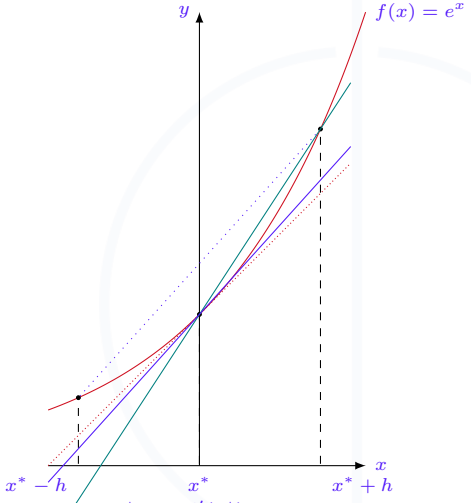

$x^* = 2, f'(x^*) = 0.3536$

$h = 0.2, \frac{f(x^*+h)-f(x^*)}{h} = 0.3453, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.35397$

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# Example: Derivative of $e^x$





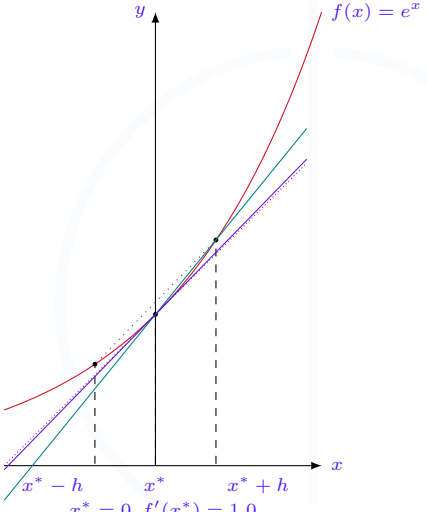

$x^* = 0, f'(x^*) = 1.0$

$h = 0.8, \frac{f(x^*+h)-f(x^*)}{h} = 1.53192, \frac{f(x^*+h)-f(x^*-h)}{2h} = 1.11$

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# Example: Derivative of $e^x$



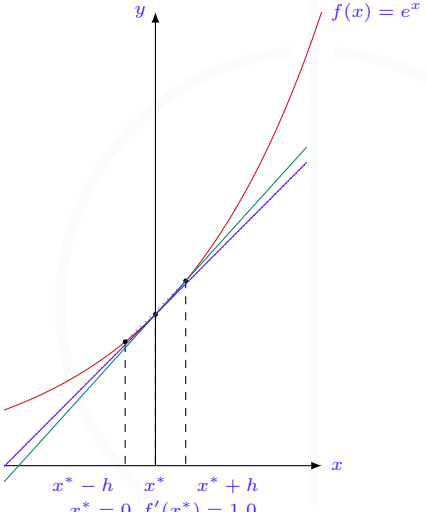



$x^* = 0, f'(x^*) = 1.0$

$h = 0.4, \frac{f(x^*+h)-f(x^*)}{h} = 1.22945, \frac{f(x^*+h)-f(x^*-h)}{2h} = 1.02667$

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# Example: Derivative of $e^x$



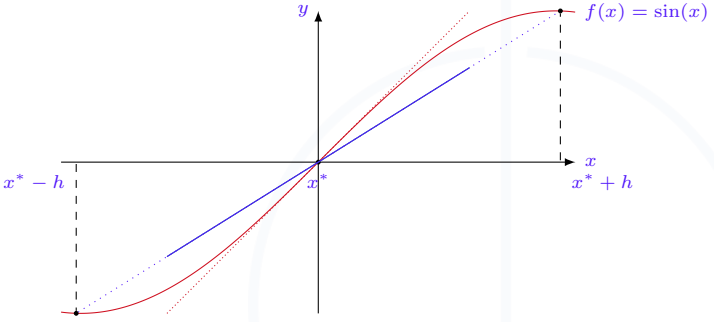

$x^* = 0, f'(x^*) = 1.0$

$h = 0.2, \frac{f(x^*+h)-f(x^*)}{h} = 1.10687, \frac{f(x^*+h)-f(x^*-h)}{2h} = 1.00624$

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# Example: Derivative of $\sin(x)$





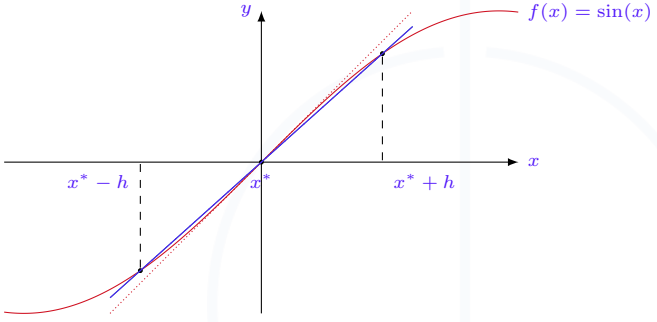

$$x^* = 0, f'(x^*) = 1.0$$

$$h = 1.6, \frac{f(x^*+h)-f(x^*)}{h} = 0.6247, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.6247$$

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# Example: Derivative of $\sin(x)$





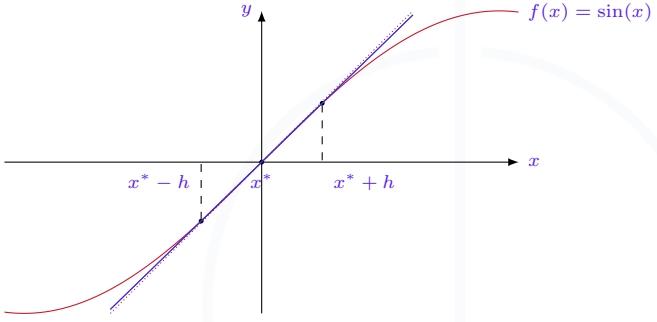

$$x^* = 0, f'(x^*) = 1.0$$

$$h = 0.8, \frac{f(x^*+h)-f(x^*)}{h} = 0.8967, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.8966$$

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# Example: Derivative of $\sin(x)$





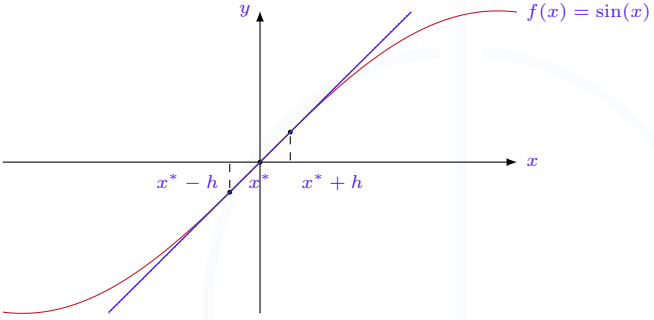

$$x^* = 0, f'(x^*) = 1.0$$

$$h = 0.4, \frac{f(x^*+h)-f(x^*)}{h} = 0.97356, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.97356$$

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# Example: Derivative of $\sin(x)$





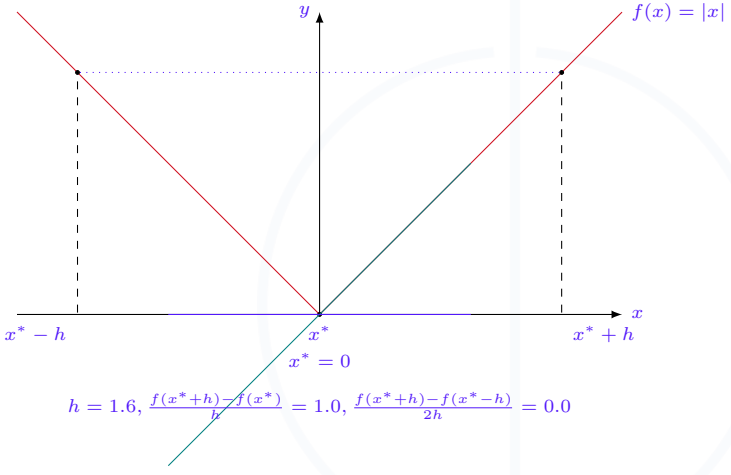

$$x^* = 0, f'(x^*) = 1.0$$

$$h = 0.2, \frac{f(x^*+h)-f(x^*)}{h} = 0.9932, \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.99324$$

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# Example: Derivative of $|x|$

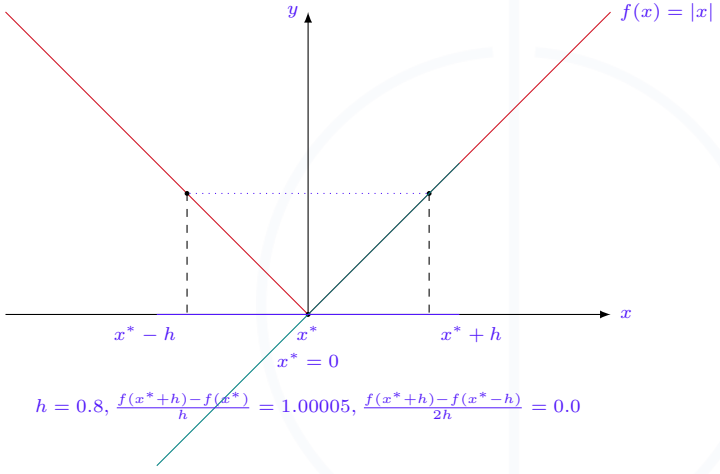





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# Example: Derivative of $|x|$



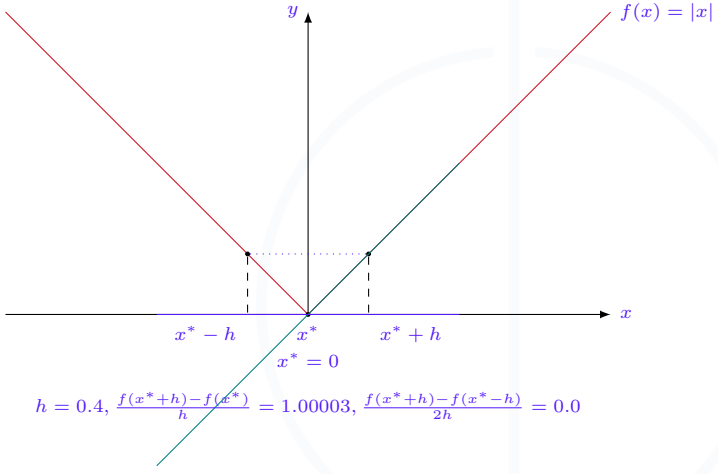



$h = 0.8,\ \frac{f(x^*+h)-f(x^*)}{h} = 1.00005,\ \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.0$

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# Example: Derivative of $|x|$



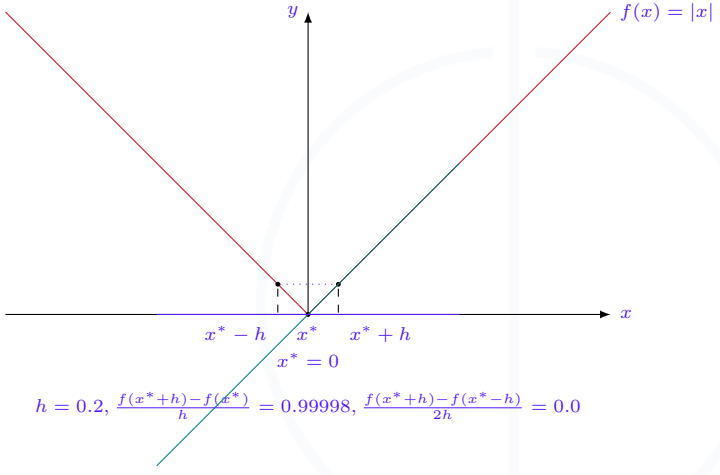



$h = 0.4,\ \frac{f(x^*+h)-f(x^*)}{h} = 1.00003,\ \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.0$

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# Example: Derivative of $|x|$






$h = 0.2,\ \frac{f(x^*+h)-f(x^*)}{h} = 0.99998,\ \frac{f(x^*+h)-f(x^*-h)}{2h} = 0.0$

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**Def** A function $f: I \to \mathbb{R}$ is differentiable at a given point $x^* \in I$ **:iff**

$$c(h) = \frac{f(x^*+h) - f(x^*)}{x^*+h-x^*} = \frac{f(x^*+h) - f(x^*)}{h}$$

is continuous at $0$ (i.e. $\lim_{h\to 0} c(h)$ exists). Then

$$f'(x^*) := (\tfrac{d}{dx} f)(x^*) := \lim_{h\to 0} c(h) = \lim_{h\to 0} \frac{f(x^*+h) - f(x^*)}{h}$$

denotes the derivative of $f$ at $x^*$.

This defines $f'$ as function in every point in which $f$ is differentiable. If $f'$ is again differentiable, we can repeat this to obtain the higher derivatives of $f$.

Taking the derivate $n$-times at $x^*$ is denoted by $\frac{d^n}{dx^n} f|_{x=x^*}$ or $f^{(n)}(x^*)$

- Equivalently one can use the symmetric difference:

$$f'(x^*) = \lim_{h\to 0} \frac{f(x^*+h) - f(x^*-h)}{2h}$$

⚠ If $f$ is differentiable at $x^*$, it is also continuous.

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# Computing derivatives: $1, x, x^2$

- $f(x) = c$ (constant) at any fixed $x^*$:

$$\frac{f(x^* + h) - f(x^*)}{h} = \frac{c - c}{2h} = \frac{0}{2h} \overset{h \neq 0}{=} 0 \xrightarrow{h \to 0} 0$$

  Thus $f'(x) = 0$.

- $f(x) = x$ at any fixed $x^*$:

$$\frac{f(x^* + h) - f(x^*)}{h} = \frac{(x^* + h) - (x^*)}{h} = \frac{h}{h} \overset{h \neq 0}{=} 1 \xrightarrow{h \to 0} 1$$

  Thus $f'(x) = 1$.

- $f(x) = x^2$ at any fixed $x^*$:

$$\frac{f(x^* + h) - f(x^*)}{h} = \frac{(x^* + h)^2 - (x^*)^2}{h} = \frac{2x^*h + h^2}{h} \overset{h \neq 0}{=} 2x^* + h \xrightarrow{h \to 0} 2x^*$$

  Thus $f'(x) = 2x$.

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# Computing derivatives: $x^{-1}, \sin(x)$

- $f(x) = \frac{1}{x}$ at any fixed $x^* \neq 0$:

$$\begin{aligned}
\tfrac{f(x^*+h)-f(x^*)}{h} &= \tfrac{(x^*+h)^{-1}-(x^*)^{-1}}{h} \\
&= \tfrac{(x^*)-(x^*+h)}{h(x^*+h)(x^*)} \\
&\overset{h\neq 0}{=} \tfrac{-1}{(x^*+h)(x^*)} \\
&\xrightarrow{h\to 0} \tfrac{-1}{(x^*)^2}
\end{aligned}$$

  Thus $f'(x) = -x^{-2}$ for $x \neq 0$.

- By means of the series or geometrically: $\lim_{h\to 0} \frac{\sin(h)}{h} = 1$. Thus:

  $f(x) = \sin(x)$ at any fixed $x^*$ (with $\sin(x^* \pm h) = \sin(x^*)\cos(h) \pm \cos(x^*)\sin(h)$):

$$\begin{aligned}
\tfrac{f(x^*+h)-f(x^*-h)}{2h} &= \tfrac{\sin(x^*+h)-\sin(x^*-h)}{2h} \\
&= \cos(x^*)\tfrac{\sin(h)}{h} \\
&\xrightarrow{h\to 0} \cos(x^*)
\end{aligned}$$

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# Computing derivatives: exp

- $f(x) = e^x$ at any fixed $x^*$:

$$\frac{e^{x^*+h} - e^{x^*}}{h} = e^{x^*} \cdot \left(1 + \frac{e^h - (1+h)}{h}\right) \xrightarrow{h \to 0} e^{x^*}$$

Thus $f'(x) = e^x$.

- Using the series:

$$e^h = \sum_{k=0}^{\infty} \frac{h^k}{k!} = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24} + \dots$$

Thus

$$\frac{e^h - (1+h)}{h} = \frac{h}{2} + \frac{h^2}{6} + \frac{h^3}{24} + \dots = h \cdot \sum_{l=0}^{\infty} \frac{h^l}{(l+2)!}$$

and

$$\left|\frac{e^h - (1+h)}{h}\right| \leq |h| \cdot \sum_{l=0}^{\infty} \frac{|h|^l}{(l+2)!} \leq |h| \cdot \sum_{l=0}^{\infty} \frac{|h|^l}{l!} = |h| \cdot e^{|h|} \xrightarrow{h \to 0} 0$$

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5 Differentiation and Integration

Derivative

Differentiation Rules

Antiderivative

Integration and Fundamental Theorem of Calculus

L'Hospital (*)

Taylor series, extrema (*)

Newton's Method (*)

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# Differentiation rules: linearity, product, composition

**Lem** Let $f\colon I \to \mathbb{R}$, $g\colon J \to \mathbb{R}$ with $I, J \subseteq \mathbb{R}$ intervals. Fix $x^* \in I$.

If $f'(x^*)$ and $g'(x^*)$ exist ($x^* \in I \cap J$), then: (rules for $\lim$)

- $(\lambda f + \mu g)'(x^*) = \lambda f'(x^*) + \mu g'(x^*)$ (for $\lambda, \mu \in \mathbb{R}$)
- $(f \cdot g)'(x^*) = f(x^*) \cdot g'(x^*) + f'(x^*) \cdot g(x^*)$
- $(\frac{f}{g})'(x^*) = (f \cdot \frac{1}{g})'(x^*) = f'(x^*) \cdot \frac{1}{g(x^*)} + f(x^*) \cdot \frac{-g'(x^*)}{g(x^*)^2}$ (if $g(x^*) \neq 0$)

If $f'(x^*)$ and $g'(f(x^*))$ exist ($f(x^*) \in J$), then:

$$(g \circ f)'(x^*) = g'(f(x^*)) \cdot f'(x^*)$$

- Examples:
  - $\frac{d}{dx} x^2 = \left(\frac{d}{dx} x\right) \cdot x + x \cdot \left(\frac{d}{dx} x\right) = 1 \cdot x + x \cdot 1 = 2x$
  - $\frac{d}{dx} \frac{1}{f(x)} \overset{g(x)=1/x}{=} \frac{-1}{f(x)^2} \cdot f'(x)$ e.g. $\frac{d}{dx} \frac{1}{cos(x)} = \frac{\sin(x)}{\cos(x)^2}$
  - $\frac{d}{dx} \tan(x) = \frac{d}{dx} \frac{\sin(x)}{\cos(x)} = \frac{\cos(x)}{\cos(x)} + \frac{\sin(x)^2}{\cos(x)^2} = 1 + \tan(x)^2$.

    (Extend $\frac{\cos(x)}{\cos(x)}$ continuously.)

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# Differentiation rules: linearity, product, composition

**Lem** Let $f\colon I \to \mathbb{R}$, $g\colon J \to \mathbb{R}$ with $I, J \subseteq \mathbb{R}$ intervals. Fix $x^* \in I$.

If $f'(x^*)$ and $g'(x^*)$ exist ($x^* \in I \cap J$), then: (rules for $\lim$)

- $(\lambda f + \mu g)'(x^*) = \lambda f'(x^*) + \mu g'(x^*)$ (for $\lambda, \mu \in \mathbb{R}$)
- $(f \cdot g)'(x^*) = f(x^*) \cdot g'(x^*) + f'(x^*) \cdot g(x^*)$
- $(\frac{f}{g})'(x^*) = (f \cdot \frac{1}{g})'(x^*) = f'(x^*) \cdot \frac{1}{g(x^*)} + f(x^*) \cdot \frac{-g'(x^*)}{g(x^*)^2}$ (if $g(x^*) \neq 0$)

If $f'(x^*)$ and $g'(f(x^*))$ exist ($f(x^*) \in J$), then:

$$(g \circ f)'(x^*) = g'(f(x^*)) \cdot f'(x^*)$$

- Example: If $f$ and $g = f^{-1}$ are differentiable:

$$1 = \frac{d}{dx}x = \frac{d}{dx}\left(f^{-1} \circ f\right)(x) = \left(\frac{d}{dx}f^{-1}\right)(f(x)) \cdot f'(x)$$

Thus if $f'(x) \neq 0$:

$$\left(\frac{d}{dx}f^{-1}\right)(f(x)) = f'(x)^{-1} = \frac{1}{f'(x)} \quad \text{resp.} \quad \left(\frac{d}{dx}f^{-1}\right)(y) = f'(f^{-1}(y))^{-1} = \frac{1}{f'(f^{-1}(y))}$$

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- More generally:

**Lem** Let $f\colon I \to \mathbb{R}$, $g\colon J \to \mathbb{R}$ with $I, J \subseteq \mathbb{R}$ intervals. Fix $x^* \in I$.
If $f$ is strictly monotone and continuous on $I$ and $f'(x^*) \neq 0$ exists,
(i.e. $f(I)$ is an interval and $f^{-1}\colon f(I) \to I$ is continuous and strictly monotone)
then $f^{-1}$ is differentiable at $y^* := f(x^*)$ with

$$(f^{-1})'(y^*) = (\tfrac{d}{dy} f^{-1})(y^*) = \frac{1}{(\frac{d}{dx} f)(x^*)} = \frac{1}{f'(x^*)} \overset{x^* = f^{-1}(y^*)}{=} \frac{1}{f'(f^{-1}(y^*))}$$

(Recall: $f^{-1} = f^\top$ results from $f$ by swapping $x$ and $y$: $(y^*, f^{-1}(y^*)) = (x^*, f(x^*))$.)

- Examples: With $f = \exp =$ for all $y \in \mathbb{R}_{>0}$:

$$\ln'(y) = \frac{1}{\exp(\ln(y))} = \frac{1}{y}$$

With $f = x^n$ for all $y \in \mathbb{R}_{>0}$ and $n > 1$:

$$\frac{d}{dy} \sqrt[n]{y} = \frac{d}{dy} y^{1/n} = \frac{1}{n(y^{1/n})^{n-1}} = \frac{1}{n} \cdot y^{1/n-1}$$

(Note that $y^{1/n}$ is not differentiable at $y = 0$.)

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$$
\begin{aligned}
& \lim_{h\to 0} \tfrac{(\lambda f+\mu g)(x^*+h)-(\lambda f+\mu g)(x^*)}{h} \\
= \; & \lim_{h\to 0} \left(\lambda \tfrac{f(x^*+h)-f(x^*)}{h} + \mu \tfrac{g(x^*+h)-g(x^*)}{h}\right) \\
= \; & \lim_{h\to 0} \lambda \tfrac{f(x^*+h)-f(x^*)}{h} + \mu \lim_{h\to 0} \tfrac{g(x^*+h)-g(x^*)}{h} \\
= \; & \lambda f'(x^*) + \mu g'(x^*)
\end{aligned}
$$

$$
\begin{aligned}
& \lim_{h\to 0} \tfrac{(fg)(x^*+h)-(fg)(x^*)}{h} \\
= \; & \lim_{h\to 0} \tfrac{f(x^*+h)g(x^*+h)-f(x^*)g(x^*+h)+f(x^*)g(x^*+h)-f(x^*)g(x^*)}{h} \\
= \; & \lim_{h\to 0} \tfrac{f(x^*+h)-f(x^*)}{h} g(x^*+h) + \lim_{h\to 0} f(x^*) \tfrac{g(x^*+h)-g(x^*)}{h} \\
= \; & f'(x^*)g(x^*) + f(x^*)g'(x^*)
\end{aligned}
$$

$$
\begin{aligned}
& \lim_{h\to 0} \tfrac{(f/g)(x^*+h)-(f/g)(x^*)}{h} \\
= \; & \lim_{h\to 0} \tfrac{f(x^*+h)/g(x^*+h)-f(x^*)/g(x^*+h)+f(x^*)/g(x^*+h)-f(x^*)/g(x^*)}{h} \\
= \; & \lim_{h\to 0} \tfrac{f(x^*+h)-f(x^*)}{h} \cdot \tfrac{1}{g(x^*+h)} + \lim_{h\to 0} \tfrac{f(x^*)}{g(x^*+h)g(x^*)} \cdot \tfrac{-g(x^*+h)+g(x^*)}{h} \\
= \; & f'(x^*)\tfrac{1}{g(x^*)} + f(x^*)\tfrac{-g'(x^*)}{g(x^*)^2}
\end{aligned}
$$

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- With

$$g(f(x^*)+\eta)-g(f(x^*))=g'(f(x^*))\eta+e_g(\eta) \quad \lim_{\eta\to 0}\frac{e_g(\eta)}{\eta}=0$$

and

$$f(x^*+h)-f(x^*)=f'(x^*)h+e_f(h) \quad \lim_{h\to 0}\frac{e_f(h)}{h}=0$$

and for $h$ small enough

$$\eta(h):=f(x^*+h)-f(x^*)=f'(x^*)h+e_f(h) \quad |\eta(h)|<C|h|$$

also

$\frac{g(f(x^*+h))-g(f(x^*))}{h}=\frac{g'(f(x^*))[f'(x^*)h+e_f(h)]}{h}+\frac{e_g(\eta(h))}{\eta(h)}\cdot\frac{\eta(h)}{h}\xrightarrow{h\to\infty} g'(f(x^*))f(x^*)$

- By assumptions $f(I)$ is an interval, $f^{-1}$ continuous and $f'(f^{-1}(y^*))\neq 0$.

With $x=f^{-1}(y)$ and $y\to y^*$ also $x\to x^*=f^{-1}(y^*)$:

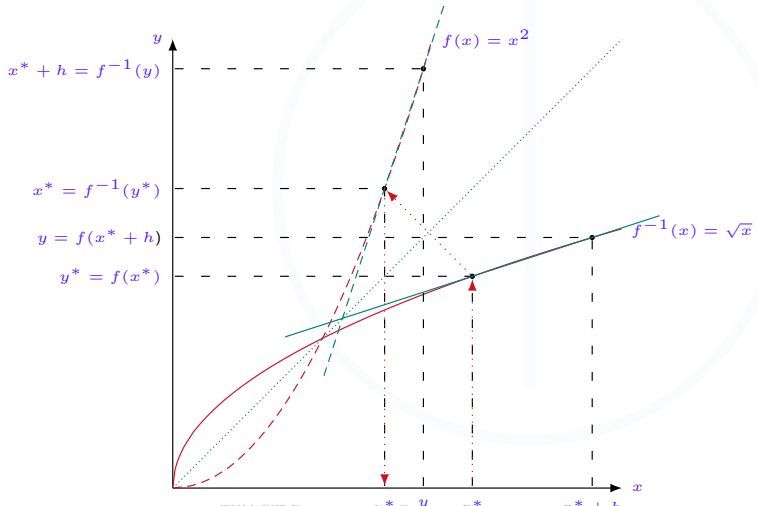
$$\frac{f^{-1}(y)-f^{-1}(y^*)}{y-y^*}=\frac{x-x^*}{f(x)-f(x^*)}=\frac{1}{\frac{f(x)-f(x^*)}{x-x^*}}\xrightarrow{y\to y^*}\frac{1}{f'(x^*)}=\frac{1}{f'(f^{-1}(y^*))}$$

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# Illustration: $(f^{-1})'(x^*) = \frac{1}{f'(y^*)} = \frac{1}{f'(f^{-1}(x^*))}$



- As $f^{-1} = f^\top$ is obtained from $f$ by swapping $x$ and $f(x)$, the tangent at $f^{-1}$ in $(x^*, y^*) = (x^*, f(x^*))$ corresponds to the tangent at $f$ in $(y^*, x^*) = (y^*, f^{-1}(y^*))$ with the slope $(f^{-1})'(x^*) = \frac{1}{f'(f^{-1}(x^*))}$.




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# Derivatives of elementary functions

Lem

- $\frac{d}{dx} x^{k+1} = (k+1)x^k$ for $k \in \mathbb{N}_0$
- $\frac{d}{dx} x^{-k} = -kx^{-k-1}$ for $k \in \mathbb{N}$ and $x \neq 0$
- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} \sin(x) = \cos(x)$
- $\frac{d}{dx} \cos(x) = -\sin(x)$
- $\frac{d}{dx} \ln x = \frac{1}{e^{\ln x}} = \frac{1}{x}$ for $x > 0$
- $\frac{d}{dx} b^x = \frac{d}{dx} e^{x \ln(b)} = \ln(b) \cdot e^{x \ln b} = \ln(b) \cdot b^x$ for $b > 0$
- $\frac{d}{dx} x^r = \frac{d}{dx} e^{r \ln(x)} = \frac{r}{x} e^{r \ln(x)} = \frac{r}{x} \cdot x^r = r \cdot x^{r-1}$ for $x > 0$

- Examples:
  - $(x^{k+1})' = (x \cdot x^k)' = 1 \cdot x^k + x \cdot (x^k)' \overset{\text{Induction}}{=} x^k + x \cdot kx^{k-1} = (k+1)x^k$
  - $(x^{-k})' = \left(\frac{1}{x^k}\right)' = -\frac{kx^{k-1}}{x^{2k}} = -kx^{-k-1}$
  - If $r \in \mathbb{R} \setminus \{0\}$ and $x > 0$ s.t. $\ln(x)$ exists:

$$\frac{d}{dx} x^r = \frac{d}{dx} \exp(r \ln(x)) = \exp(r \ln(x)) \cdot \frac{r}{x} = r x^{r-1}$$

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# Derivatives of elementary functions

Lem

- $\frac{d}{dx} x^{k+1} = (k+1)x^k$ for $k \in \mathbb{N}_0$
- $\frac{d}{dx} x^{-k} = -kx^{-k-1}$ for $k \in \mathbb{N}$ and $x \neq 0$
- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} \sin(x) = \cos(x)$
- $\frac{d}{dx} \cos(x) = -\sin(x)$
- $\frac{d}{dx} \ln x = \frac{1}{e^{\ln x}} = \frac{1}{x}$ for $x > 0$
- $\frac{d}{dx} b^x = \frac{d}{dx} e^{x \ln(b)} = \ln(b) \cdot e^{x \ln b} = \ln(b) \cdot b^x$ for $b > 0$
- $\frac{d}{dx} x^r = \frac{d}{dx} e^{r \ln(x)} = \frac{r}{x} e^{r \ln(x)} = \frac{r}{x} \cdot x^r = r \cdot x^{r-1}$ for $x > 0$

- The derivatives of $\sin$ and $\cos$ can be obtained from their representations as series. (Alternatively, using their geometric definitions argue that $\lim_{h \to \infty} \frac{\sin(h)}{h} = 0$ and $\lim_{h \to 0} \frac{\cos(h)}{h} = 1$, then use the addition theorems to reduce the derivative at $x^*$ to the derivative at $0$.)
- Again, just rules which the computer can apply for us without understanding what $\sin$ or $\sin'$ actually means. It can simply ("mindlessly") replace $\sin'$ by $\cos$ and thus "compute" with these "symbols", see e.g. SymPy.

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5 Differentiation and Integration

- Derivative
- Differentiation Rules
- Antiderivative
- Integration and Fundamental Theorem of Calculus
- L'Hospital (*)
- Taylor series, extrema (*)
- Newton's Method (*)

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# Antiderivative

**Def** Let $f\colon I \to \mathbb{R}$ with $I \subseteq \mathbb{R}$ interval. $F\colon I \to \mathbb{R}$ is an antiderivative of $f$ **:iff** $F'$ exists and coincides with $f$ on $I$.

**Lem** Reading the differentiation rules backwards yields:

- If $F_1' = f = F_2'$ on $I$, then $F_1 - F_2 \equiv c$ on $I$ for some constant $c \in \mathbb{R}$.
- If $F' = f$ and $G' = g$, then $(\lambda F + \mu G)' = \lambda f + \mu g$ ($\lambda, \mu \in \mathbb{R}$; linearity).
- If $H' = f'g$, then $(fg - H)' = fg'$ (assuming $f', g'$ exist on $I$'; partial integration).
- If $G' = g$, then $(G \circ f)' = (g \circ f) \cdot f'$ (assuming $g'$ exists on $I$; substitution rule).

In particular, the antiderivative is not uniquely determined.

- Proof sketch:
  - Intuition: as $D := F_1 - F_2$ is differentiable on $I$ with $D' \equiv 0$, it is constant.
    Formally: pick any two $x_1 < x_2 \in I$; by the mean value theorem there is some $\xi \in (x_1, x_2)$ s.t. $D'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$; as $D'(\xi) = 0$ we obtain $D(x_1) = D(x_2)$.
  - Rest: simply use the linearity, product rule, and chain rule, respectively, to check that the derivative on the LHS evaluates to the the RHS under the stated assumptions.

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# Antiderivative

**Def** Let $f\colon I \to \mathbb{R}$ with $I \subseteq \mathbb{R}$ interval. $F\colon I \to \mathbb{R}$ is an antiderivative of $f$ **:iff** $F'$ exists and coincides with $f$ on $I$.

**Lem** Reading the differentiation rules backwards yields:

- If $F_1' = f = F_2'$ on $I$, then $F_1 - F_2 \equiv c$ on $I$ for some constant $c \in \mathbb{R}$.
- If $F' = f$ and $G' = g$, then $(\lambda F + \mu G)' = \lambda f + \mu g$ ($\lambda, \mu \in \mathbb{R}$; linearity).
- If $H' = f'g$, then $(fg - H)' = fg'$ (assuming $f', g'$ exist on $I$; partial integration).
- If $G' = g$, then $(G \circ f)' = (g \circ f) \cdot f'$ (assuming $g'$ exists on $I$; substitution rule).

In particular, the antiderivative is not uniquely determined.

- Examples: An antiderivative of
  - $f(x) = 1$ is $F(x) = x + c$ (for every $c \in \mathbb{R}$).
  - $3\cos(x) + 5\sin(x)$ is e.g. $3\sin(x) - 5\cos(x) + 1$.
  - $\ln(x)$ is $x\ln(x) - x$ for $x \in (0, \infty)$.
    (Start with $1 \cdot \ln(x)$ and guess $f(x) = \ln(x)$, $g'(x) = 1$, $g(x) = x$ s.t. $H' = f'g = 1$.)

(For the substitution rule see the next slide.)

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# Substitution rule: applications

**Lem** Let $f$ be differentiable on $I$.

- ▷ If $f(x) > 0$ on $I$, then $\ln(f(x))$ is an antiderivative of $\frac{f'(x)}{f(x)}$ on $I$.
  (Consider $(G(f(x)))'$ for $g(x) = \frac{1}{x}$, $G(x) = \ln(x)$.)
  If $f(x) < 0$ on $I$, then $\ln(-f(x))$ antiderivative of $\frac{-f'(x)}{-f(x)} = \frac{f'(x)}{f(x)}$.
- ▷ $\exp(f(x))$ is an antiderivative of $f'(x)e^{f(x)}$ on $I$.
  (Consider $(G(f(x)))'$ for $G(x) = g(x) = \exp(x)$)
- ▷ $a^{-1}f(ax+b)$ is an antiderivative of $f'(ax+b)$ ($a, b \in \mathbb{R}$, $a \neq 0$).
  (Consider $(f(g(x)))'$ for $g(x) = ax + b$.)

▷ Example: $\ln(\cos(x)^{-1})$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$ is an antiderivative of

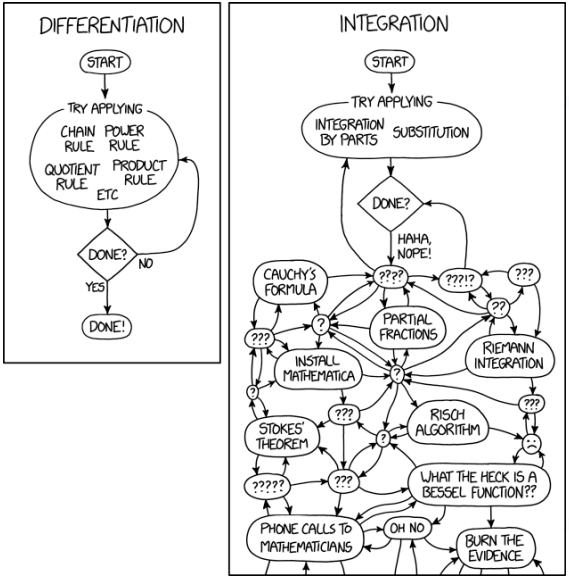
$$(\ln(\cos(x)^{-1}))' = (-\ln(\cos(x)))' = -\frac{-\sin(x)}{\cos(x)} = \tan(x)$$

▷ Example: $\frac{b^x}{\ln(b)}$ ($b > 0$) on $\mathbb{R}$ is an antiderivative of

$$\left(\frac{b^x}{\ln(b)}\right)' = \frac{(\exp(\ln(b)x))'}{\ln(b)} = \frac{\ln(b)\cdot\exp(\ln(b)x)}{\ln(b)} = b^x$$

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# Moving backwards is sometimes harder




[xkcd]

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# Frequently used antiderivatives

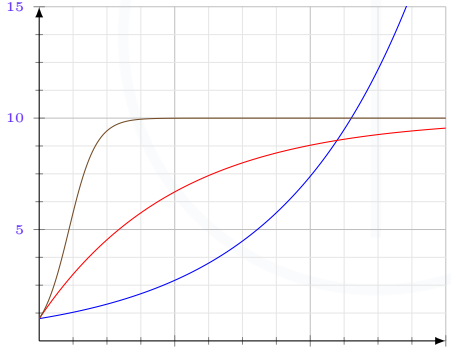
| $f$ | $F$ | domain |
|---|---|---|
| $x^a$ | $\frac{x^{a+1}}{a+1}$ | $x > 0$ $(a \neq -1)$ |
| $x^{-1}$ | $\ln x$ | $x > 0$ |
| $(1-x^2)^{-1}$ | $\frac{1}{2}\ln\left\lvert\frac{x+1}{x-1}\right\rvert$ | $\lvert x\rvert \neq 1$ |
| $e^x$ | $e^x$ | $x \in \mathbb{R}$ |
| $b^x$ | $\frac{b^x}{\ln(b)}$ | $x \in \mathbb{R}$ $(b > 0)$ |
| $\sin(x)$ | $-\cos(x)$ | $x \in \mathbb{R}$ |
| $\cos(x)$ | $\sin(x)$ | $x \in \mathbb{R}$ |
| $\tan(x)$ | $-\ln(\cos(x))$ | $x \in (-\pi/2, \pi/2)$ |
| $\ln(x)$ | $x\ln(x) - x$ | $x > 0$ |
| $\sin(x)^2$ | $\frac{1}{2}(x - \sin(x)\cos(x))$ | $x \in \mathbb{R}$ |
| $\cos(x)^2$ | $\frac{1}{2}(x + \sin(x)\cos(x))$ | $x \in \mathbb{R}$ |
| $\sin(x)^{-2}$ | $-\cot(x)$ | $x \in (0, \pi)$ |
| $\cos(x)^{-2}$ | $\tan(x)$ | $x \in (-\pi/2, \pi/2)$ |

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# Growth (*)

**Lem** Assume a growth rate $\gamma > 0$. One can show that for given initial value $f_0 = f(0)$ the following functions are the unique solution of the respective differential equation (check that they are some solution):

- (natural) $f(t) = f_0 e^{\gamma t}$ of $f'(t) = \gamma f(t)$.
- (bounded) $f(t) = F - (F - f_0)e^{-\gamma t}$ of $f'(t) = \gamma(F - f(t))$ $(0 < f_0 \leq F)$.
- (logistic) $f(t) = \frac{F}{1-(1-\frac{F}{f_0})e^{-\gamma F t}}$ of $f'(t) = \gamma f(t)(F - f(t))$ $(0 < f_0 \leq F)$.




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5 Differentiation and Integration

- Derivative
- Differentiation Rules
- Antiderivative
- Integration and Fundamental Theorem of Calculus
- L'Hospital (*)
- Taylor series, extrema (*)
- Newton's Method (*)

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**Def** An antiderivative $F$ of $f$ is also called an indefinite integral and denoted by

$$F(x) = \int f(x)dx$$

i.e. "$\int f(x)dx$" stands for "determine an antiderivative of $f$".

- ▷ Note that $\int f(t)dt = \int f(x)dx$: the variable name is bound by $dx$ resp. $dt$, and thus can be changed locally.

**Def** If $f\colon [a,b] \to \mathbb{R}$ ($a < b$), then

$$\int_a^b f(x)dx$$

is called definite integral of $f$ on $[a,b]$.

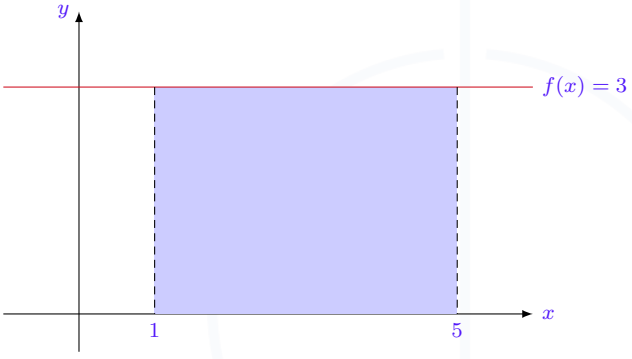
If $f$ is continuous and $f(x) \geq 0$ on $[a,b]$, then $\int_a^b f(x)dx$ is defined as the oriented area of the region delimited by the graph of $f$, the $x$-axis and the interval bounds $a$ and $b$.

I.e. "$\int_a^b f(x)dx$" stands for "compute the enclosed area".

("oriented" means that $\int_b^a f(x)dx := -\int_a^b f(x)dx$.)

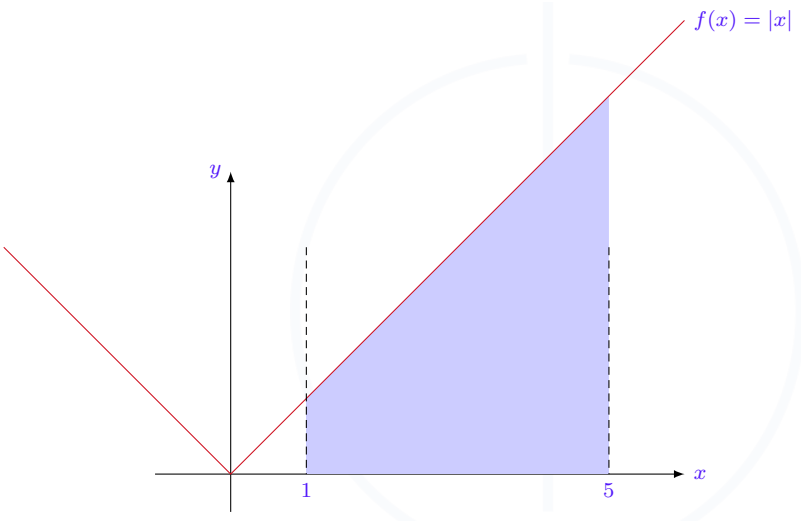
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# Definite integral: constant function




$$\int_1^5 f(x)dx \stackrel{!}{=} 3 \cdot (5-1) = 12$$

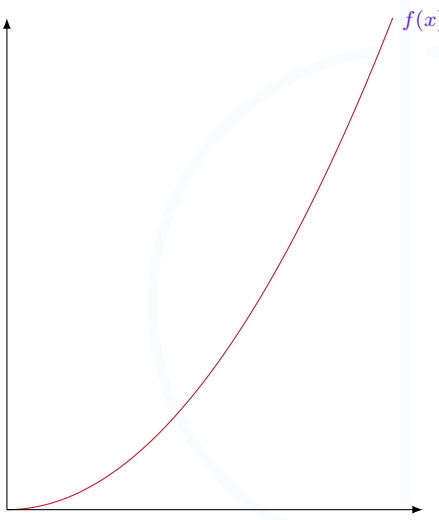
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$$\int_1^5 f(x)dx \stackrel{!}{=} \frac{1}{2} \cdot 5 \cdot 5 - \frac{1}{2} \cdot 1 \cdot 1 = 12$$

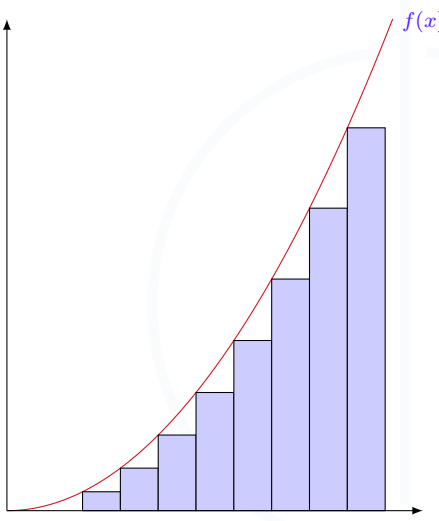
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# Definite integral: quadratic function



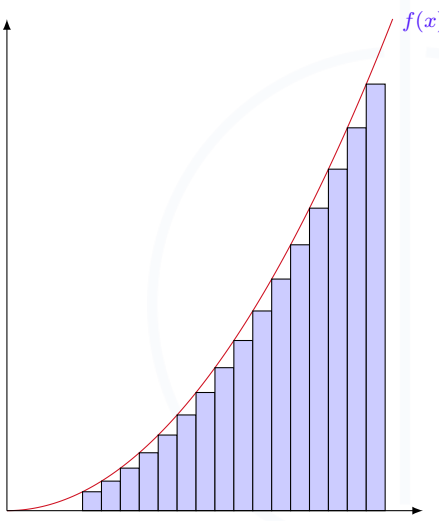

$$\int_1^5 \frac{x^2}{4} dx = ?$$

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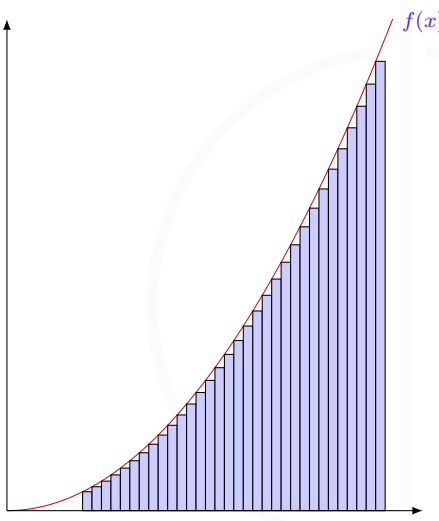

$$\int_1^5 \frac{x^2}{4} dx \approx \sum_{i=0}^{7} f(1 + i/2) \cdot \frac{1}{2} = 8.875$$

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$$\int_1^5 \frac{x^2}{4} dx \approx \sum_{i=0}^{15} f(1 + i/4) \cdot \frac{1}{4} = 9.59375$$

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$$\int_1^5 \frac{x^2}{4} dx \approx \sum_{i=0}^{31} f(1 + i/8) \cdot \frac{1}{8} = 9.9609375$$

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- In general, one has to show that the approximations of the area $\int_a^b f(\xi)d\xi$ by

  $$\sum_{i=0}^{n-1} f(\xi_i)(x_{i+1} - x_i) \quad \text{(with } a = x_0 < x_1 < \ldots < x_{n-1} < x_n = b \text{ and } \xi_i \in [x_i, x_{i+1}]\text{)}$$

  all converge to the same value for $\Delta := \max(x_{i+1} - x_i) \to 0$ independent on how we partition the interval $[a, b]$ s.t. the area is indeed well-defined.

  This leads to the Darboux integral resp. Riemann integral.

- EG E.g. consider $f\colon [0, 1] \to \mathbb{R}$ with $f(x) = 1$ if $x \in \mathbb{Q}$ else $f(x) = 0$.

  If all $\xi_i$ are rational (irrational), the sum will be equal to $b - a$ $(0)$.

  I.e. not every function is "Riemann/Darboux-integrable".

  (There exist several formalizations of the concept of integration which allow also to integrate such functions e.g. the Lebesgue integral which is important for probability theory.)

  - At least if $f\colon [a, b] \to \mathbb{R}$ is continuous, $\int_a^b f(\xi)d\xi$ can be shown to be well-defined.

    (Then always $|\max f([x_i, x_{i+1}]) - \min f([x_i, x_{i+1}])| \to 0$ for $|x_{i+1} - x_i| \to 0$.)

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- In general, one has to show that the approximations of the area $\int_a^b f(\xi)d\xi$ by

$$\sum_{i=0}^{n-1} f(\xi_i)(x_{i+1} - x_i) \quad \text{(with } a = x_0 < x_1 < \ldots < x_{n-1} < x_n = b \text{ and } \xi_i \in [x_i, x_{i+1}]\text{)}$$

all converge to the same value for $\Delta := \max(x_{i+1} - x_i) \to 0$ independent on how we partition the interval $[a, b]$ s.t. the area is indeed well-defined.

This leads to the Darboux integral resp. Riemann integral.

**Thm** Fundamental theorem of calculus

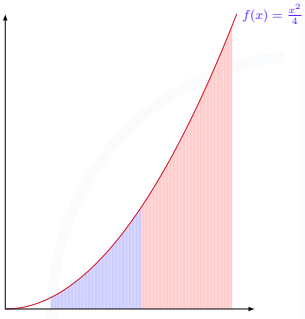
If $f\colon [a, b] \to \mathbb{R}$ is continuous, then the sums always converge to the same value, i.e. $\int_a^b f(\xi)d\xi$ is well-defined as the limit of these approximations.

The limit of theses sums define a continuous function $F\colon [a, b] \to \mathbb{R}$ with $F(x) := \int_a^x f(\xi)d\xi$ which is also differentiable on $(a, b)$ with $F' = f$.

For any antiderivative $F$ of $f$ on $[a, b]$, we have $\int_a^b f(\xi)d\xi = F(b) - F(a)$.

- ▷ Main idea for last part: $F(x_{i+1}) - F(x_i) \approx f(\xi_i) \cdot (x_{i+1} - x_i)$, s.t. $\sum_{i=0}^{n-1} f(\xi_i)(x_{i+1} - x_i) \approx \sum_{i=0}^{n-1} (F(x_{i+1}) - F(x_i)) = F(b) - F(a)$.
(E.g. Newtonian mechanics: $v = s'$, $s = s_0 + \int_0^t v dt$; cf. also Euler's method.)

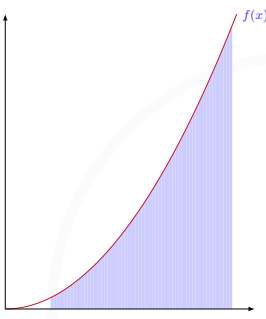
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- If $f\colon [1,b] \to \mathbb{R}$ is continuous, also the area $A\colon [1,5] \to \mathbb{R}, t \mapsto \int_1^t f(x)dx$ is a well-defined function.
- The change of the area $A(t+h) - A(t) = \int_t^{t+h} f(x)dx$ is roughly the rectangular area $f(\xi)h$ for any $\xi \in [t, t+h]$ if $h$ is small (continuity).
- As $f$ is continuous, $\lim_{h\to 0} \frac{A(t+h)-A(t)}{h} = f(t)$, i.e. $A' = f$ at least on $(a,b)$.
- For any antiderivative $F$ of $f$ we have $A = F + c$ for some constant, hence:
  $\int_a^b f(x)dx = A(b) = A(b) - A(a) = (F(b)+c) - (F(a)+c) = F(b) - F(a)$.

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# Definite integral: quadratic function

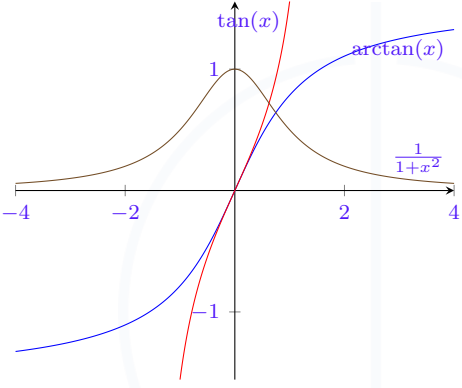



- $f(x) = 0.25x^2$ on $[1, 5]$ with $F(x) = \frac{1}{12}x^3$: $\int_1^5 f(x)dx = [\frac{x^3}{12}]_1^5 = 10.\overline{3}$
- For comparison: $\sum_{i=0}^{n-1} f(a + i\frac{b-a}{n}) \cdot \frac{b-a}{n}$:

| $n$ | 4 | 8 | 16 | 32 | 64 |
|---|---|---|---|---|---|
| $\sum$ | 7.5, | 8.875 | 9.59375 | 9.9609375 | 10.146484375 |

Alternative approximation e.g. using trapezoids (i.e. $\frac{f(x)+f(x+h)}{2} \cdot h$) might converge faster, but all converge to $F(b) - F(a)$ for $h \to 0$.

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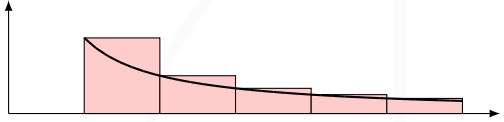

$$\tan'(x) = \frac{\cos(x)}{\cos(x)} + \frac{\sin(x)^2}{\cos(x)^2} = 1 + \tan(x)^2 \qquad x \in (-\pi/2, \pi/2)$$

$$(\arctan(x))' = \frac{1}{(\tan)'(\arctan(x))} = \frac{1}{1+x^2}$$

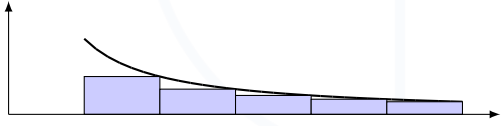
$$\int_0^1 \frac{1}{1+x^2} dx = \int_0^1 \arctan'(x) dx = [\arctan(x)]_0^1 = \frac{\pi}{4}$$

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$$\ln(n) + \frac{1}{n} < \sum_{k=1}^{n} \frac{1}{k} < 1 + \ln(n)$$



$$\sum_{k=1}^{n} \frac{1}{k} = \frac{1}{n} + \sum_{k=1}^{n-1} \frac{1}{k} \geq \frac{1}{n} + \int_{1}^{n} \frac{1}{x} dx$$



$$\sum_{k=1}^{n} \frac{1}{k} = 1 + \sum_{k=2}^{n} \frac{1}{k} \leq 1 + \int_{1}^{n} \frac{1}{x} dx$$

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**Def** If $f$ is only continuous on $[a,b)$ ($-\infty < a < b \leq \infty$), then

$$\int_a^b f(x)\,dx := \lim_{t\to b} \int_a^t f(x)\,dx$$

is called an improper integral if the limit exists. Analogously for $f\colon (a,b] \to \mathbb{R}$.

In case of $f\colon (a,b) \to \mathbb{R}$ consider the sum of the improper integrals on $(a,c]$ and $[c,b)$ for any $c \in (a,b)$

**Lem** Partial integration and substitution assuming continuous integrands:

If $f, g\colon [a,b] \to \mathbb{R}$ are continuously differentiable, then

$$\int_a^b f'(x)g(x)\,dx = [f\cdot g]_a^b - \int_a^b f(x)g'(x)dx$$

If $f\colon I \to \mathbb{R}$ and $g\colon [a,b] \to I$ are both continuously differentiable, then

$$\int_a^b f'(g(x))\cdot g'(x)\,dx = [f\circ g]_a^b = [f]_{g(a)}^{g(b)} = \int_{g(a)}^{g(b)} f'(x)\,dx$$

(Think of $g(x) \in [a,b]$ as position and $g'(x)$ as the speed at which we move through $[a,b]$.)

Note that this can be read as rule in both directions.

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# Examples

**EG** With $f(x) = x, g'(x) = e^{-x}, f'(x) = 1, g(x) = -e^{-x}$

$$\int_0^b x \cdot e^{-x}\, dx = [x(-e^{-x})]_0^b - \int_0^b (-e^{-x})\, dx = -be^{-b} - [e^{-x}]_0^b = 1 - (b+1)e^{-b} \xrightarrow{b\to\infty} 1$$

**EG** $\int_0^t \cos(x)dx = \sin(t)$ for $t \in [0, \infty)$, but $\int_0^\infty \cos(x)dx$ is undefined.

**EG** Consider

$$\int_0^a \sin(2x+1)\, dx$$

Set $g(x) := 2x + 1$ with $g'(x) = 2$.

$$\int_0^a \sin(g(x)) \cdot \tfrac{2}{2}\, dx = \tfrac{1}{2} \int_0^a \sin(g(x)) \cdot g'(x)\, dx = \int_1^{2a+1} \sin(x)\, dx = \ldots$$

**Lem** In general for a linear transformation $g(x) = cx + d$ with $c \neq 0$:

$$\int_a^b f(cx+d)dx = \tfrac{1}{c} \int_{ca+d}^{cb+d} f(x)\, dx$$

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# Example: integration by substitution

EG $f\colon [0,1] \to \mathbb{R}, x \mapsto \sqrt{1-x^2}$ describes a quarter circle. Hence, we expect

$$\int_0^1 \sqrt{1-x^2}dx = \pi/4$$

As $g\colon [0,\pi/2] \to [0,1], x \mapsto \sin(x)$ is continuously differentiable:

$$\begin{aligned}
& \int_0^1 \sqrt{1-x^2}dx \\
= \; & \int_{g(0)}^{g(\pi/2)} f(x)dx \\
= \; & \int_0^{\pi/2} f(g(y)) \cdot g'(y)dy \\
= \; & \int_0^{\pi/2} \sqrt{1-\sin(y)^2} \cdot \cos(y)dy \\
= \; & \int_0^{\pi/2} \sqrt{\cos(y)^2} \cdot \cos(y)dy \\
= \; & \int_0^{\pi/2} |\cos(y)| \cdot \cos(y)dy \\
= \; & \int_0^{\pi/2} \cos(y)^2dy && (\cos(x) \geq 0 \text{ on } [0,\pi/2]\ ) \\
= \; & \tfrac{1}{2}\left[x + \sin(x)\cos(x)\right]_0^{\pi/2} && (\text{cf. table; or compare to } \textstyle\int \sin(x)^2dx) \\
= \; & \pi/4
\end{aligned}$$

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# Example: area between two functions, volume of rotation symmetric objects

EG Area enclosed by two continuous functions $f, g$ with antiderivative $F, G$.

Assuming $c \leq f(x) \leq g(x)$ for $x \in [a, b]$, the area is

$$\int_a^b (g(x)-c)dx - \int_a^b (f(x)-c)dx = [G-c]_a^b - [F-c]_a^b = [G-F]_a^b = \int_a^b (f-g)(x)dx$$

If $f - g$ has a root $r \in [a, b]$, split the integral.

EG Volume of e.g. rotation symmetric (rotation around $x$ axis) objects.

Instead of inscribing rectangles of height $f(x)$ and width $h = \frac{b-a}{n}$, use discs of radius $f(x)$ and width $h$ and thus volume $\pi f(x)^2 h$:

$$\sum_{i=0}^{n} \pi \cdot f(a+ih)^2 \cdot h \xrightarrow{n\to\infty} \pi \int_a^b f(x)^2 dx$$

E.g. volume of a sphere with radius $r$:

$$\pi \int_{-r}^{r} \left(\sqrt{r^2 - x^2}\right)^2 dx = 2\pi \int_0^r (r^2 - x^2)dx = 2\pi[r^2 x - x^3/3]_0^r = \tfrac{4\pi}{3} \cdot r^3$$

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# Example: Gamma function (*)

EG The gamma function is defined by the improper integral

$$\Gamma(z) := \int_0^\infty e^{-x} x^{z-1} dx$$

One can show that $\Gamma(z)$ is well-defined for all $z \in \mathbb{R}_{>0}$.

We only consider $n \in \mathbb{N}$ and show that $\Gamma(n) = (n-1)!$ for all $n \in \mathbb{N}$.

I.B.: For $n = 1$

$$\Gamma(1) = \lim_{b\to\infty} \int_0^b e^{-x} dx = \lim_{b\to\infty} (1 - e^{-b}) = 1$$

I.S.: Fix any $n \in \mathbb{N}$ and assume (I.A.) $\Gamma(n) = (n-1)!$.

We show that then also (I.C.) $\Gamma(n+1) = n!$ using partial integration:

$$\int_0^b e^{-x} x^{(n+1)-1} dx = [-e^{-x} x^n]_0^b + n \int_0^b e^{-x} x^{n-1} dx \xrightarrow[n\geq 1]{b\to\infty} n\Gamma(n) = n!$$

(See also the previous example for $n = 2$.)

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5 Differentiation and Integration

- Derivative
- Differentiation Rules
- Antiderivative
- Integration and Fundamental Theorem of Calculus
- L'Hospital (*)
- Taylor series, extrema (*)
- Newton's Method (*)

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**Thm** L'Hospital's rule (actually J. Bernoulli's rule):

Let $f, g\colon (a,b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ on $(a,b)$. If

$\lim_{a<x\to a} f(x) = \lim_{a<x\to a} g(x) = 0$

or $\lim_{a<x\to a} g(x) = \infty$ (resp. $\lim_{a<x\to a} g(x) = -\infty$)

and $\lim_{a<x\to a} \frac{f'(x)}{g'(x)}$ exists, then then

$$\lim_{a<x\to a} \frac{f(x)}{g(x)} = \lim_{a<x\to a} \frac{f'(x)}{g'(x)}$$

(also for $a = -\infty$; "$a < x \to a$": "from above")

Analogously for $x \to b \in \mathbb{R} \cup \{\infty\}$ with $x < b$ ("from below").

**EG** $\lim_{x\to 0} \frac{1-\cos(x)}{x} = \lim_{x\to 0} \frac{\sin(x)}{1} = 0$

**EG** $\lim_{x\to 0} \frac{1-\cos(x/2)}{1-\cos(x)} = \lim_{x\to 0} \frac{\frac{1}{2}\sin(x/2)}{\sin(x)} = \lim_{x\to 0} \frac{\frac{1}{4}\cos(x/2)}{\cos(x)} = \frac{1}{4}$

(as the last limit exists, the second and thus also the first exist and all coincide)

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Let $f, g\colon (a,b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ on $(a,b)$. If

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(also for $a = -\infty$; "$a < x \to a$": "from above")

Analogously for $x \to b \in \mathbb{R} \cup \{\infty\}$ with $x < b$ ("from below").

**EG** $\lim_{0<x\to 0} x\ln(x) = \lim_{0<x\to 0} \frac{\ln(x)}{1/x} = \lim_{0<x\to 0} \frac{1/x}{-1/x^2} = \lim_{0<x\to 0} -x = 0$

$\lim_{0<x\to 0} x^x = \lim_{0<x\to 0} e^{x\ln(x)} = e^{\lim_{0<x\to 0} x\ln(x)} = e^0 = 1$

⚠ $\lim_{n\to\infty}(1+\frac{x}{n})^n = e^{\lim_{n\to\infty} \frac{\ln(1+x/n)}{1/n}} = e^{\lim_{n\to\infty} \frac{(1+x/n)^{-1}\cdot(-x/n^2)}{-1/n^2}} = e^x$

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**Thm** L'Hospital's rule (actually J. Bernoulli's rule):

Let $f, g\colon (a,b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ on $(a,b)$. If

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(also for $a = -\infty$; "$a < x \to a$": "from above")

Analogously for $x \to b \in \mathbb{R} \cup \{\infty\}$ with $x < b$ ("from below").

▷ Proof idea: In the special case $f(a) = g(a) = 0$

$$\frac{f(x)}{g(x)} \overset{f(a)=g(a)=0}{=\!=} \frac{f(x)-f(a)}{g(x)-g(a)} \overset{x\geq a}{=\!=} \frac{\frac{f(x)-f(a)}{x-a}}{\frac{g(x)-g(a)}{x-a}} \xrightarrow{x\to a} \frac{f'(a)}{g'(a)}$$

(The general proof requires the mean value theorem; see analysis.)

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- Newton's Method (*)

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**Def** Let $f\colon I \to \mathbb{R}$ be $n$-times differentiable at $x^* \in I$ ($I \subseteq \mathbb{R}$ interval).

Then the $n$-th Taylor polynomial at $x^*$ is defined as

$$[T_{n;x^*} f](x) := \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x^*) \cdot (x - x^*)^k$$

so that its coincides with $f$ at $x^*$ up to the $n$-th derivative:

$$[T_{n;x^*} f]^{(k)}(x) = f^{(k)}(x^*) \text{ for } 0 \leq k \leq n$$

- Examples:
  - For $n = 0$ we have the constant approximation at $x^*$:

    $$f(x) \approx f(x^*)$$

    For $n = 1$ we obtain the linearization of $f$ at $x^*$:

    $$f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

    For $n = 2$ we obtan a parabola as approximation of $f$ at $x^*$:

    $$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2} f''(x^*)(x - x^*)$$

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**Def** Let $f\colon I \to \mathbb{R}$ be $n$-times differentiable at $x^* \in I$ ($I \subseteq \mathbb{R}$ interval).

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so that its coincides with $f$ at $x^*$ up to the $n$-th derivative:

$$[T_{n;x^*} f]^{(k)}(x) = f^{(k)}(x^*) \text{ for } 0 \leq k \leq n$$

- Examples:
  - With $\frac{d}{dx} e^x = e^x$ and $e^0 = 1$ we obtain:

$$[T_{n;0} \exp](x) = 1 + x + \tfrac{1}{2}x^2 + \ldots + \tfrac{1}{n!}x^n$$

With $\frac{d}{dx} \sin(x) = \cos(x)$, $\frac{d}{dx} \cos(x) = -\sin(x)$, $\sin(0) = 0$ and $\cos(0) = 1$:

$$\begin{aligned} [T_{2n+1;0} \sin](x) &= x - \tfrac{1}{6}x^3 + \tfrac{1}{120}x^5 + \ldots + \tfrac{(-1)^{2n+1}}{(2n+1)!}x^{2n+1} \\ [T_{2n;0} \cos](x) &= 1 - \tfrac{1}{2}x^2 + \tfrac{1}{24}x^4 + \ldots + \tfrac{(-1)^{2n}}{(2n)!}x^{2n} \end{aligned}$$

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Thm Analogously to the approximation error of the linearization we have

$$e_{n;x^*}(x) := \frac{f(x) - [T_{n;x^*} f](x)}{(x - x^*)^n} \xrightarrow{x \to x^*} 0$$

i.e. the error decreases asymptotically faster than $(x - x^*)^n$ for $x \to x^*$.

▷ (*) Proof idea:

E.g. for $n = 2$:

$$\begin{aligned} & \lim_{x \to x^*} \frac{f(x) - (f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2} f''(x^*)(x - x^*)^2)}{(x - x^*)^2} \\ \overset{\text{l'Hospital}}{=} & \lim_{x \to x^*} \frac{f'(x) - (f'(x^*) + f''(x^*)(x - x^*))}{2(x - x^*)} \\ = & \frac{1}{2} \left( \lim_{x \to x^*} \left( \frac{f'(x) - f'(x^*)}{x - x^*} \right) - f''(x^*) \right) = 0 \end{aligned}$$

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Thm Analogously to the approximation error of the linearization we have

$$e_{n;x^*}(x) := \frac{f(x) - [T_{n;x^*}f](x)}{(x-x^*)^n} \xrightarrow{x \to x^*} 0$$

i.e. the error decreases asymptotically faster than $(x-x^*)^n$ for $x \to x^*$.

Cor If $p$ is a polynomial of degree at most $n$ with $\lim_{x\to x^*} \frac{f(x)-p(x)}{(x-x^*)^n} = 0$, then also

$$0 = \lim_{x\to x^*} \frac{[T_{n;x^*}f](x)-f+f-p(x)}{(x-x^*)^n} = \lim_{x\to x^*} \frac{[T_{n;x^*}f](x)-p(x)}{(x-x^*)^n}$$

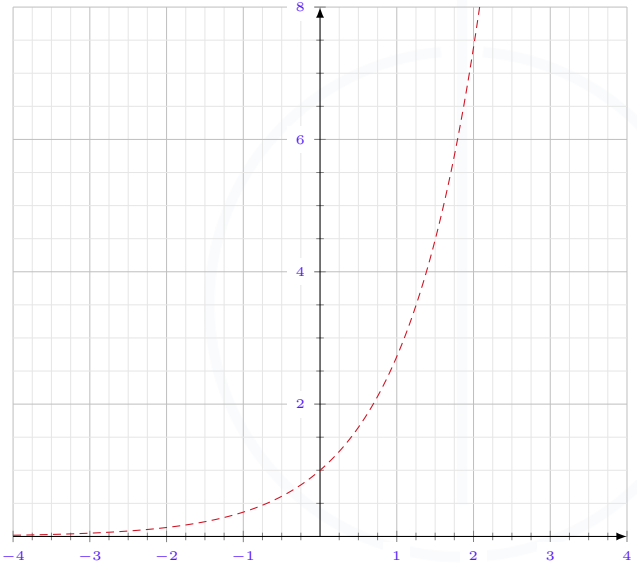
and thus $p = [T_{n;x^*}f]$ (i.e. $T_{n;x^*}f$ is the "best" polynomial of degree $\leq n$.)

▷ Application: If $f'(x^*) = \ldots = f^{(n-1)}(x^*) = 0$ and $f^{(n)}(x^*) \neq 0$, then locally

$$f(x^*+h) = f(x^*) + \left(\frac{f^{(n)}(x^*)}{n!} + \frac{e_{n;x^*}(x^*+h)}{h^n}\right) \cdot h^n \approx f(x^*) + \frac{f^{(n)}(x^*)}{n!}h^n$$

which allows to decide whether we have a local extremum ($n$ even) or a saddle point ($n$ odd).

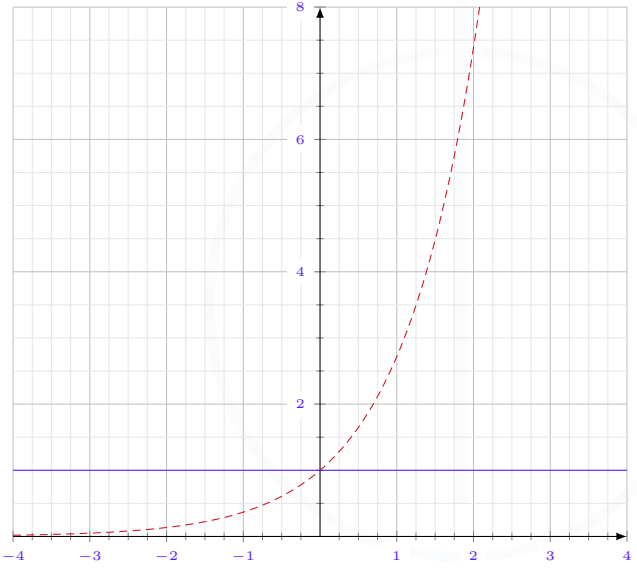
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$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 + \tfrac{1}{24}x^4 + \dots$$

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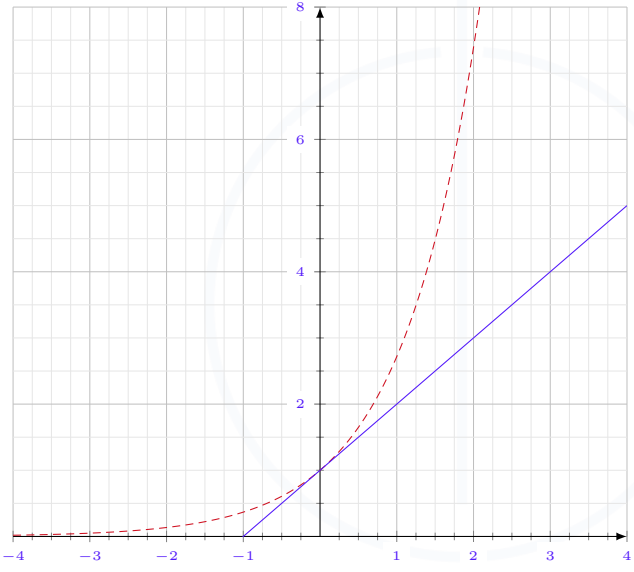
# Example: Taylor expansion of exp at 0




$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 + \tfrac{1}{24}x^4 + \dots$$

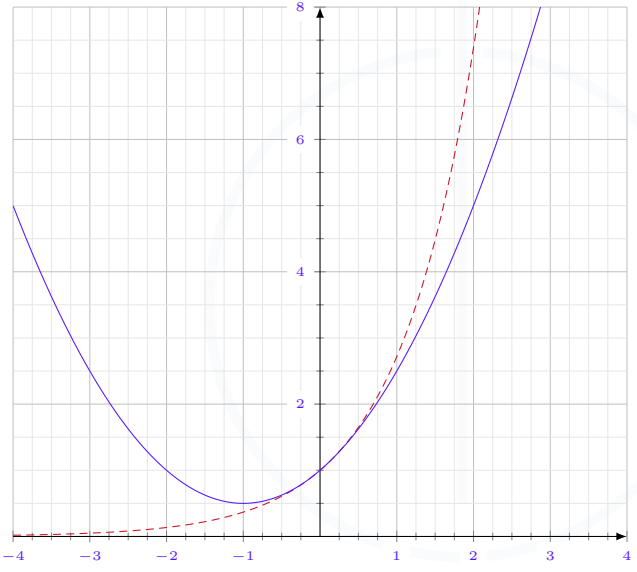
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# Example: Taylor expansion of $\exp$ at $0$




$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 + \tfrac{1}{24}x^4 + \dots$$

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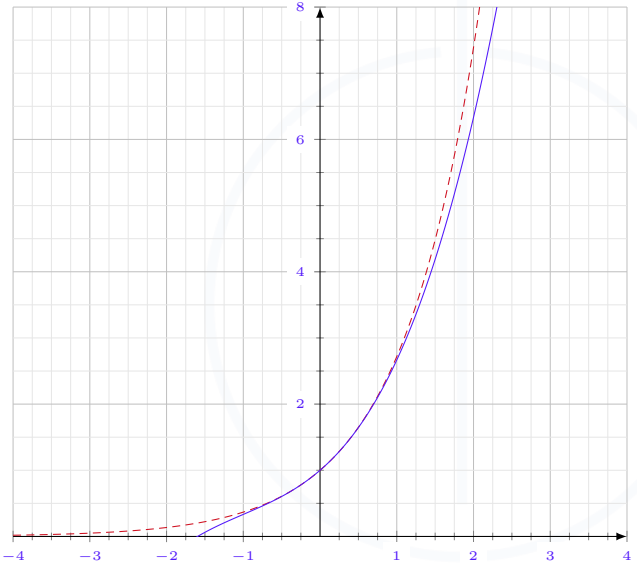



$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 + \tfrac{1}{24}x^4 + \dots$$

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# Example: Taylor expansion of $\exp$ at $0$



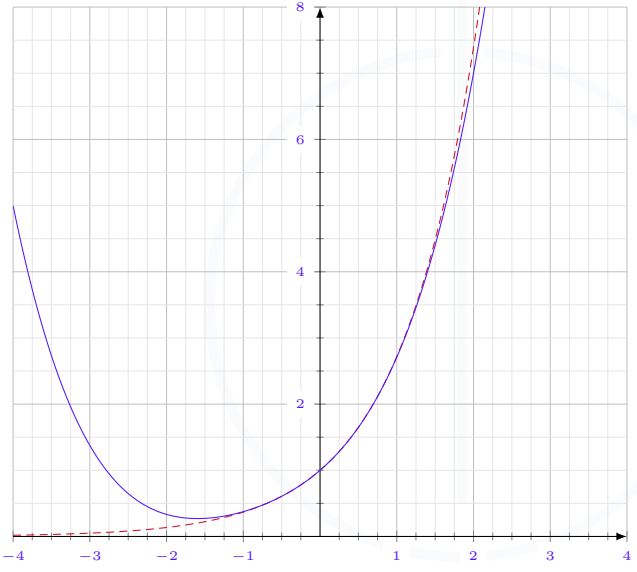



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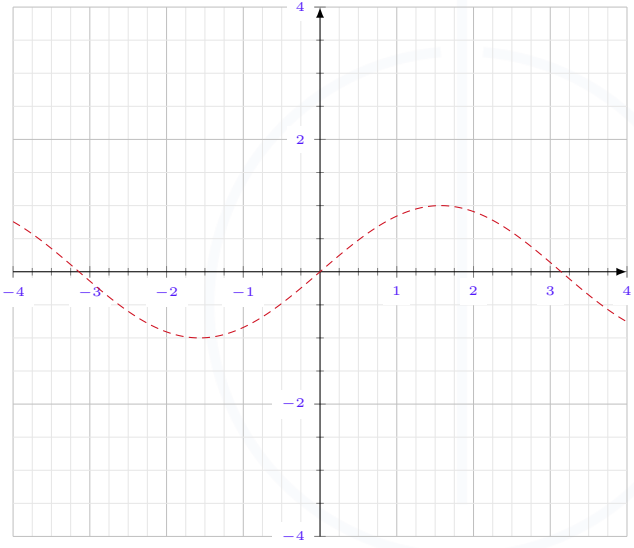



$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 + \tfrac{1}{24}x^4 + \dots$$

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# Example: Taylor expansion of $\sin$ at $0$



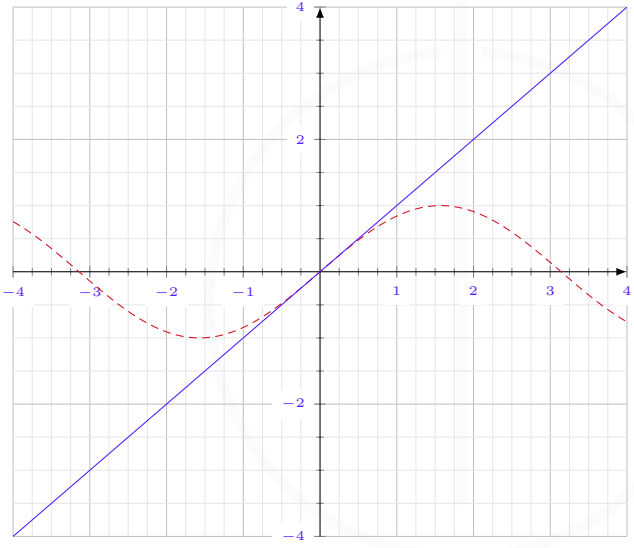



$$\sin(x) = x - \tfrac{1}{6}x^3 + \tfrac{1}{120}x^5 + \dots$$

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# Example: Taylor expansion of $\sin$ at $0$



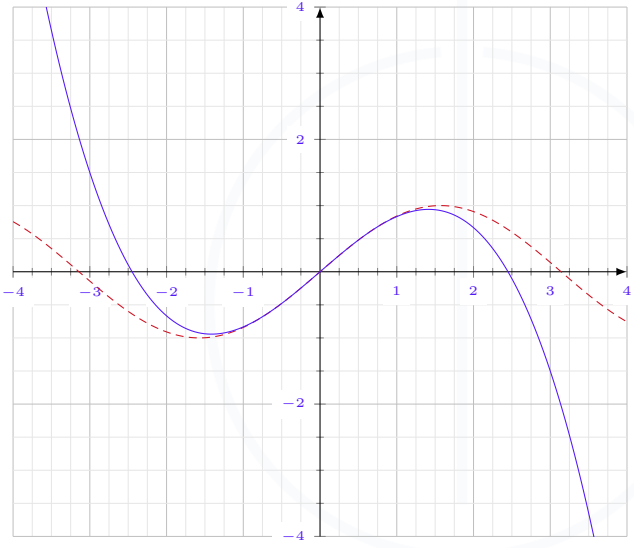



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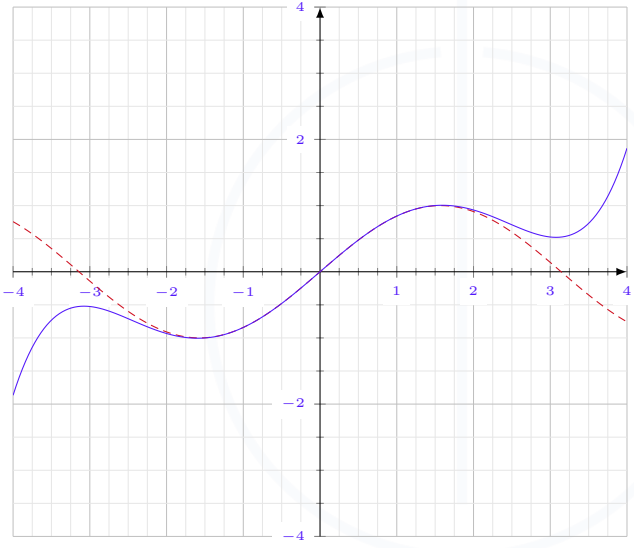



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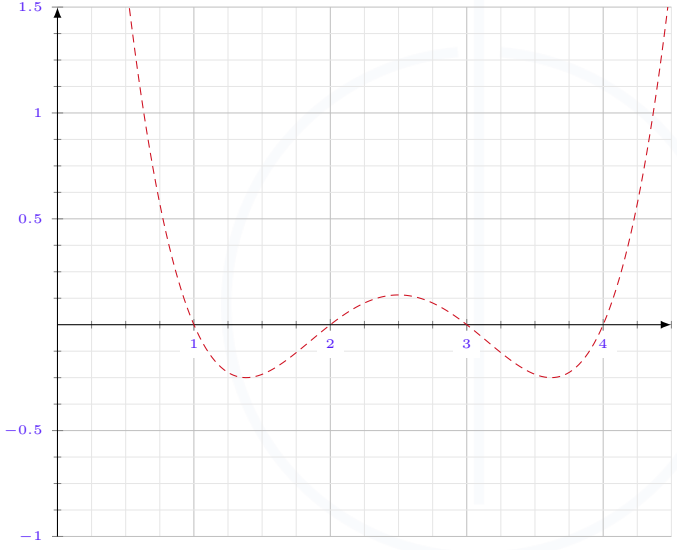



$$\sin(x) = x - \tfrac{1}{6}x^3 + \tfrac{1}{120}x^5 + \dots$$

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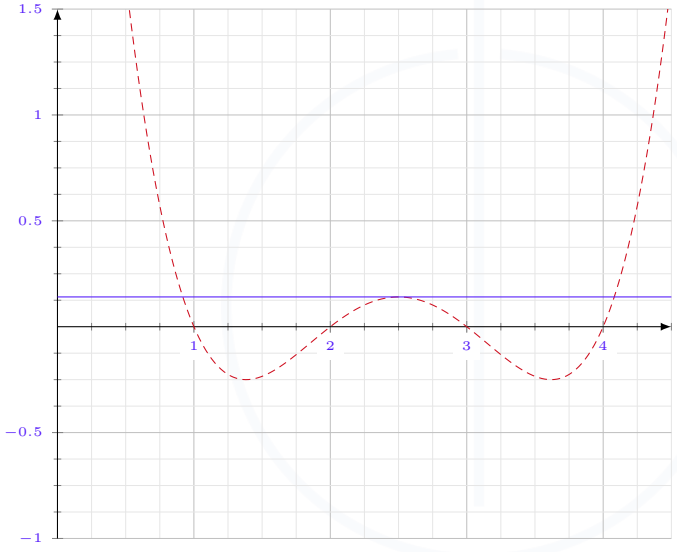
# Example: Taylor expansion of a polynom





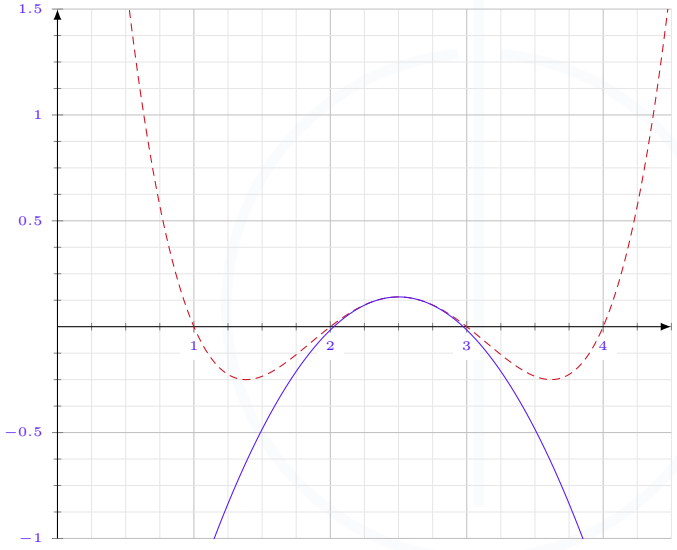

$$0.25(x-1)(x-2)(x-3)(x-4) = 0.140625 + 0.625(x-2.5)^2 + 0.25(x-2.5)^4$$

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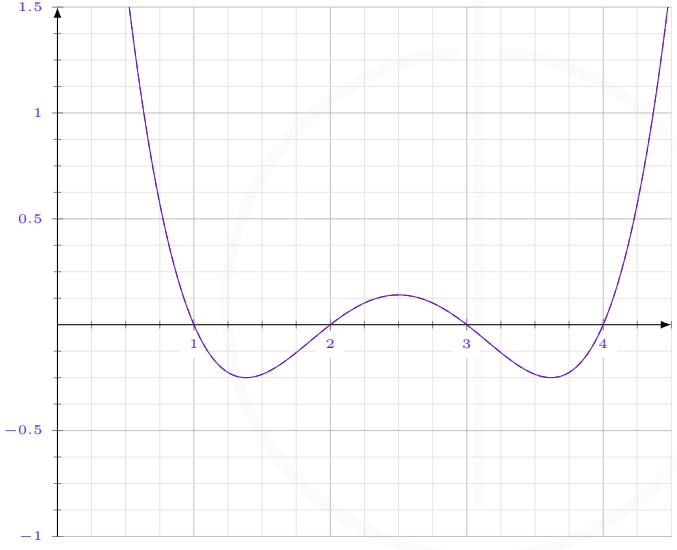

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**Lem** Let $f$ be continuous on $[a,b]$ and differentiable on $(a,b)$. Then:

- (necessary) If $\xi \in (a,b)$ is a local extremum, then $f'(\xi) = 0$.

  i.e. there ex. $\delta > 0$ s.t. for all $x \in (\xi - \delta, \xi + \delta)$:

  $f(\xi) \geq f(x)$ (local maximum) resp. $f(\xi) \leq f(x)$ (local minimum)

  (sufficient) If $f'' = \frac{d}{dx} f'$ exists, and

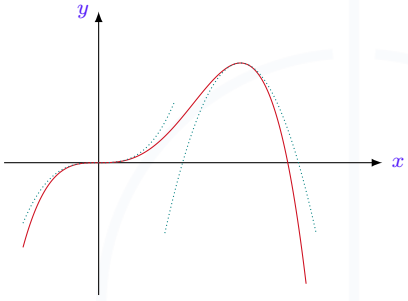
  - $f'(\xi) = 0$ and $f''(\xi) > 0$, then $f$ has a local minimum at $\xi$.
  - $f'(\xi) = 0$ and $f''(\xi) < 0$, then $f$ has a local maximum at $\xi$.

  (Alternatively: $f'$ changes its sign at $\xi$.)

- $f$ is mon. increasing (resp. decreasing) on $[a,b]$ **iff** $f'(\xi) \geq 0$ (resp. $f'(\xi) \leq 0$) for all $\xi \in (a,b)$.
- $f$ is constant on $[a,b]$ **iff** $f'(\xi) = 0$ for all $\xi \in (a,b)$.
- If $f'(\xi) > 0$ (resp. $f'(\xi) < 0$) for all $\xi \in (a,b)$,
  then $f$ is stricty mon. increasing (resp. decreasing).
  (Only sufficient: consider e.g. $f(x) = x^3$, use e.g. higher derivatives/Taylor.)

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# Example: extrema of a differentiable function




- $f(x) = 5x^3(1-x) = 5x^3 - 5x^4$:
  - $f'(x) = 5x^2(3-4x)$, $f''(x) = 30x(1-2x)$, $f'''(x) = 30(1-4x)$
  - Candidates for local extrema: $f'(x) = 0$ **iff** $x = 0$ or $x = \frac{3}{4}$
  - (global) maximum at $x = \frac{3}{4}$: $f(x) = \frac{135}{256} - \frac{45}{8}(x - \frac{3}{4})^2 + \ldots$
  - Saddle point at $x = 0$: $f(x) = 5(x-0)^3 - 5(x-0)^4$
  - Strictly mon. increasing on $(-\infty, \frac{3}{4}]$: $f'''(0) > 0$.
  - Strictly mon. decreasing on $[\frac{3}{4}, \infty)$.

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# Newton's method

- Many applications lead to systems of equations as formal descriptions of some aspect of system.
- If the equations are non-linear, then a standard approach is to approximate them by linear equations and solve these instead. The classical example is that of Newton's method:

**Def** Newton's method

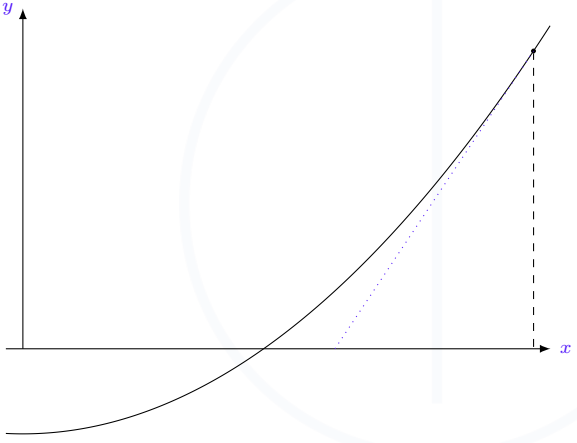
- Given: $f\colon \mathbb{R} \to \mathbb{R}$ differentiable on $\mathbb{R}$; initial approximation $x_0$
- Wanted: Approximation of a zero of $0 = f(x)$
- Ansatz: For $k = 0, 1, \ldots$ iterate until $f(x_k) \approx 0$
  1. Approximate $f$ at $x_k$ by the line $g_k(x) := f'(x_k) \cdot (x - x_k) + f(x_k) = [T_{1;x_k} f](x)$
  2. Choose $x_{k+1}$ as the root of $g_k(x)$: $x_{k+1} := x_k - f(x_k)/f'(x_k)$.
  3. Set $k := k + 1$

⚠ In general, Newton's method will not converge; it's convergence depends on the choice of $x_0$. But Newton's method can converge must faster than bisection, e.g. when approximating principal roots $\sqrt[n]{c}$, i.e. the unique positive root of $f(x) = x^n - c$ (as function on $\mathbb{R}$ for some fixed $c \in \mathbb{R}_{>0}$).

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# Newton's method: Example

- $f(x) = \frac{1}{4}x^2 - \frac{1}{2}$ with $f(x) = 0$ for $x = \sqrt{2} \approx 1,4142$
- Approximations:

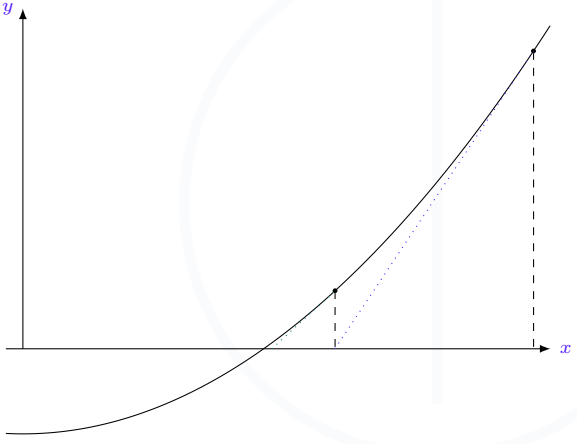



$x_0 \approx 3, x_1 \approx 1.8334$

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# Newton's method: Example

- $f(x) = \frac{1}{4}x^2 - \frac{1}{2}$ with $f(x) = 0$ for $x = \sqrt{2} \approx 1,4142$
- Approximations:

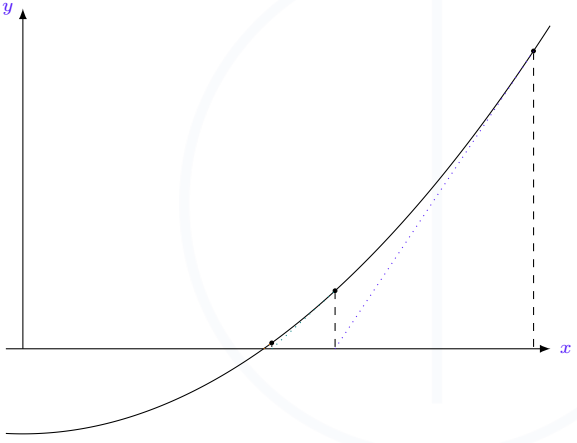



$x_1 \approx 1.834, x_2 \approx 1.46227$

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# Newton's method: Example

- $f(x) = \frac{1}{4}x^2 - \frac{1}{2}$ with $f(x) = 0$ for $x = \sqrt{2} \approx 1,4142$
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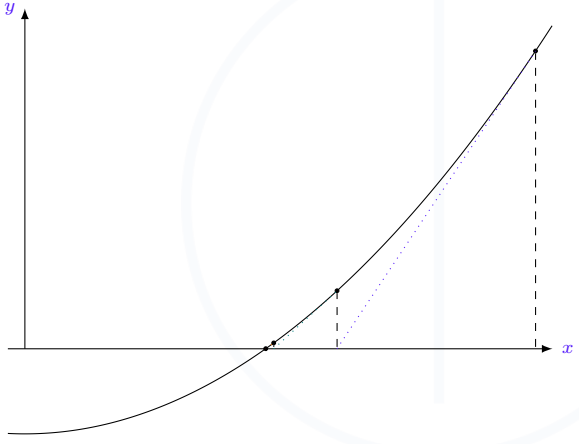



$x_2 \approx 1.462, x_3 \approx 1.41502$

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# Newton's method: Example

- $f(x) = \frac{1}{4}x^2 - \frac{1}{2}$ with $f(x) = 0$ for $x = \sqrt{2} \approx 1,4142$
- Approximations:




$x_3 \approx 1.415, x_4 \approx 1.41425$

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**Lem** Let $c \in \mathbb{R}_{>0}$ and $n \in \mathbb{N}$ with $n \geq 2$.

Then Newton's method applied to $f(x) = x^n - c$, i.e.

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = x_i - \frac{x_i^n - y}{n x_i^{n-1}} = \frac{(n-1)x_i^n + c}{n x_i^{n-1}}$$

converges to $\sqrt[n]{c}$ starting from any $x_0 >$ (e.g. $x_0 = c$).

- ▷ Note: if $c$ is rational, all approximations are also rational.

▷ Example: $x^n = 2$ for $n \in \{2, 3, 4, 5\}$ with always $x_0 = 2$.

| $n$ | $x_0$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $\sqrt[k]{2}$ |
|---|---|---|---|---|---|---|
| 2 | 2 | 1.5 | 1.41667 | 1.41422 | 1.41421 | 1.414213... |
| 3 | 2 | 1.5 | 1.2963 | 1.26093 | 1.25992 | 1.259921... |
| 4 | 2 | 1.5625 | 1.30295 | 1.20325 | 1.18945 | 1.189207... |
| 5 | 2 | 1.625 | 1.35736 | 1.20373 | 1.1535 | 1.148698... |

(Use Python to compare Newton and bisection for approximating $\sqrt[n]{a}$.)

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